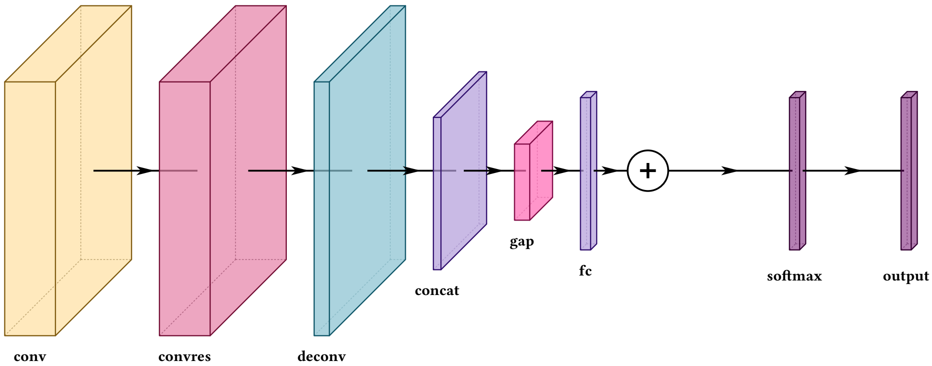
VINNT

Vision Illustration of Neural Networks in Typst



Version 0.2.0 · J.C. Vaught

Every figure in this manual is compiled from the code printed beside it.

CHAPTER 0

Contents

CHAPTER 1

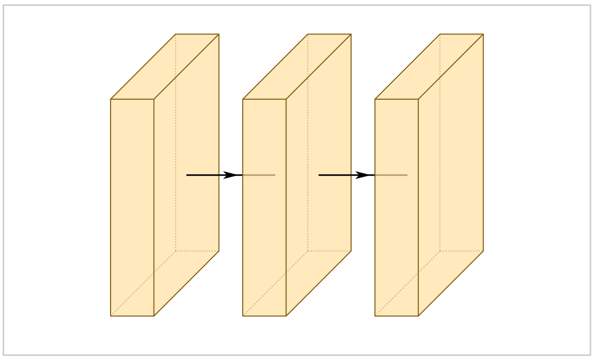
What this is

VINNT draws layered neural network architectures as isometric block diagrams, the kind that open a paper's method section. It is a Typst package built on CeTZ.

You describe the network as an array of layers. The package works out the geometry, the spacing, the arrows between blocks, the routing of any skip connection you add, and the legend.

It also draws the other canonical figure — the neuron-and-edge diagram, a circle per unit — with `draw-mlp`, which has a chapter of its own (chapter 14).

```
#draw-network((
  conv(),
  conv(),
  conv(),
))
```



Three convolutions, no options at all. Sizes, spacing and arrows are defaults.

1.1 How this manual is organised

Every option is shown on its own, usually across several figures, because one picture of an option tells you it exists and three tell you what it does. Where an option has a failure mode, the failure is shown next to the fix. Each chapter ends with a composite figure built from what the chapter covered, and chapter 15 is nothing but composites.

Complete published architectures — AlexNet, VGG, ResNet, U-Net, FCN-8, YOLO26-n and six RGB-IR fusion variants — are not here. They are whole readable files in `examples/` in the repository, and they are better read as files than quoted as fragments.

1.2 Installing

```
#import "@preview/vinnt:0.2.0": draw-network, conv, pool, input
```

Every example in this manual assumes an import like that one. Import `draw-network` plus whichever constructors you use, or take everything with `: *`, which is what the examples here do.

> VINNT is not yet published on Typst Universe. Until it is, point the import at a local copy: `#import "path/to/src/lib.typ": *`, which is what every example in this manual does.

1.3 The shape of a figure

```
#draw-network(
  layers,
  connections: (),
  groups: (),
  ...
)
```

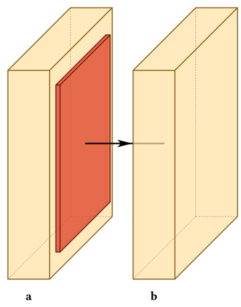
Everything in this manual is either an option on a layer, an option on a connection or a group, or one of those figure-wide settings.

CHAPTER 2

Writing a layer

A layer is a constructor call. Every type has one, named after it, and its arguments are the options:

```
#draw-network((
  conv(label: "a"),
  pool(),
  conv(label: "b"),
))
```



A constructor's signature is exactly the set of options that type accepts, so an editor can list them while you type, and Typst rejects a misspelled argument by name before the package ever sees it. A plain dictionary in the layer list is an error rather than a second way to write the same thing, so every figure reads the same. When options arrive as a dictionary anyway — shared settings, an imported dump — spread it into the constructor: `conv(..opts)`.

2.1 Options that do not exist are errors

An option a layer type does not read is rejected, not ignored:

```
error: unknown layer option "hieght" on layer 3 (type "conv").
Did you mean "height"? Options accepted here: bandfill, channels,
connection-label, depth, fill, height, image, label, ...
```

This matters more than it sounds. A key that is quietly ignored produces a figure that is merely wrong, and a wrong figure reads as the package being broken rather than as the typo it is. The check covers layer options, connection options, group options, unknown layer types, a layer with no `type`, and a connection or group naming a layer that has no such `name`.

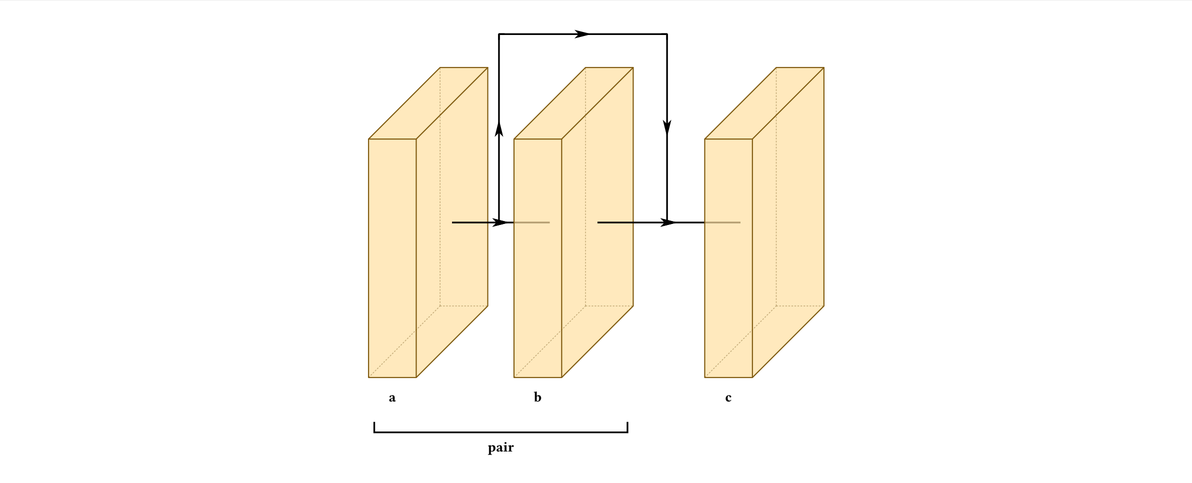
> The suggestion searches the options that type actually accepts, so `widths` on a `pool` is caught as readily as `hieght` on a `conv`. The first is spelled correctly and still means nothing there.

2.2 Naming a layer

`name` default `none`

An identifier for this layer. Never printed. Connections and groups find a layer by it, and nothing else does, so a layer with no name cannot be referred to.

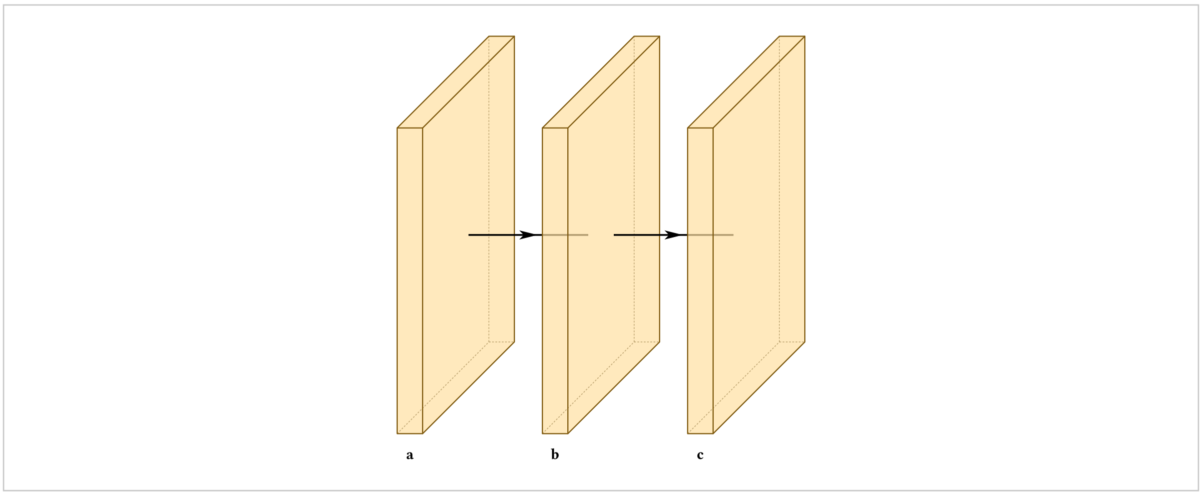
```
#draw-network(
  (
    conv(name: "a", label: "a"),
    conv(name: "b", label: "b"),
    conv(name: "c", label: "c"),
  ),
  connections: (connection(from: "a", to: "c"),),
  groups: (group(from: "a", to: "b", label: "pair"),),
)
```



2.3 The list is just an array

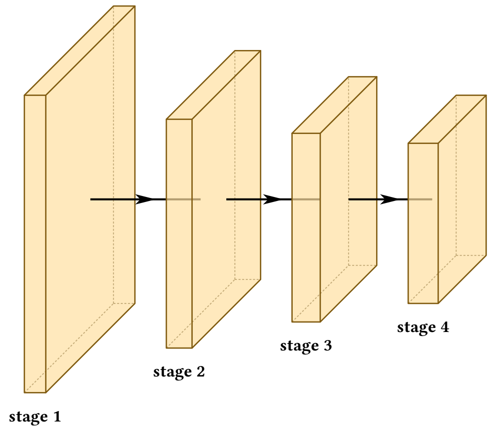
Nothing requires you to write layers out one by one. The list is an array and a constructor is a function, so ordinary Typst builds both. Sharing options across several layers is a spread:

```
#let big = (height: 6, depth: 6, widths: (0.5,))

#draw-network((
  conv(label: "a", ..big),
  conv(label: "b", ..big),
  conv(label: "c", ..big),
))
```



And generating layers is a `map`:

```
#let stage(n) = conv(
  label: "stage " + str(n),
  shape: (32 * n, 128 / n, 128 / n),
)

#draw-network(range(1, 5).map(stage))
```



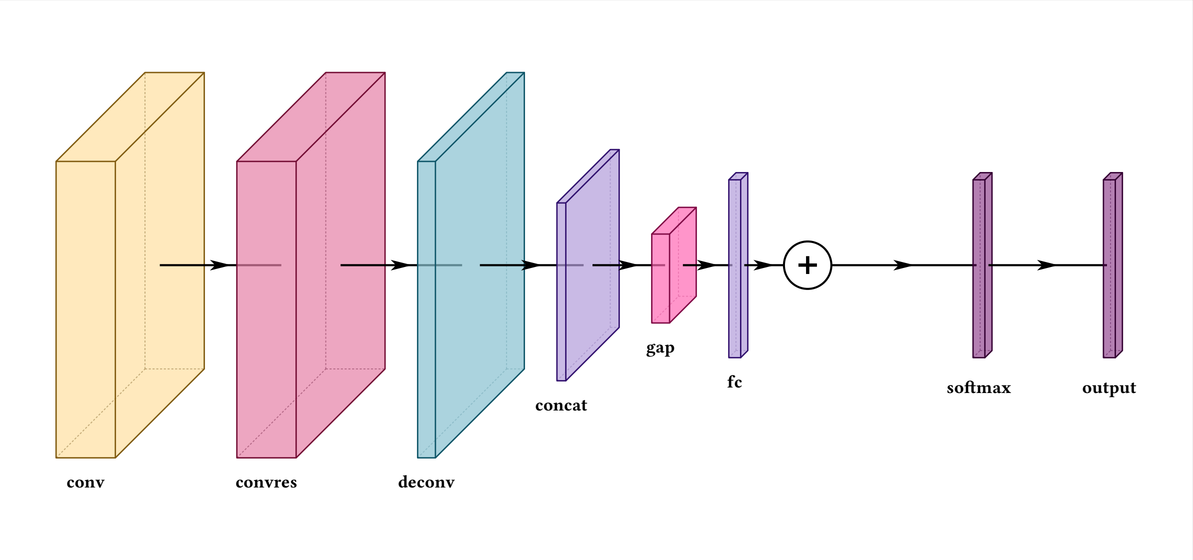
This is the difference between a figure you maintain and a figure you rewrite. Section 15.5 takes it further.

CHAPTER 3

Layer types

Fourteen types are predefined. Each carries a colour, a default size, a default legend entry, and in three cases a behaviour.

```
#draw-network((
  conv(label: "conv"),
  convres(label: "convres"),
  deconv(label: "deconv"),
  concat(label: "concat"),
  gap(label: "gap"),
  fc(label: "fc"),
  sum(),
  softmax(label: "softmax"),
  output(label: "output", offset: 2),
))
```



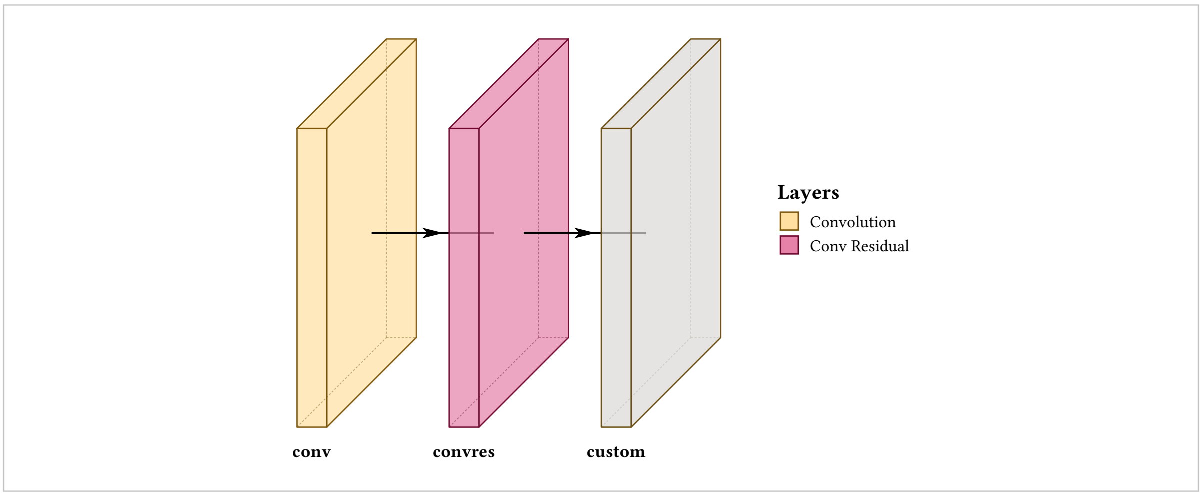
Also predefined: `input`, `pool`, `unpool`, `convsoftmax`, `custom`.

Type governs colour and defaults, not capability. Any type can be resized, recoloured, relabelled and given an image. Choose by what the block **means**: the type decides what the legend calls it and what colour a reader will come to associate with it across your figures.

3.1 The convolution family

`conv` and `convres` differ only in colour and legend text. `custom` is the same block with no meaning attached, and is the one you extend (chapter 9).

```
#draw-network(
  (
    conv(label: "conv", widths: (0.5,)),
    convres(label: "convres", widths: (0.5,)),
    custom(label: "custom", width: 0.5),
  ),
  show-legend: true,
)
```



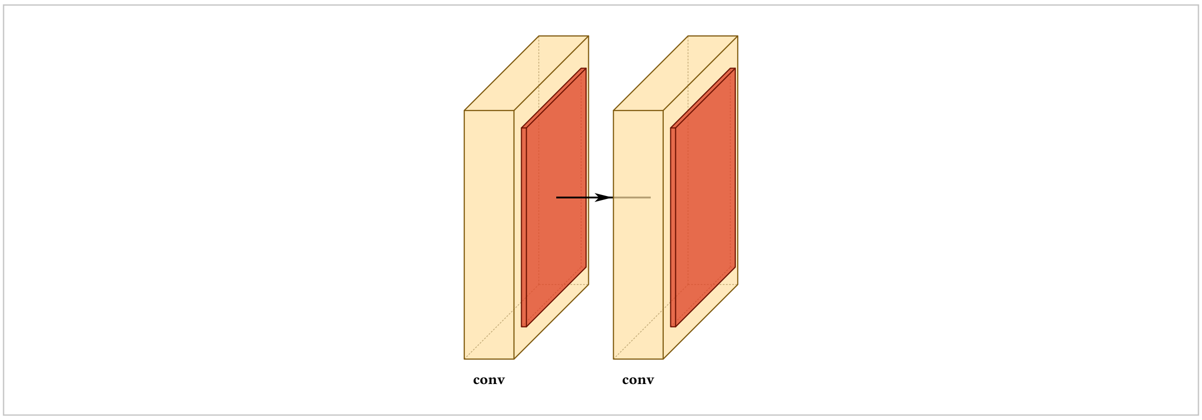
3.2 Pooling attaches to the block in front

offset default `auto`

read specially by `pool` and `unpool`

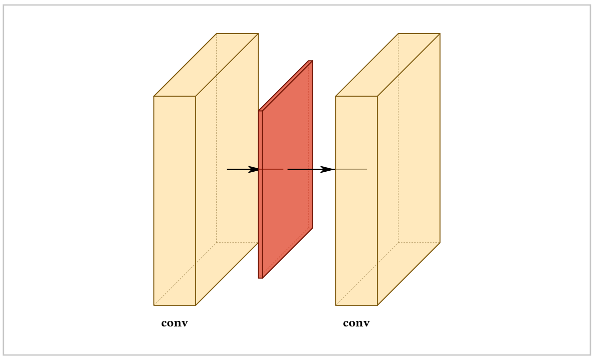
On `pool` and `unpool`, leaving `offset` unset attaches the layer to the block before it, with no arrow between them. That is what makes a convolution-plus-pool read as one stage rather than two.

```
#draw-network((
  conv(label: "conv"),
  pool(),
  conv(label: "conv"),
  pool(),
))
```



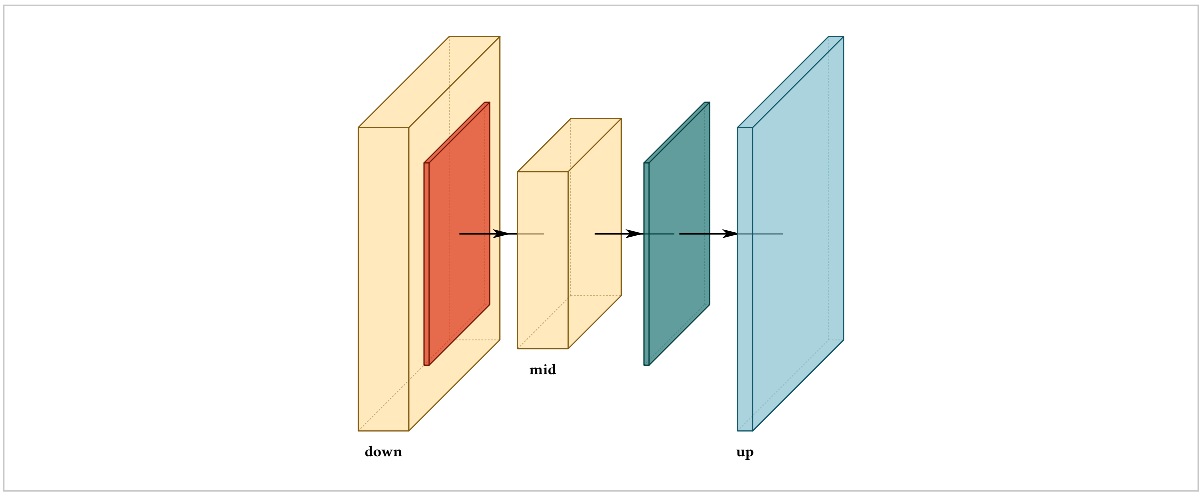
Give one an `offset` and it detaches, and the axis arrow appears:

```
#draw-network((
  conv(label: "conv"),
  pool(offset: 1.5),
  conv(label: "conv"),
))
```



An encoder-decoder uses both, `pool` on the way down and `unpool` on the way up:

```
#draw-network((
  conv(label: "down", height: 6, depth: 6),
  pool(height: 4, depth: 4),
  conv(label: "mid", height: 3.5, depth: 3.5),
  unpool(height: 4, depth: 4, offset: 1.5),
  deconv(label: "up", height: 6, depth: 6),
))
```



3.3 The sum node

`sum` is a node, not a block. It has a `radius` where other types have a width, and a `symbol` you can replace.

`radius` default `0.4`

`sum`

Size of the disc.

`symbol` default `+`

`sum`

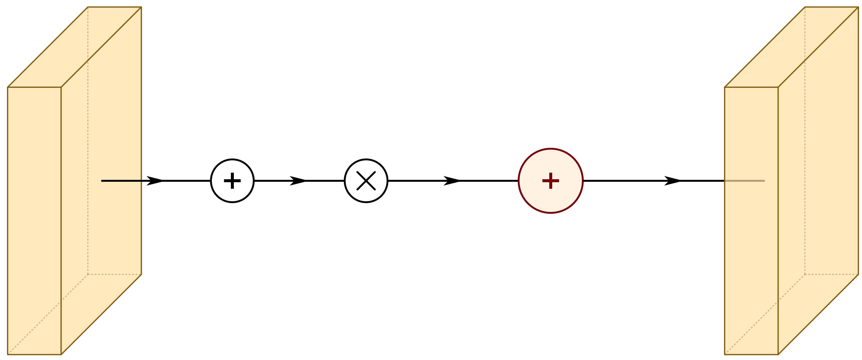
Any content, centred in the disc.

`stroke` default `black`

`sum`

Outline colour.

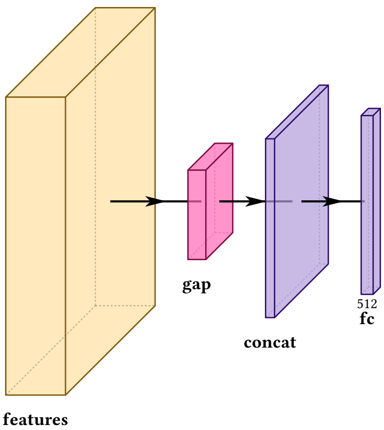
```
#draw-network((
  conv(),
  sum(),
  sum(symbol: [$times$]),
  sum(radius: 0.6, fill: rgb("#FFF2E3"), stroke: rgb("#73000A")),
  conv(),
))
```



3.4 Pooling to a vector, and joining

`gap` collapses a feature map to a vector and is small by default. `concat` joins two paths, and is the usual target for a fan of arriving routes (section 10.6).

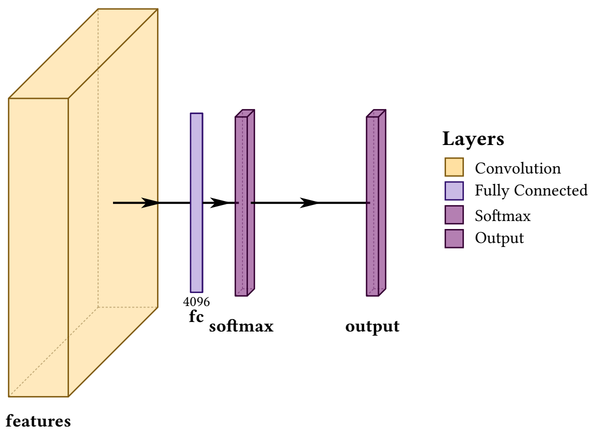
```
#draw-network((
  conv(label: "features", height: 5, depth: 5),
  gap(label: "gap"),
  concat(label: "concat"),
  fc(label: "fc", channels: (512,)),
))
```



3.5 Classifier heads

`fc`, `softmax`, `convsoftmax` and `output` are the tail of a classifier. `fc` with `depth: 0` draws as a flat rectangle, which is what a vector looks like.

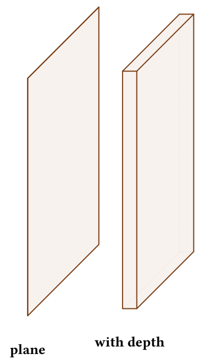
```
#draw-network(
  (
    conv(label: "features"),
    fc(label: "fc", channels: (4096,), depth: 0),
    softmax(label: "softmax"),
    output(label: "output", offset: 2),
  ),
  show-legend: true,
)
```



3.6 The input layer

`input` defaults to a flat plane with no thickness, which is what an image is. Give it a `width` and a `depth` to make it a block instead. It is also the one type with no outgoing arrow by default (section 10.9).

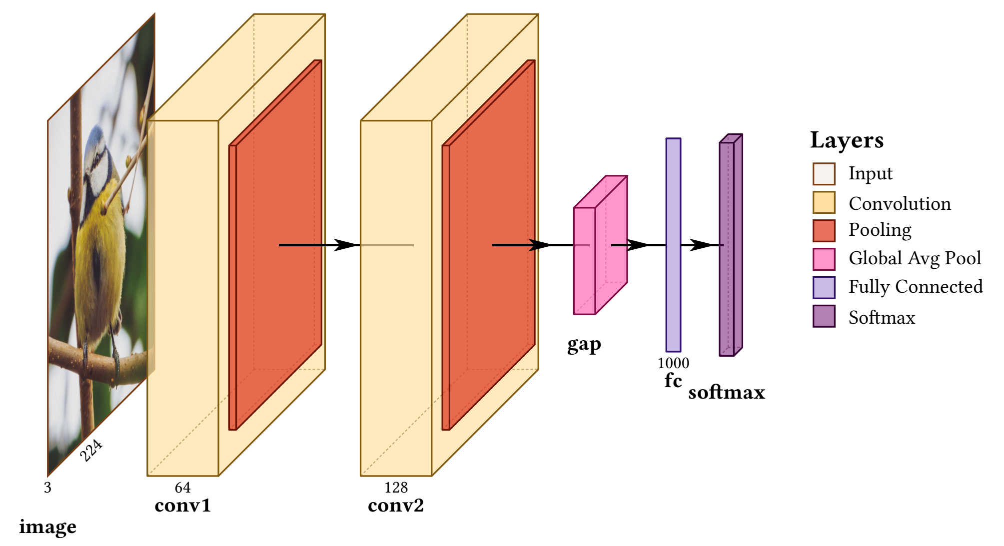
```
#draw-network((
  input(label: "plane"),
  input(label: "with depth", depth: 4, width: 0.3, offset: 2),
))
```



3.7 Putting the types together

Nothing below sets a size, a colour or a gap. It is types, labels and channel counts only, and it is already a readable figure.

```
#draw-network(
  (
    input(image: "default", label: "image", channels: (3, 224)),
    conv(label: "conv1", channels: (64,), offset: 1.4),
    pool(),
    conv(label: "conv2", channels: (128,)),
    pool(),
    gap(label: "gap"),
    fc(label: "fc", channels: (1000,), depth: 0),
    softmax(label: "softmax"),
  ),
  show-legend: true,
)
```



CHAPTER 4

Sizing a block

A block has three dimensions. `height` and `depth` are the spatial extent of the tensor; thickness is `width` or `widths`, and conventionally means channel count.

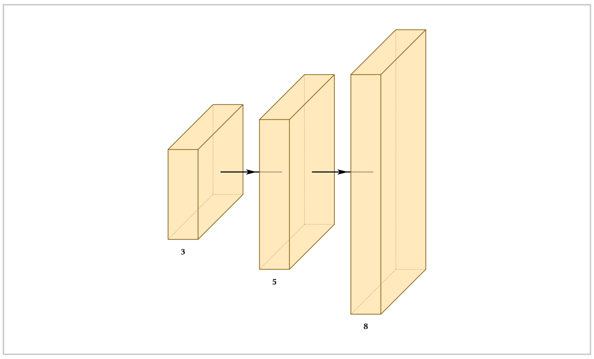
height default `5, varies by type`

Vertical extent, in canvas units.

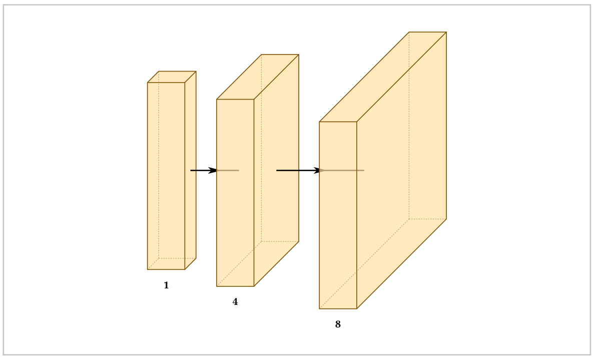
depth default `5, varies by type`

Extent along the projection. `0` draws a flat rectangle.

```
#draw-network((
  conv(label: "3", height: 3),
  conv(label: "5", height: 5),
  conv(label: "8", height: 8),
))
```




```
#draw-network((
  conv(label: "1", depth: 1),
  conv(label: "4", depth: 4),
  conv(label: "8", depth: 8),
))
```



4.1 Thickness

width default `varies`

`input`, `deconv`, `concat`, `convsoftmax`, `softmax`, `output`, `custom`

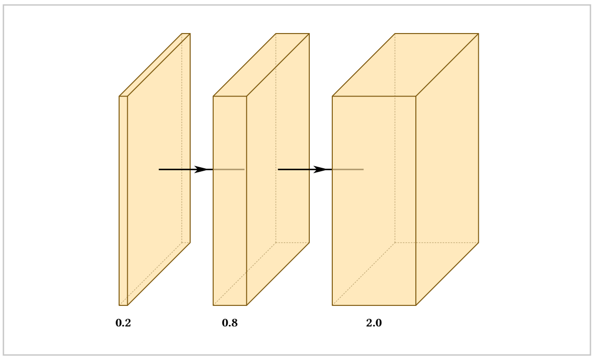
A single thickness.

widths default `(1,)`

`conv`, `convres`, `custom`

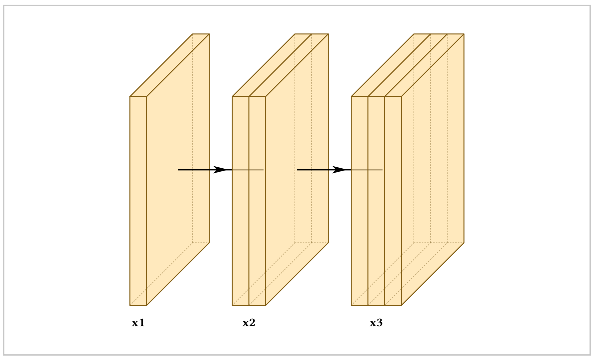
An array of thicknesses, drawing one band per entry inside a single block.

```
#draw-network((
  conv(label: "0.2", widths: (0.2,)),
  conv(label: "0.8", widths: (0.8,)),
  conv(label: "2.0", widths: (2.0,)),
))
```



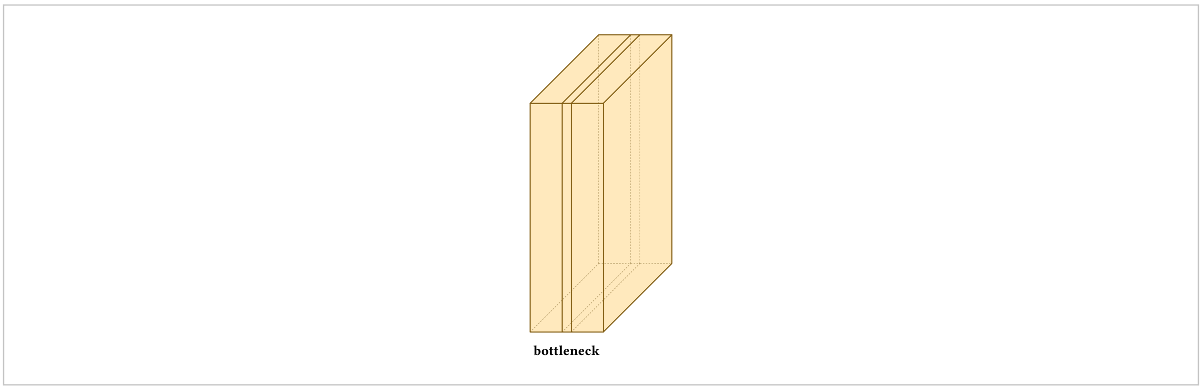
`widths` is how a stage of two or three convolutions at one resolution is normally drawn: one block, banded, rather than three blocks in a row.

```
#draw-network((
  conv(label: "x1", widths: (0.4,)),
  conv(label: "x2", widths: (0.4, 0.4)),
  conv(label: "x3", widths: (0.4, 0.4, 0.4)),
))
```



The bands need not be equal, which is how you draw a bottleneck:

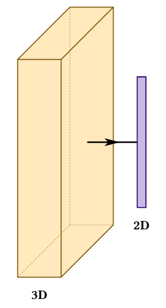
```
#draw-network((
  conv(label: "bottleneck", widths: (0.7, 0.2, 0.7)),
))
```



> `width` and `widths` are different options and not interchangeable. No type accepts both except `custom`, and using the wrong one is an error naming the other.

4.2 Flat layers

```
#draw-network((
  conv(label: "3D", depth: 4),
  fc(label: "2D", depth: 0),
))
```



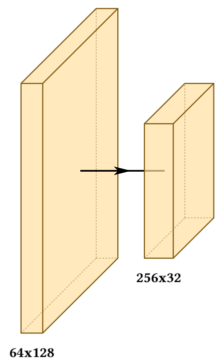
4.3 Sizing from the tensor shape

Stating three numbers per block and keeping them consistent across a pyramid is the tedious part of drawing one of these, and the part that rots when the model changes.

shape default `none`

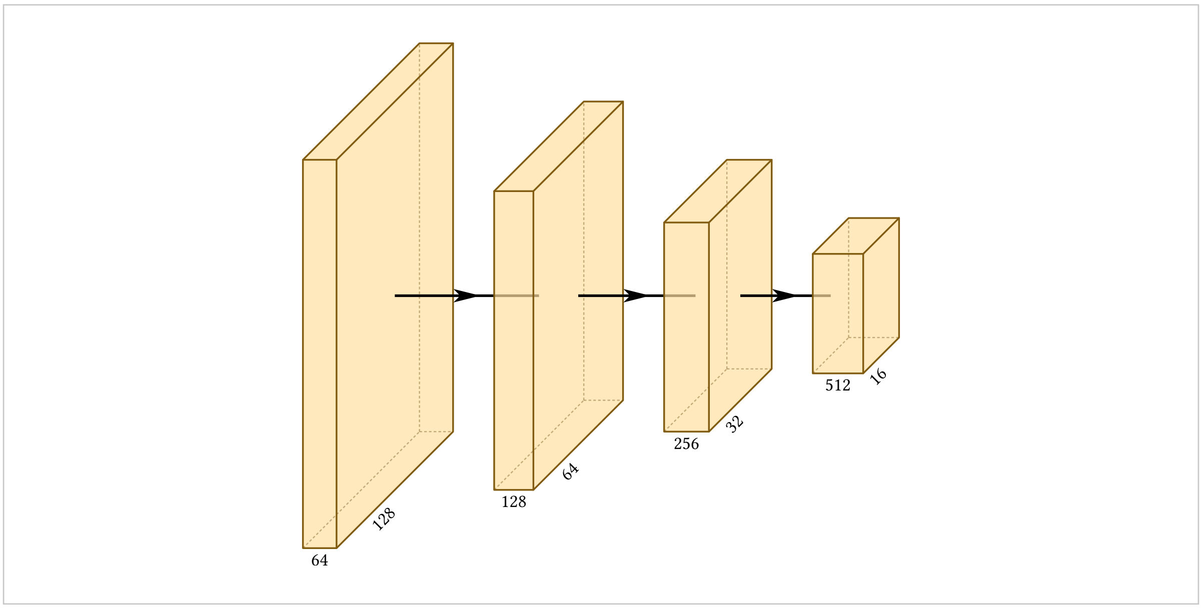
The tensor this layer produces, as `(channels, height, width)`. Height, depth and width are derived from it.

```
#draw-network((
  conv(shape: (64, 128, 128), label: "64x128"),
  conv(shape: (256, 32, 32), label: "256x32"),
))
```



Across a pyramid it is the whole geometry:

```
#draw-network((
  conv(shape: (64, 128, 128), channels: (64, 128)),
  conv(shape: (128, 64, 64), channels: (128, 64)),
  conv(shape: (256, 32, 32), channels: (256, 32)),
  conv(shape: (512, 16, 16), channels: (512, 16)),
))
```



Both axes are logarithmic:

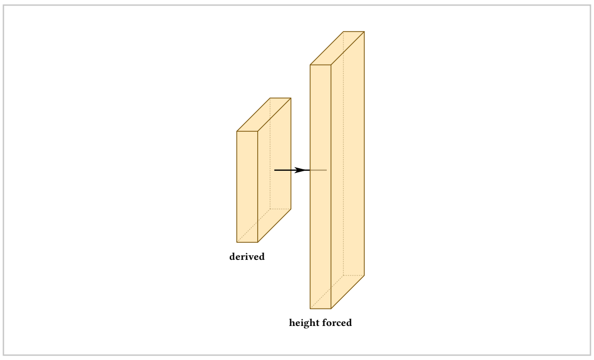
$$\text{height}, \text{depth} = 1.2 \log_2(\text{spatial}) - 3.2, \quad \text{width} = 0.075 \log_2(\text{channels})$$

Linear spatial extent does not work. Across a 640-to-20 pyramid it puts the smallest block at a quarter of a unit against 8 for the input, which is invisible.

Shape supplies defaults, not values

Anything you state explicitly wins, field by field, so a block can take its width and depth from its shape while its height is forced:

```
#draw-network((
  conv(label: "derived", shape: (256, 40, 40)),
  conv(
    label: "height forced",
    shape: (256, 40, 40),
    height: 7,
  ),
))
```



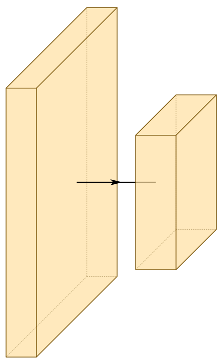
Changing the mapping

`shape-scale` default `(spatial: (1.2, -3.2), channels: (0.075, 0.0))`

`draw-network`

Slope and intercept for each axis, applied to the base-2 logarithm. The constants are absolute rather than normalised across a figure, so the same shape gives the same size everywhere and two figures stay comparable.

```
#draw-network(
  (conv(shape: (64, 128, 128)), conv(shape: (256, 32, 32))),
  shape-scale: (
    spatial: (2.0, -6.0),
    channels: (0.15, 0.0),
  ),
)
```

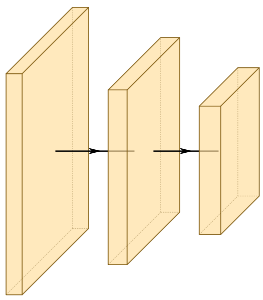


> Only `conv`, `convres` and `custom` take a derived width, because only those three use thickness to mean channel count. The others have a fixed thickness that says something else, and keep it.

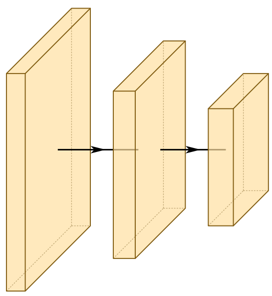
4.4 By hand against by shape

The same three blocks, written both ways. The first drifts the moment the model changes; the second cannot.

```
#draw-network((
  conv(height: 6.2, depth: 6.2, widths: (0.45,)),
  conv(height: 4.9, depth: 4.9, widths: (0.53,)),
  conv(height: 3.6, depth: 3.6, widths: (0.60,)),
))
```



```
#draw-network((
  conv(shape: (64, 128, 128)),
  conv(shape: (128, 64, 64)),
  conv(shape: (256, 32, 32)),
))
```



CHAPTER 5

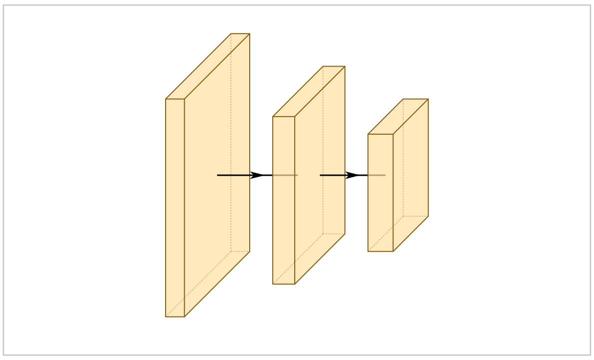
Spacing

offset

default auto

The gap between this layer and the one before it. `auto` computes it from the drawing; a number states it directly.

```
#draw-network((
  conv(shape: (64, 128, 128)),
  conv(shape: (128, 64, 64)),
  conv(shape: (256, 32, 32)),
))
```

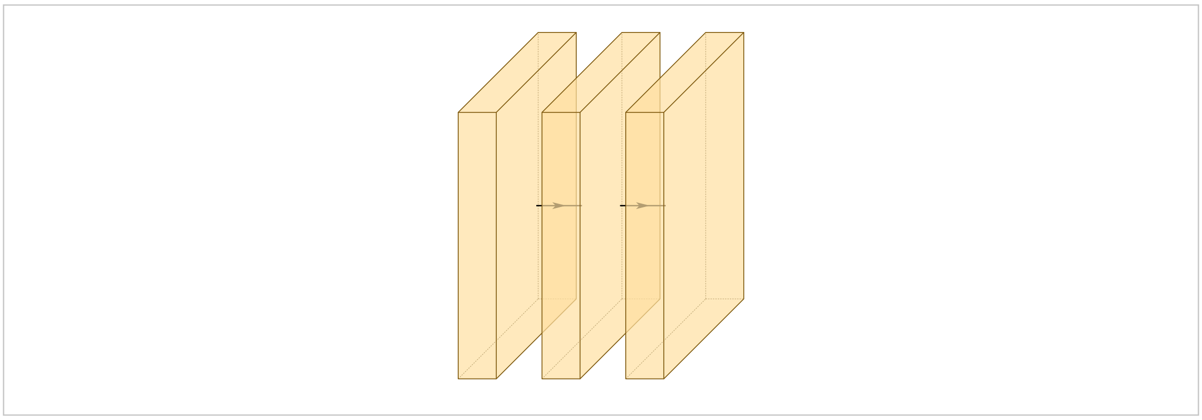


Automatic spacing covers the previous block's isometric lean, adds a fixed strip of white space, and widens where a connection descends into the gap.

5.1 Why a constant offset fails

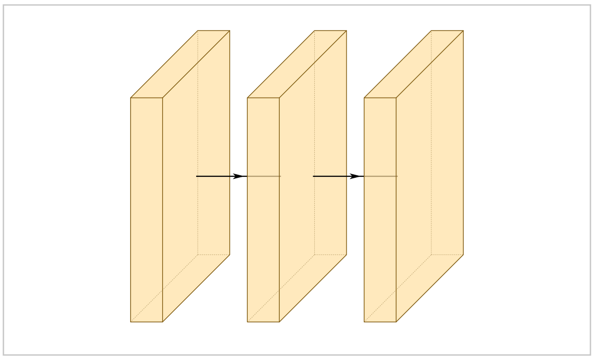
A block drawn in isometric projection leans right by `depth × depth-multiplier`. A gap narrower than that lean buys no visible space at all: the next block is drawn inside the previous one's sheared face, and the arrow between them disappears behind it.

```
#draw-network((
  conv(height: 7, depth: 7),
  conv(height: 7, depth: 7, offset: 1.2),
  conv(height: 7, depth: 7, offset: 1.2),
))
```



`offset: 1.2` against a depth-7 block, whose lean is 2.1.

```
#draw-network((
  conv(height: 7, depth: 7),
  conv(height: 7, depth: 7),
  conv(height: 7, depth: 7),
))
```



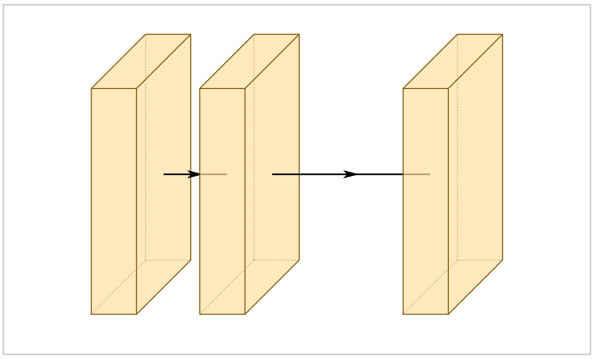
The same blocks with the offsets removed.

That is why `auto` is the default. A constant that works for one size of block fails for the next, and a figure that computes its own offsets ends up restating the size pyramid twice, with nothing to catch the two copies drifting apart.

5.2 Spacing by hand

A number overrides the automatic gap for one layer, and means what it always meant:

```
#draw-network((
  conv(depth: 4),
  conv(depth: 4, offset: 1.4),
  conv(depth: 4, offset: 3.5),
))
```

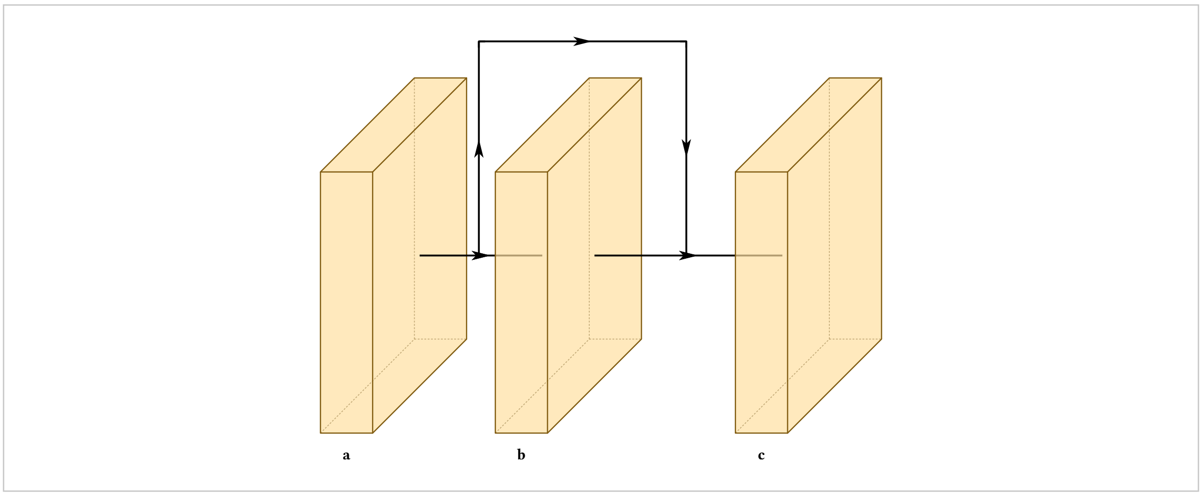


Two helpers are exported for figures that do compute their own geometry:

```
depth-shear(6)
min-clear-offset(6)
```

A connection descending between two blocks arrives at the midpoint of the arrow joining them, so it clears the lean only when half the offset exceeds it. That is what `min-clear-offset` returns, and what automatic spacing applies for you:

```
#draw-network(
  (
    conv(name: "a", label: "a", depth: 6),
    conv(name: "b", label: "b", depth: 6),
    conv(name: "c", label: "c", depth: 6),
  ),
  connections: (connection(from: "a", to: "c"),),
)
```



Both helpers take a `depth-multiplier` argument, which must match the one passed to `draw-network`. The default multiplier of `0.3` has no exact binary representation, so `depth-shear(4.5)` is `1.3499999999999999`. Compare with a tolerance, if you compare at all.

5.3 The white space constant

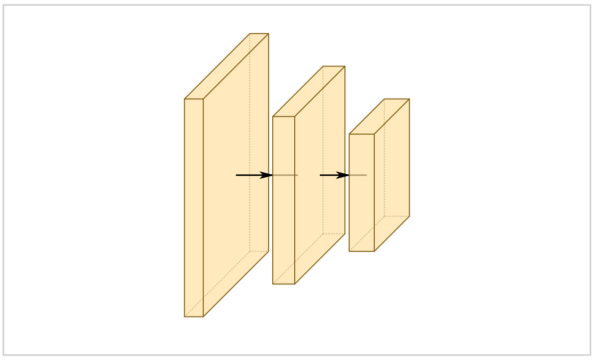
`auto-gap`

default `0.55`

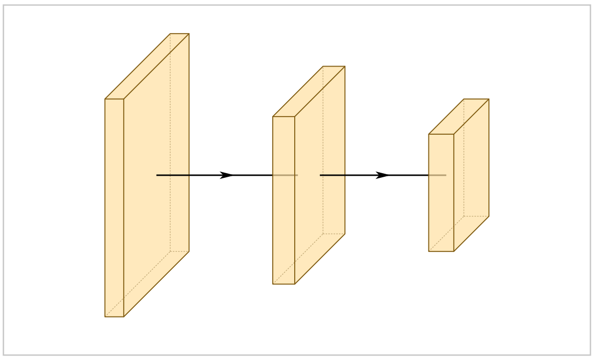
`draw-network`

How much visible white space an automatic offset adds, once the lean is covered.

```
#draw-network(
  (
    conv(shape: (64, 128, 128)),
    conv(shape: (128, 64, 64)),
    conv(shape: (256, 32, 32)),
  ),
  auto-gap: 0.1,
)
```



```
#draw-network(
  (
    conv(shape: (64, 128, 128)),
    conv(shape: (128, 64, 64)),
    conv(shape: (256, 32, 32)),
  ),
  auto-gap: 2.0,
)
```



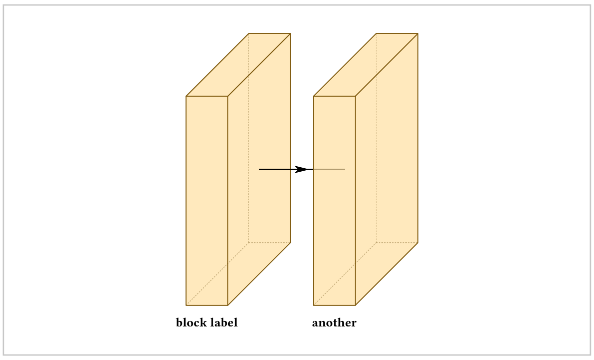
CHAPTER 6

Labels

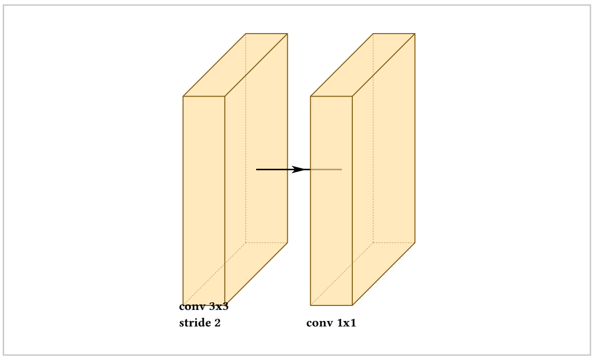
`label` default `none`

Text under the block. Any content, so markup and line breaks work.

```
#draw-network((
  conv(label: "block label"),
  conv(label: "another"),
))
```




```
#draw-network((
  conv(label: [conv 3x3 \ stride 2]),
  conv(label: [conv 1x1]),
))
```

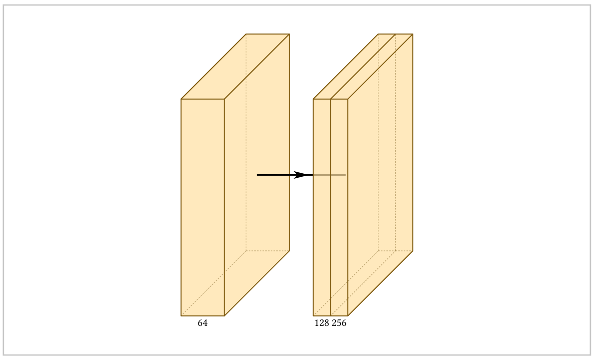


6.1 Channel numbers

`channels` default `none`

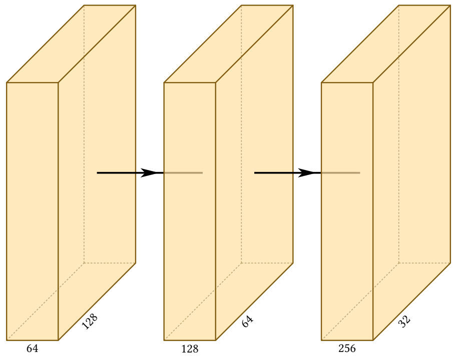
Numbers along the top of the block, one per band. At most one extra entry is allowed, and becomes the diagonal axis label. Strings work as well as numbers.

```
#draw-network((
  conv(channels: (64,)),
  conv(channels: (128, 256), widths: (0.4, 0.4)),
))
```



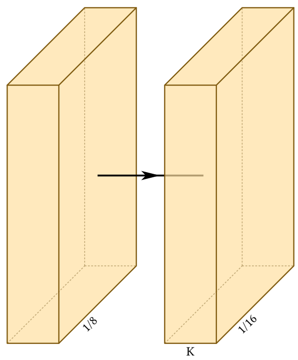
The extra entry is where spatial resolution usually goes:

```
#draw-network((
  conv(channels: (64, 128)),
  conv(channels: (128, 64)),
  conv(channels: (256, 32)),
))
```



An empty string labels the axis and nothing else:

```
#draw-network((
  conv(channels: ("", "1/8")),
  conv(channels: ("K", "1/16")),
))
```



> More than one extra entry is an error from inside the package, reported as an array index out of bounds rather than as a message about `channels`. If you see that, count your bands.

6.2 Axis names

`xlabel` default `none`

`conv`, `convres`, `custom`

`ylabel` default `none`

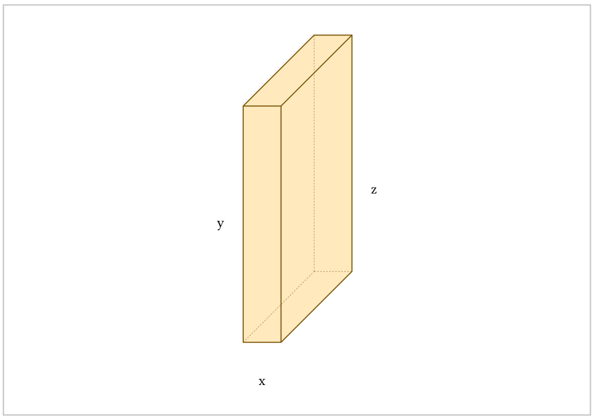
`conv`, `convres`, `custom`

`zlabel` default `none`

`conv`, `convres`, `custom`

Names for the three axes of one block, for the figure that has to explain the projection itself.

```
#draw-network((
  conv(
    xlabel: "x", ylabel: "y", zlabel: "z",
    widths: (0.8,),
  ),
))
```

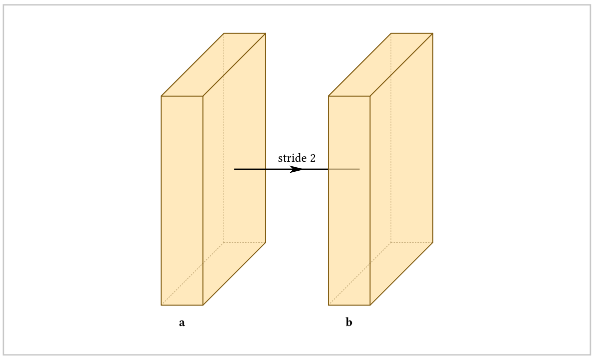


6.3 Labelling the arrow between two blocks

`connection-label` default `none`

Text on the axis arrow **after** the layer that declares it.

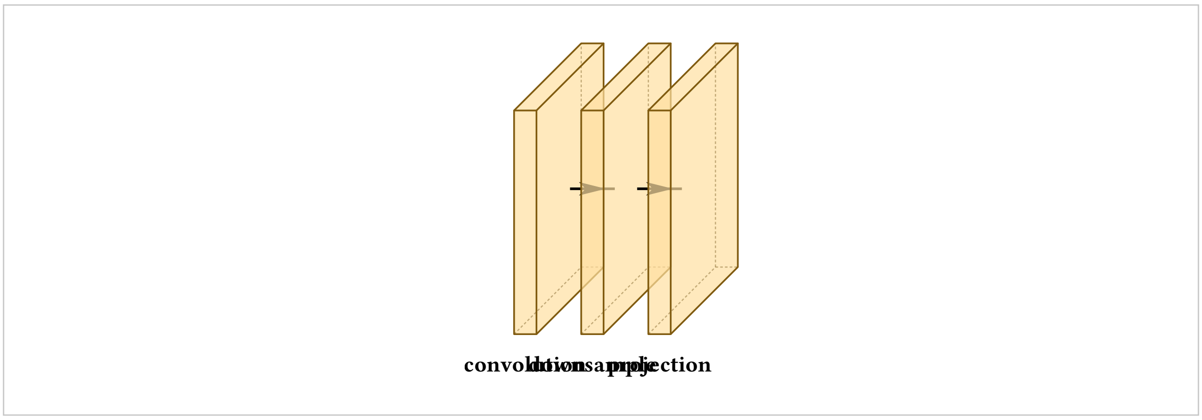
```
#draw-network((
  conv(label: "a", connection-label: "stride 2"),
  conv(label: "b", offset: 3),
))
```



6.4 When labels collide

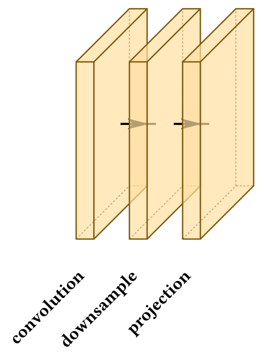
Labels sit at a fixed spot beneath each block, so long labels on closely spaced blocks overlap:

```
#let b = (height: 3, depth: 3, widths: (0.3,), offset: 0.6)
#draw-network((
  conv(label: "convolution", ..b),
  conv(label: "downsample", ..b),
  conv(label: "projection", ..b),
))
```



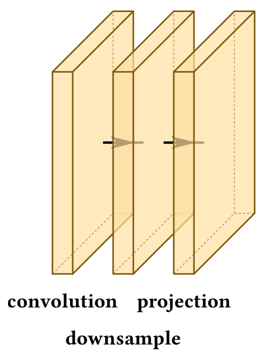
There are three fixes, and which to reach for depends on how crowded the row is. Rotating trades horizontal space, which is scarce, for vertical space, which is usually free:

```
#let b = (
  height: 3, depth: 3, widths: (0.3,), offset: 0.6,
  label-orient: "diagonal",
)
#draw-network((
  conv(label: "convolution", ..b),
  conv(label: "downsample", ..b),
  conv(label: "projection", ..b),
))
```



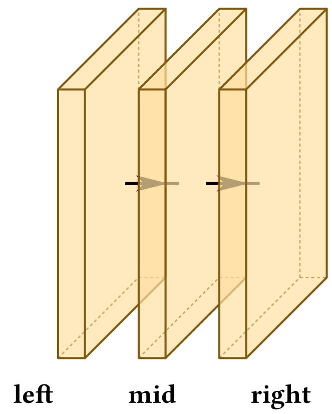
Dropping one label to a second row clears a single collision without rotating anything:

```
#let b = (height: 3, depth: 3, widths: (0.3,), offset: 0.6)
#draw-network((
  conv(label: "convolution", ..b),
  conv(label: "downsample", label-dy: -0.55, ..b),
  conv(label: "projection", ..b),
))
```



Anchoring the outer labels by their facing edges fans them sideways:

```
#let b = (height: 3, depth: 3, widths: (0.3,), offset: 0.6)
#draw-network((
  conv(label: "left", label-anchor: "base-east",
       label-dx: -0.2, ..b),
  conv(label: "mid", ..b),
  conv(label: "right", label-anchor: "base-west",
       label-dx: 0.2, ..b),
))
```

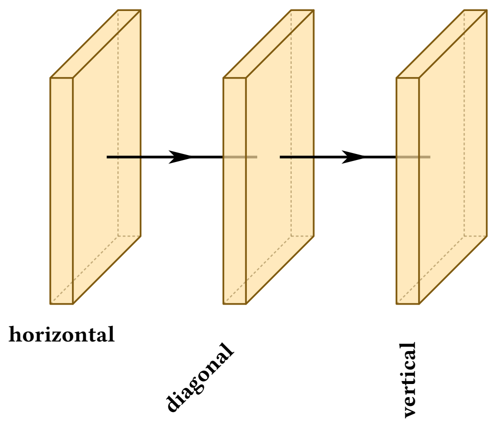


6.5 The placement options

`label-orient` default `"horizontal"`

`"horizontal"`, `"diagonal"` or `"vertical"`. Each preset pairs its angle with the anchor that places it correctly, so a diagonal label points its end at its own block and a vertical one hangs centred beneath it. Any other value is an error.

```
#let b = (height: 3, depth: 3, widths: (0.3,), offset: 2)
#draw-network((
  conv(label: "horizontal", ..b),
  conv(label: "diagonal", label-orient: "diagonal", ..b),
  conv(label: "vertical", label-orient: "vertical", ..b),
))
```



`label-dx` default `0`

Horizontal shift.

`label-dy` default `0`

Vertical shift.

`label-anchor`

default `set by the preset`

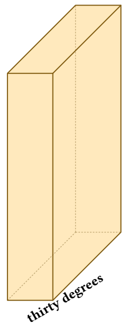
Which point of the text is pinned to the label position. Labels anchor on their baseline rather than their bounding box, so a label with descenders sits level with one without. Use `"base-east"` and `"base-west"` to anchor horizontally while keeping that alignment.

`label-angle`

default `0deg`

An arbitrary angle. Setting both this and `label-orient` is an error, since the right anchor for an arbitrary rotation depends on where you want the text.

```
#draw-network((
  conv(
    label: "thirty degrees",
    label-angle: 30deg,
    label-anchor: "base-west",
  ),
))
```



CHAPTER 7

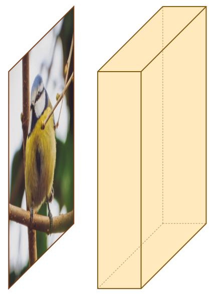
Images inside blocks

`image`

default none

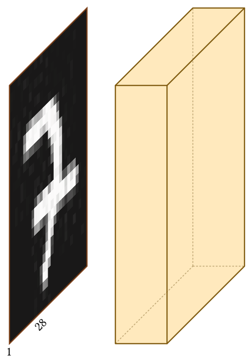
Content drawn on the block's face, sheared to match the projection. The string `"default"` uses the bundled photograph. Any content works, not only images.

```
#draw-network((
  input(image: "default"),
  conv(),
))
```



Any Typst image:

```
#let mnist = "../../examples/networks/mnist-img-sample.jpg"
#draw-network((
  input(image: image(mnist), channels: (1, 28)),
  conv(),
))
```

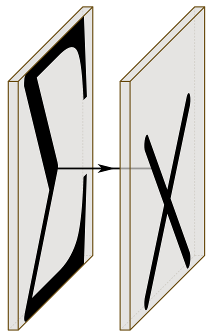


Nothing restricts this to the input. Putting a picture on a later block is how a figure shows what a stage produces:

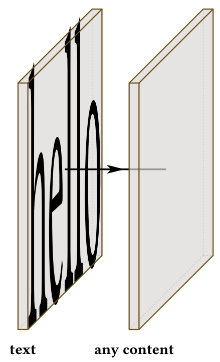
```
#draw-network((
  input(image: "default", label: "in"),
  conv(image: "default", label: "features",
       depth: 4, widths: (0.4,), offset: 2),
))
```

7.1 Content that is not an image

```
#draw-network((
  custom(image: [#text(20pt)[$Sigma$]], label: "sum"),
  custom(image: [#text(20pt)[$times$]], label: "scale"),
))
```



```
#draw-network((
  custom(image: [hello], label: "text"),
  custom(image: [#rect(width: 60%, height: 60%, fill: red)],
         label: "any content"),
))
```



CHAPTER 8

Colour

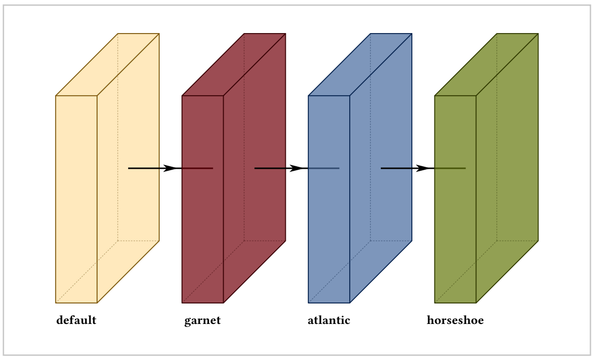
fill default `set by the type and palette`

Block colour. Edge and band colours are derived from it, so a recoloured block stays internally consistent.

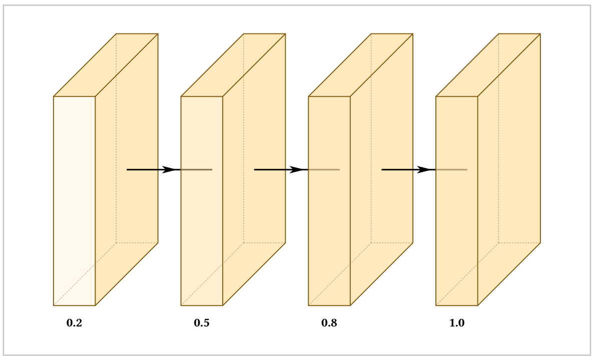
opacity default `0.7, varies by type`

How solid the block is.

```
#draw-network((
  conv(label: "default"),
  conv(label: "garnet", fill: rgb("#73000A")),
  conv(label: "atlantic", fill: rgb("#466A9F")),
  conv(label: "horseshoe", fill: rgb("#65780B")),
))
```




```
#draw-network((
  conv(label: "0.2", opacity: 0.2),
  conv(label: "0.5", opacity: 0.5),
  conv(label: "0.8", opacity: 0.8),
  conv(label: "1.0", opacity: 1.0),
))
```



8.1 Activation bands

A darker band at the end of a convolution conventionally marks the activation.

show-relu default `false`

draw-network; conv, convres, custom

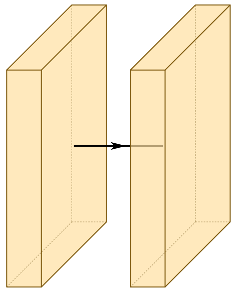
Turn bands on for a whole figure with the `draw-network` argument, and override on any one layer.

bandfill default `derived from fill`

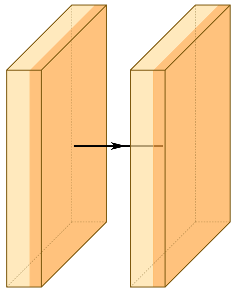
conv, convres, custom

Band colour. Undeclared, it is derived from the layer's own fill, so a recoloured block gets a matching band rather than one from the palette.

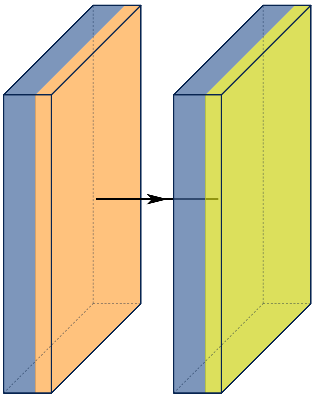
```
#draw-network(
  (conv(widths: (0.8,)), conv(widths: (0.8,))),
  show-relu: false,
)
```



```
#draw-network(
  (conv(widths: (0.8,)), conv(widths: (0.8,))),
  show-relu: true,
)
```



```
#draw-network(
  (
    conv(label: "derived band", widths: (0.8,),
         fill: rgb("#466A9F")),
    conv(label: "stated band", widths: (0.8,),
         fill: rgb("#466A9F"), bandfill: rgb("#CED318")),
  ),
  show-relu: true,
)
```



8.2 Palettes

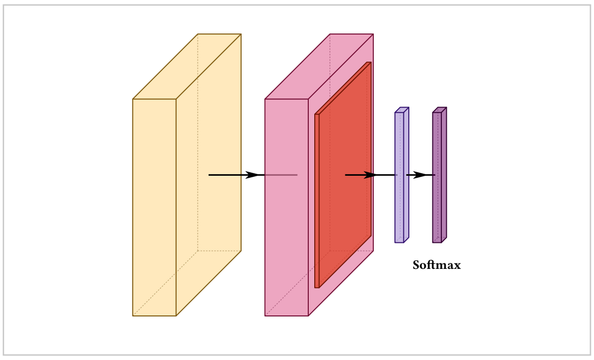
`palette`

default `"warm"`

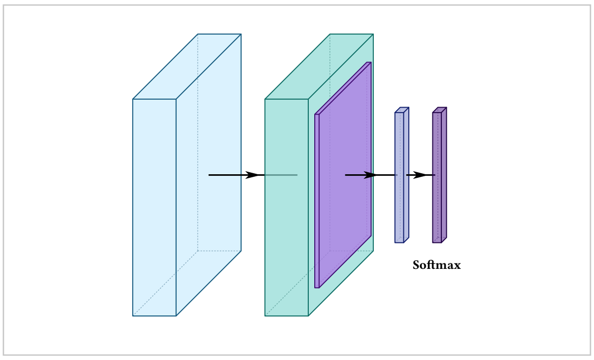
`draw-network`

`"warm"` or `"cold"`.

```
#draw-network(
  (conv(), convres(), pool(), fc(), softmax()),
  palette: "warm",
)
```

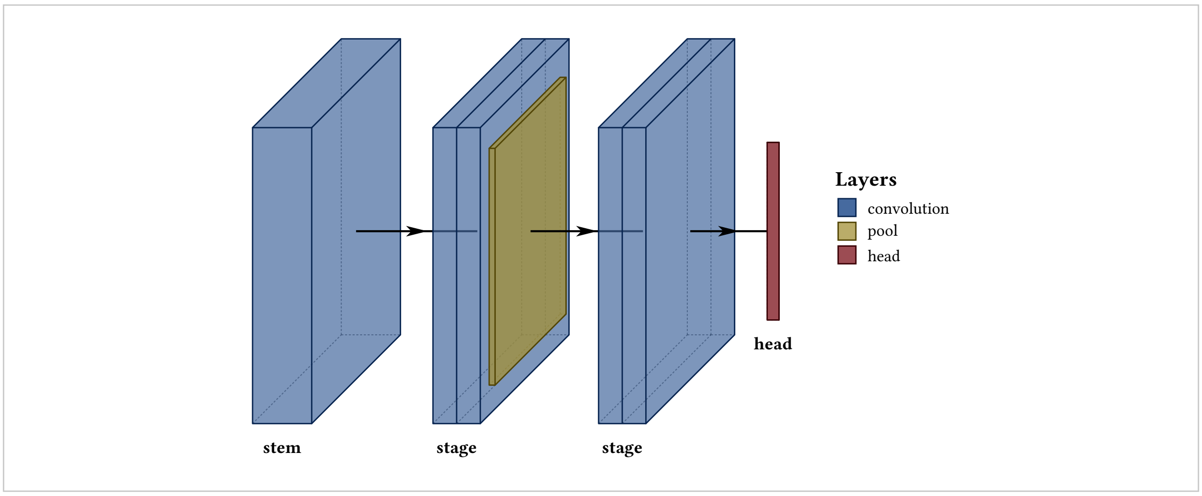



```
#draw-network(
  (conv(), convres(), pool(), fc(), softmax()),
  palette: "cold",
)
```



A palette sets defaults only. Any `fill` you state wins, so a figure can sit on a palette and still recolour three blocks — or ignore the palette entirely and put the whole figure on a scheme of your own:

```
#let garnet = rgb("#73000A")
#let atlantic = rgb("#466A9F")
#let honeycomb = rgb("#A49137")

#draw-network(
  (
    conv(label: "stem", fill: atlantic, legend: "convolution"),
    conv(label: "stage", fill: atlantic, widths: (0.4, 0.4)),
    pool(fill: honeycomb, legend: "pool"),
    conv(label: "stage", fill: atlantic, widths: (0.4, 0.4)),
    fc(label: "head", fill: garnet, depth: 0, legend: "head"),
  ),
  show-legend: true,
)
```

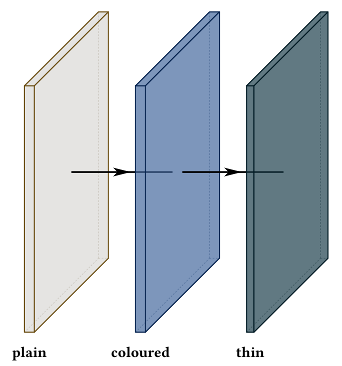


CHAPTER 9

Blocks of your own

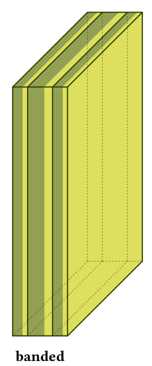
`custom` is the generic block: no meaning attached, every option available.

```
#draw-network((
  custom(label: "plain"),
  custom(label: "coloured", fill: rgb("#466A9F")),
  custom(label: "thin", width: 0.15, fill: rgb("#1F414D")),
))
```



It takes bands and an activation band exactly as a convolution does:

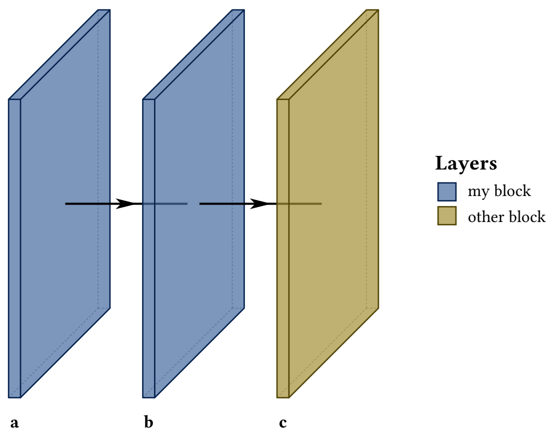
```
#draw-network(
  (
    custom(
      label: "banded",
      widths: (0.3, 0.5, 0.3),
      fill: rgb("#65780B"),
      bandfill: rgb("#CED318"),
      show-relu: true,
    ),
  ),
)
```



`legend` default `the type's default name`

What this layer is called in the legend. Layers sharing a legend name appear once, so a colour used by four blocks produces one entry.

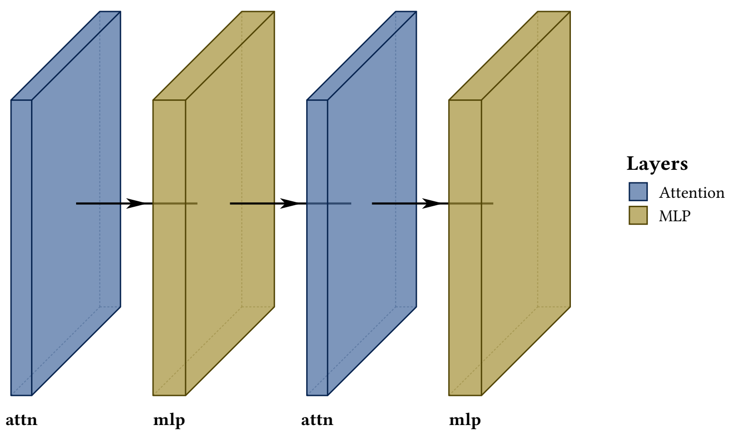
```
#draw-network(
  (
    custom(label: "a", fill: rgb("#466A9F"), legend: "my block"),
    custom(label: "b", fill: rgb("#466A9F")),
    custom(label: "c", fill: rgb("#A49137"), legend: "other block"),
  ),
  show-legend: true,
)
```



9.1 Building a vocabulary

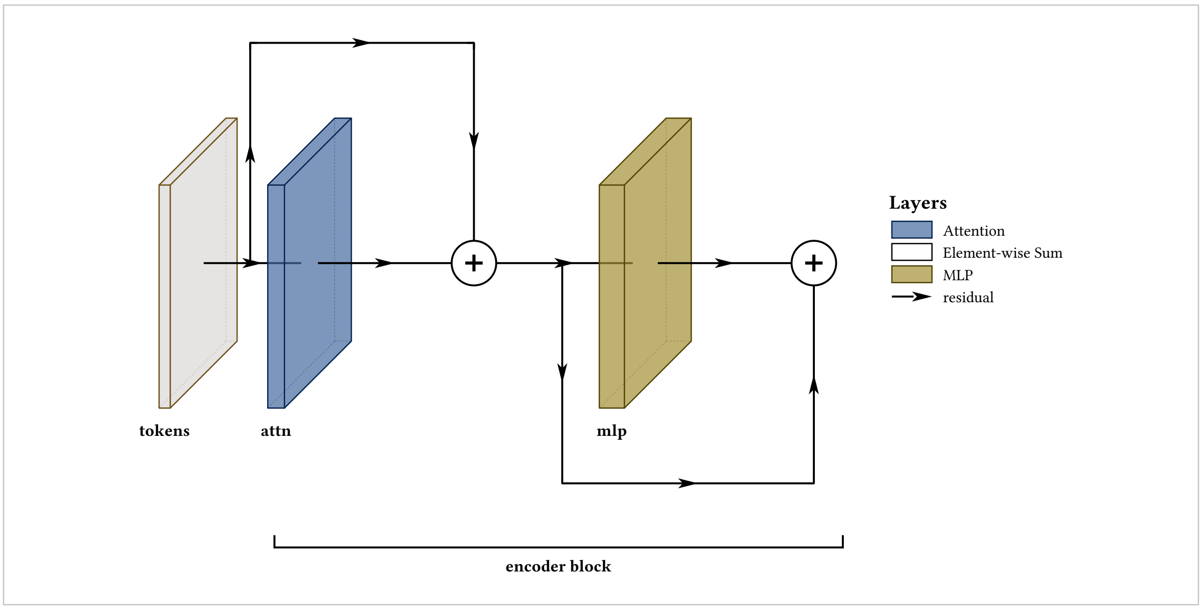
A `custom` layer with its options fixed is a function returning a layer, so your own block types are a few lines. This is the intended way to draw something the package has no name for — attention, a gate, a state-space block — rather than repurposing a type that means something else to a reader.

```
#let attention(..a) = custom(
  fill: rgb("#466A9F"), width: 0.35, legend: "Attention", ..a,
)
#let mlp(..a) = custom(
  fill: rgb("#A49137"), width: 0.55, legend: "MLP", ..a,
)

#draw-network(
  (attention(label: "attn"), mlp(label: "mlp"),
   attention(label: "attn"), mlp(label: "mlp")),
  show-legend: true,
)
```



Once the vocabulary exists, a figure is written in it:

```
#let attn(..a) = custom(fill: rgb("#466A9F"), width: 0.3,
  height: 4, depth: 4, legend: "Attention", ..a)
#let mlp(..a) = custom(fill: rgb("#A49137"), width: 0.45,
  height: 4, depth: 4, legend: "MLP", ..a)

#draw-network(
  (
    custom(name: "in", label: "tokens", width: 0.2, height: 4, depth: 4),
    attn(name: "a", label: "attn"),
    sum(name: "s1"),
    mlp(name: "m", label: "mlp"),
    sum(name: "s2"),
  ),
  connections: (
    connection(from: "in", to: "s1", legend: "residual"),
    connection(from: "s1", to: "s2"),
  ),
  groups: (group(from: "a", to: "s2", label: "encoder block"),),
  show-legend: true,
)
```



CHAPTER 10

Connections

Arrows along the main axis are drawn for you. Anything else — a skip, a residual add, a lateral feed, an auxiliary path — goes in `connections`, as a list of routes between two named layers.

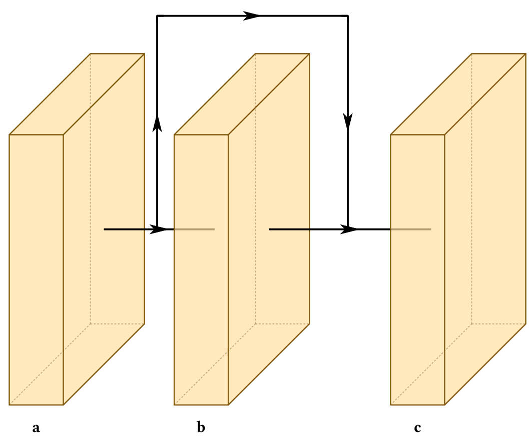
from default `required`

Name of the layer the route leaves.

to default `required`

Name of the layer it arrives at.

```
#draw-network(
  (
    conv(name: "a", label: "a"),
    conv(name: "b", label: "b"),
    conv(name: "c", label: "c"),
  ),
  connections: (connection(from: "a", to: "c"),),
)
```

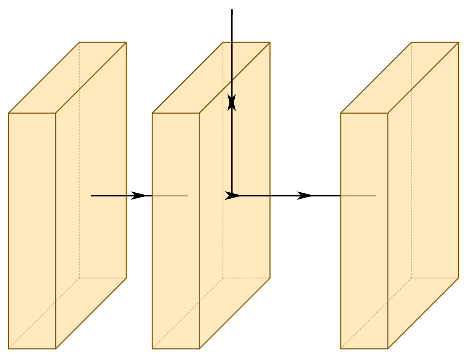


10.1 Routing modes

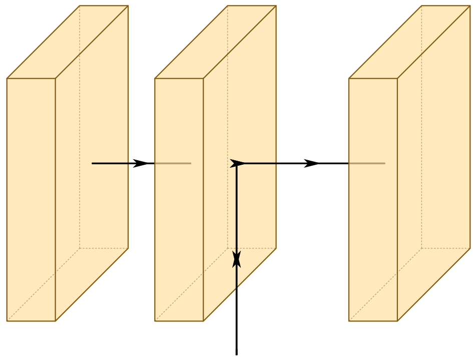
mode default `"air"`

`"air"` runs over the top of the stack, `"flat"` underneath, `"depth"` along the projection's own axis.

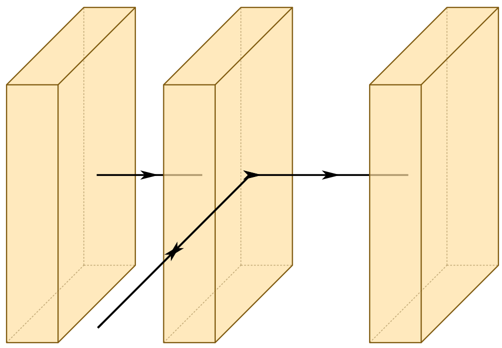
```
#draw-network(
  (conv(name: "a"), conv(), conv(name: "c")),
  connections: (connection(from: "a", to: "c", mode: "air"),),
)
```



```
#draw-network(
  (conv(name: "a"), conv(), conv(name: "c")),
  connections: (connection(from: "a", to: "c", mode: "flat"),),
)
```



```
#draw-network(
  (conv(name: "a"), conv(), conv(name: "c")),
  connections: (connection(from: "a", to: "c", mode: "depth"),),
)
```

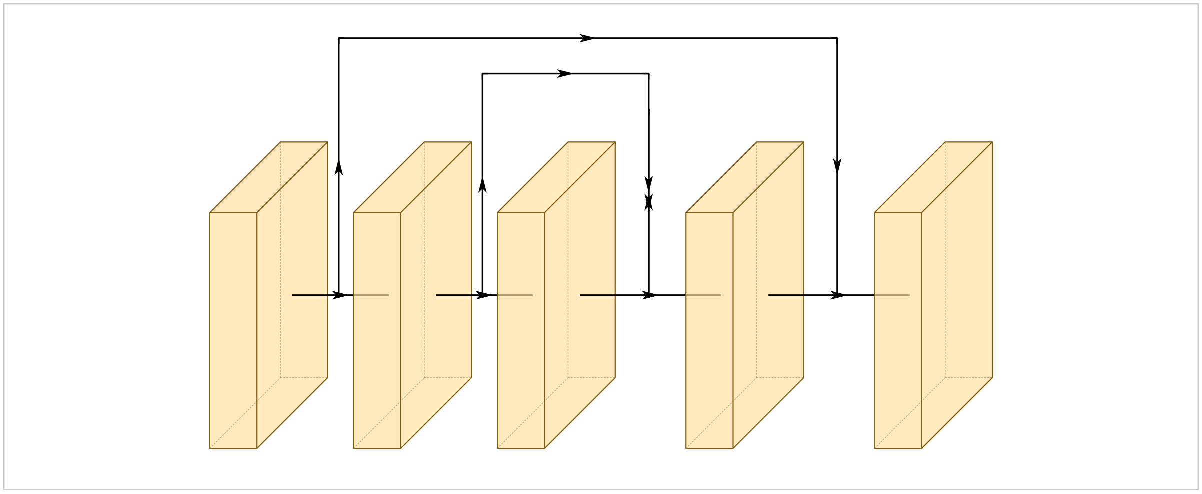


10.2 Lane heights

`pos` default `auto`

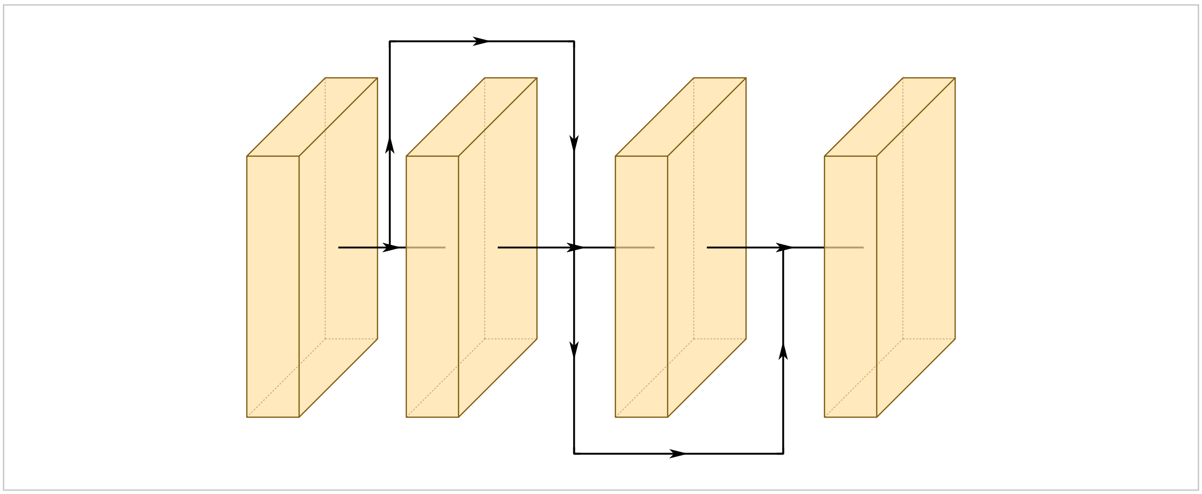
How far the route sits from the centre axis. `auto` ranks routes by how far they reach, so a route spanning more blocks sits higher and a longer route always arcs over a shorter one instead of crossing it.

```
#draw-network(
  (
    conv(name: "a"), conv(name: "b"), conv(name: "c"),
    conv(name: "d"), conv(name: "e"),
  ),
  connections: (
    connection(from: "a", to: "e"),
    connection(from: "b", to: "d"),
    connection(from: "c", to: "d"),
  ),
)
```



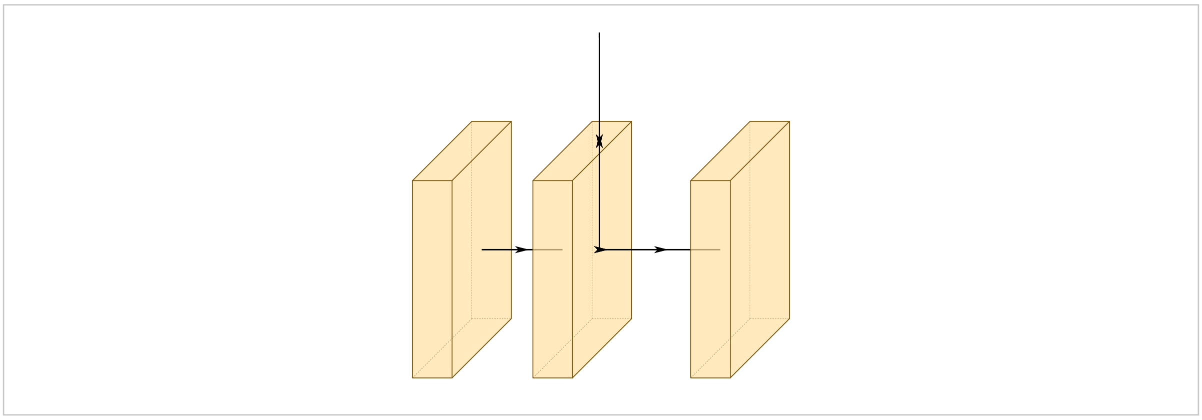
Routes of equal reach share a height. Where two of them overlap, the second goes to the opposite side of the axis at the same height, rather than being pushed further out than its reach warrants:

```
#draw-network(
  (
    conv(name: "a"), conv(name: "b"),
    conv(name: "c"), conv(name: "d"),
  ),
  connections: (
    connection(from: "a", to: "c"),
    connection(from: "b", to: "d"),
  ),
)
```



A number overrides all of it for one route:

```
#draw-network(
  (conv(name: "a"), conv(), conv(name: "c")),
  connections: (connection(from: "a", to: "c", pos: 5.5),),
)
```



lane-unit default `0.75`

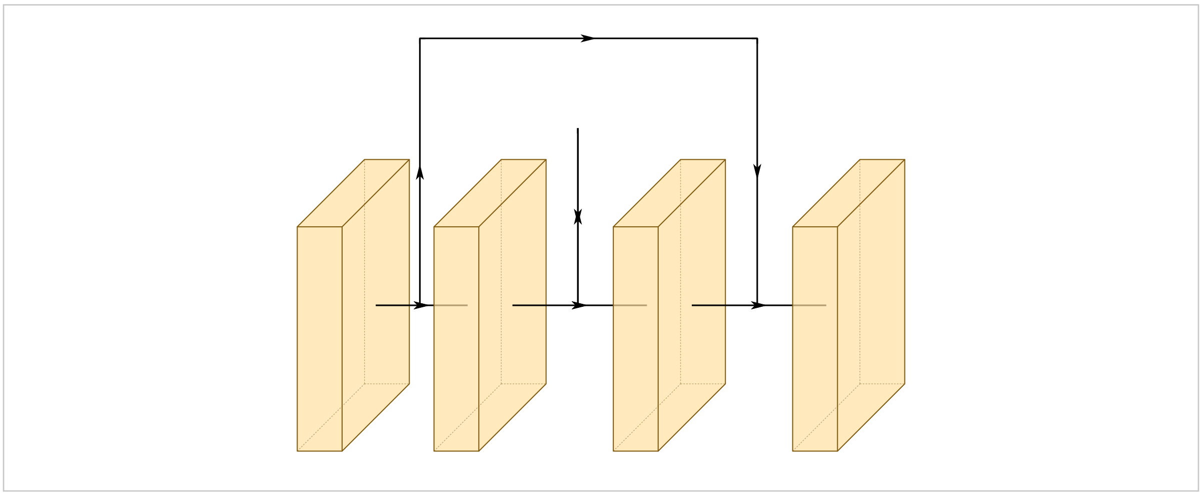
draw-network

Spacing between automatic heights.

clearance default `0.7`

How far the lowest automatic route sits from the blocks.

```
#draw-network(
  (
    conv(name: "a"), conv(name: "b"),
    conv(name: "c"), conv(name: "d"),
  ),
  connections: (
    connection(from: "a", to: "d"),
    connection(from: "b", to: "c"),
  ),
  lane-unit: 2.0,
)
```

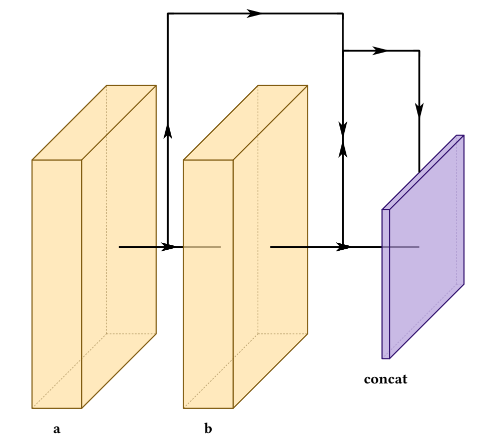


10.3 Where a route arrives

touch-layer default `false`

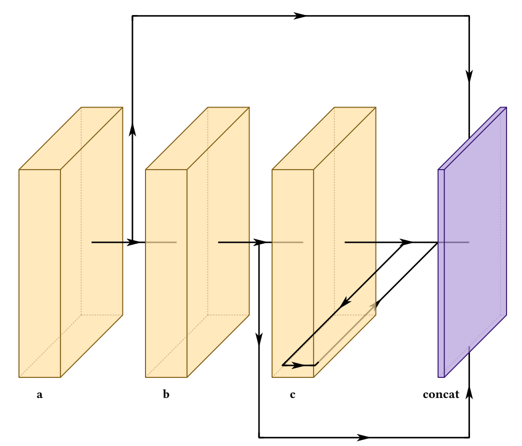
Arrive on the target block itself rather than on the axis arrow in front of it. The edge follows the routing mode: air lands on the top edge, flat on the bottom, depth on the left.

```
#draw-network(
  (
    conv(name: "a", label: "a"),
    conv(name: "b", label: "b"),
    concat(name: "cat", label: "concat", depth: 5),
  ),
  connections: (
    connection(from: "a", to: "cat"),
    connection(from: "b", to: "cat", touch-layer: true),
  ),
)
```



All three modes, arriving on the same block:

```
#let t = (touch-layer: true)
#draw-network(
  (
    conv(name: "a", label: "a"),
    conv(name: "b", label: "b"),
    conv(name: "c", label: "c"),
    concat(name: "cat", label: "concat", depth: 5, height: 5),
  ),
  connections: (
    connection(from: "a", to: "cat", mode: "air", ..t),
    connection(from: "b", to: "cat", mode: "flat", ..t),
    connection(from: "c", to: "cat", mode: "depth", ..t),
  ),
)
```



A route arriving on the bottom or left edge has its final stretch drawn behind the block, because those edges are on the far side and the route genuinely passes underneath before reaching them. Blocks are semi-transparent, so it shows faintly rather than disappearing.

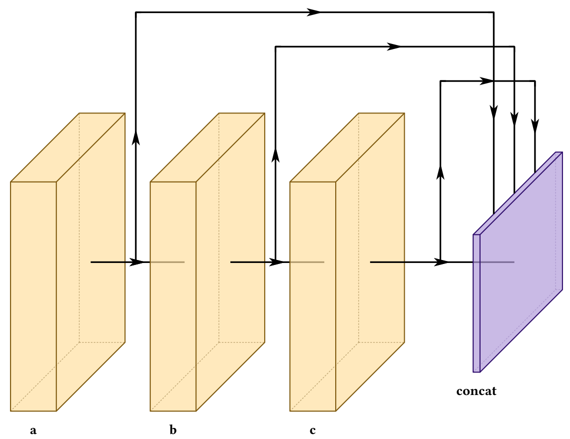
10.4 Fanning several routes into one block

Two routes in the same mode would otherwise land on the same point, with their arrowheads stacked.

`arrive-offset` default `auto`

Position along the arrival edge. `auto` spaces a whole fan evenly and leaves a lone arrival centred.

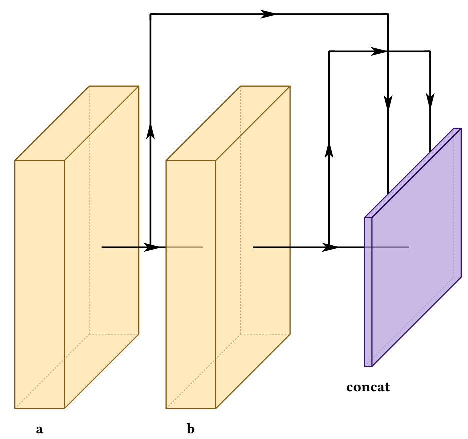
```
#draw-network(
  (
    conv(name: "a", label: "a"), conv(name: "b", label: "b"),
    conv(name: "c", label: "c"),
    concat(name: "cat", label: "concat", depth: 6),
  ),
  connections: (
    connection(from: "a", to: "cat", touch-layer: true),
    connection(from: "b", to: "cat", touch-layer: true),
    connection(from: "c", to: "cat", touch-layer: true),
  ),
)
```



Routes are grouped by the edge they land on, then spread across it: `k` routes divide the edge into `k + 1` intervals and sit at the interior boundaries, so the outermost pair is inset rather than sitting on the corners. They are ordered by where each route starts, so a fan does not cross itself, and adding a route respaces the rest.

A number places one arrival yourself, measured along the edge:

```
#draw-network(
  (
    conv(name: "a", label: "a"), conv(name: "b", label: "b"),
    concat(name: "cat", label: "concat", depth: 6),
  ),
  connections: (
    connection(from: "a", to: "cat", touch-layer: true, arrive-offset: -0.6),
    connection(from: "b", to: "cat", touch-layer: true, arrive-offset: 0.6),
  ),
)
```



The offset runs **along the edge**, not in x. The top and bottom edges of a block's west side follow the isometric depth direction, so shifting horizontally would walk the arrival off the block. The edge is $\sqrt{2} \cdot \text{depth} \cdot \text{depth-multiplier}$ for top and bottom arrivals and the block height for left ones, so a deeper block accommodates a wider fan.

10.5 Telling routes apart

A residual add, a concat feed and an auxiliary supervision path are different things, and by default they all look the same.

`color` default `palette`

Paint for the line and its arrowheads.

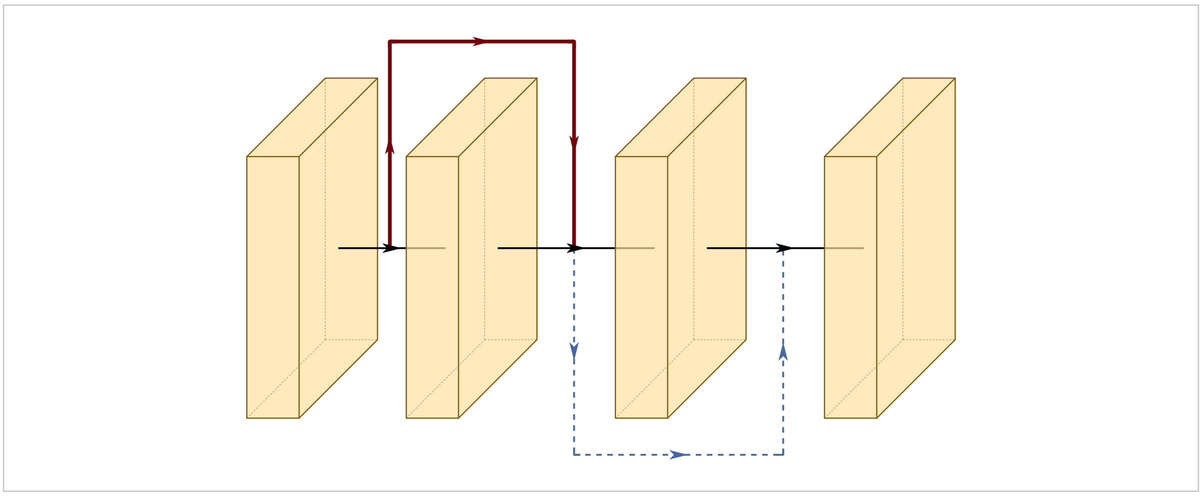
`dash` default `none`

Any Typst dash pattern, e.g. `"dashed"`, `"dotted"`.

`thickness` default `1`

Multiplies the palette stroke width, so it composes with `stroke-thickness` rather than fighting it.

```
#draw-network(
  (
    conv(name: "a"), conv(name: "b"),
    conv(name: "c"), conv(name: "d"),
  ),
  connections: (
    connection(from: "a", to: "c", color: rgb("#73000A"), thickness: 2),
    connection(from: "b", to: "d", color: rgb("#466A9F"), dash: "dashed"),
  ),
)
```



Arrowheads take the line colour, since a coloured line with black arrowheads reads as a bug rather than a choice.

10.6 Routes in the legend

`legend` default none

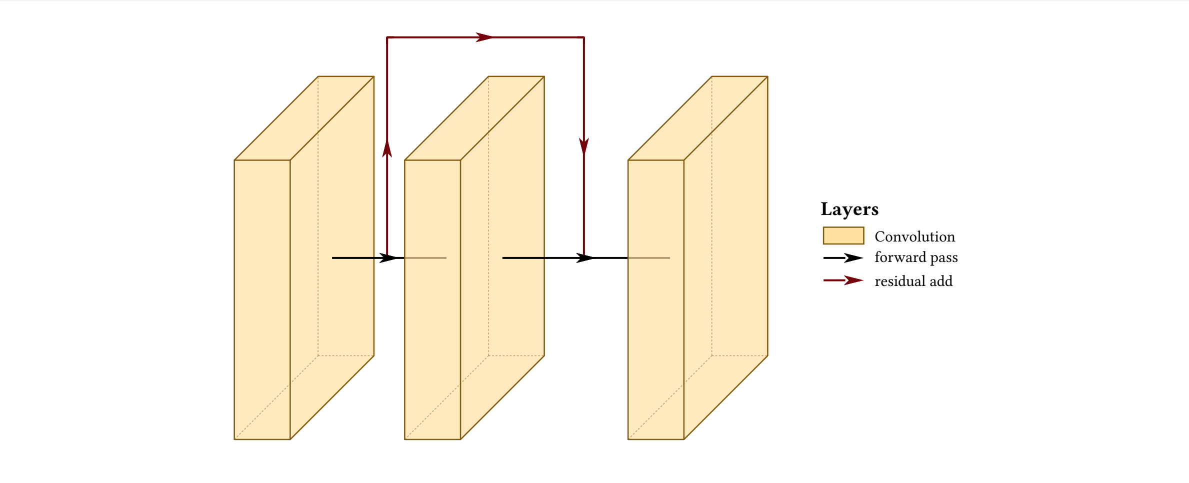
Adds a line-sample legend entry rather than a colour swatch, since what distinguishes a route is its stroke and not a fill. Routes sharing a name appear once.

`main-legend` default none

`draw-network`

Names the automatic axis arrows. Without it a legend can explain every skip in a figure and say nothing about the arrows carrying the forward pass.

```
#draw-network(
  (conv(name: "a"), conv(name: "b"), conv(name: "c")),
  connections: (
    connection(from: "a", to: "c", color: rgb("#73000A"),
     legend: "residual add"),
  ),
  show-legend: true,
  main-legend: "forward pass",
)
```

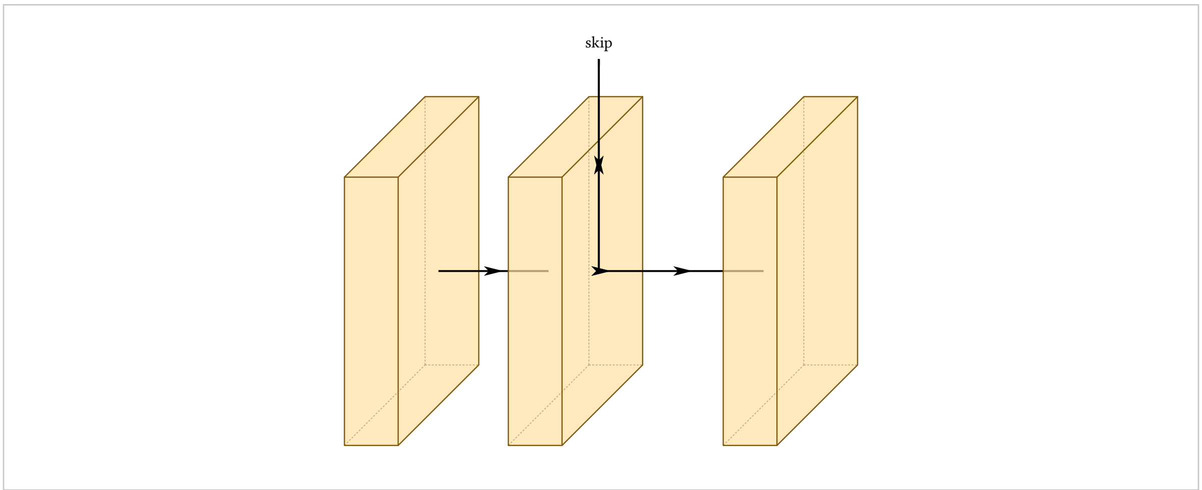


10.7 Labelling a route

label default none

Text on the route.

```
#draw-network(
  (conv(name: "a"), conv(), conv(name: "c")),
  connections: (connection(from: "a", to: "c", label: "skip"),),
)
```

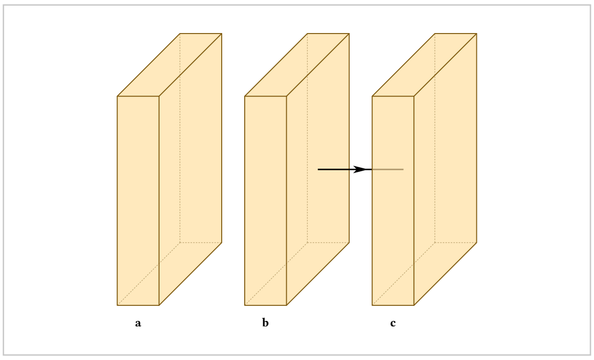


10.8 Suppressing an axis arrow

show-connection default `true; false on input`

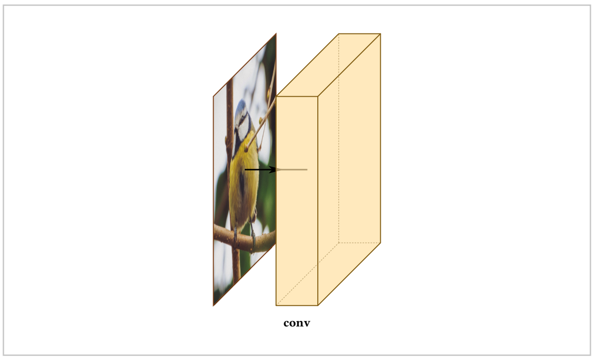
Whether to draw the axis arrow **after** this layer.

```
#draw-network((
  conv(label: "a", show-connection: false),
  conv(label: "b"),
  conv(label: "c"),
))
```



The input layer is the one that defaults to `false`, which is why an image usually sits detached. Set it `true` to join it up:

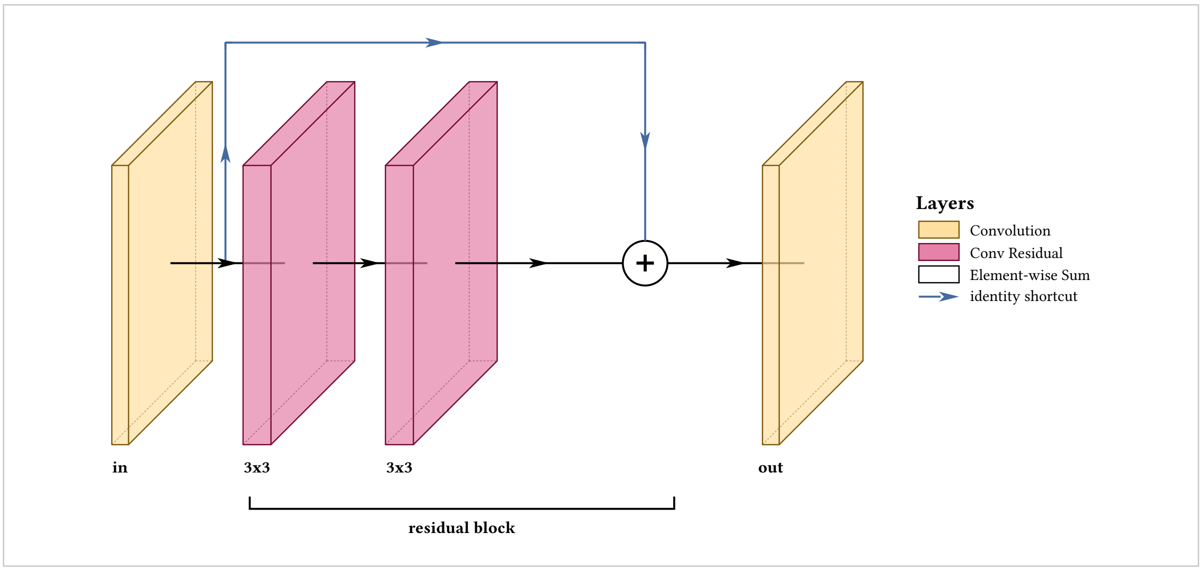
```
#draw-network((
  input(image: "default", show-connection: true),
  conv(label: "conv", offset: 1.5),
))
```



10.9 Two composites

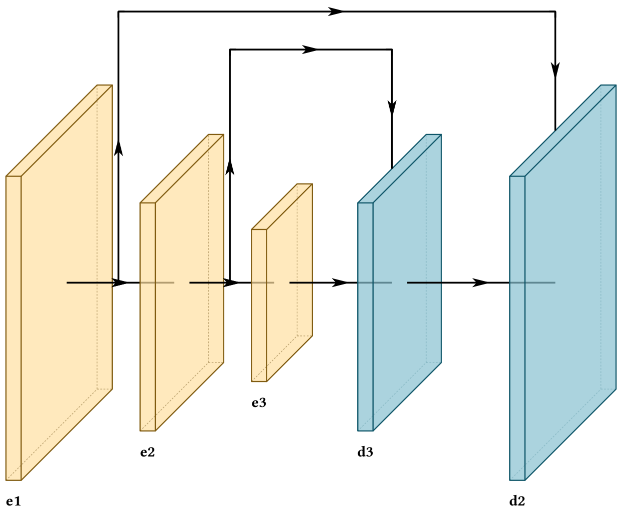
A residual block is a `sum` node plus one route:

```
#draw-network(
  (
    conv(name: "in", label: "in", widths: (0.3,)),
    convres(name: "c1", label: "3x3", widths: (0.5,)),
    convres(name: "c2", label: "3x3", widths: (0.5,)),
    sum(name: "add"),
    conv(name: "out", label: "out", widths: (0.3,)),
  ),
  connections: (
    connection(from: "in", to: "add", color: rgb("#466A9F"),
     legend: "identity shortcut"),
  ),
  groups: (group(from: "c1", to: "add", label: "residual block"),),
  show-legend: true,
)
```



An encoder-decoder is a size pyramid plus two routes arriving on their targets:

```
#let enc = (widths: (0.3,))
#draw-network(
  (
    conv(name: "e1", label: "e1", height: 6, depth: 6, ..enc),
    conv(name: "e2", label: "e2", height: 4.5, depth: 4.5, ..enc),
    conv(name: "e3", label: "e3", height: 3, depth: 3, ..enc),
    deconv(name: "d3", label: "d3", height: 4.5, depth: 4.5),
    deconv(name: "d2", label: "d2", height: 6, depth: 6),
  ),
  connections: (
    connection(from: "e2", to: "d3", touch-layer: true),
    connection(from: "e1", to: "d2", touch-layer: true),
  ),
)
```



Groups and repeats

Backbone, neck and head are the phrases anyone uses out loud to explain one of these figures. `groups` draws them, as labelled brackets under a span of layers.

`from` default `required`

First layer of the span, by name.

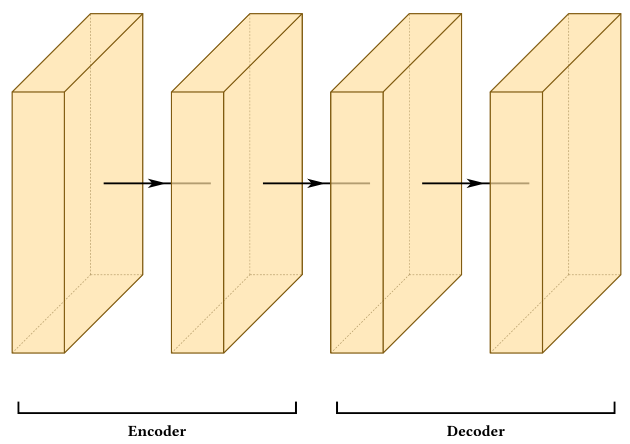
`to` default `required`

Last layer of the span, inclusive. May be the same layer.

`label` default `none`

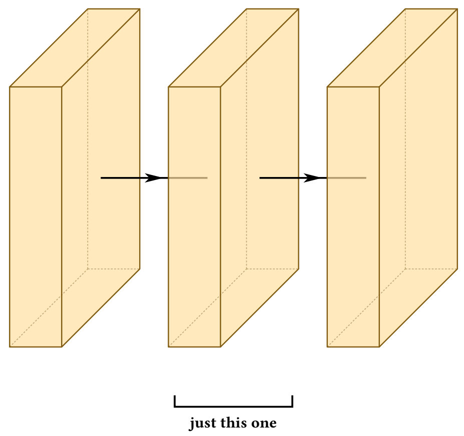
Bracket text.

```
#draw-network(
  (
    conv(name: "e1"), conv(name: "e2"),
    conv(name: "d1"), conv(name: "d2"),
  ),
  groups: (
    group(from: "e1", to: "e2", label: "Encoder"),
    group(from: "d1", to: "d2", label: "Decoder"),
  ),
)
```



A group of one is allowed and common, since a head is often a single block:

```
#draw-network(
  (conv(name: "a"), conv(name: "b"), conv(name: "c")),
  groups: (group(from: "b", to: "b", label: "just this one"),),
)
```



The bracket covers the drawn footprint rather than the front faces, so it sits under the whole block including its lean, and its ends are inset slightly so two adjacent groups read as two rather than as one continuous rule.

11.1 Tinting and stacking

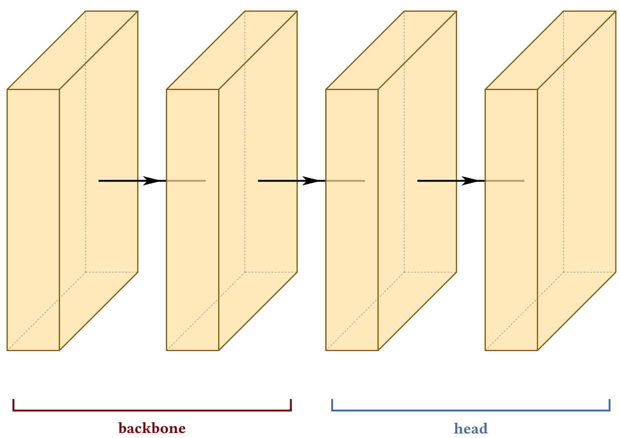
`color` default `grey`

Tints one bracket.

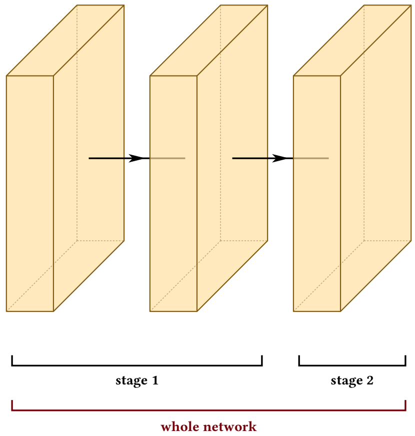
`offset` default `1.15`

Pushes the bracket further down, which is how you stack a group that encloses other groups onto its own row.

```
#draw-network(
  (conv(name: "a"), conv(name: "b"), conv(name: "c"), conv(name: "d")),
  groups: (
    group(from: "a", to: "b", label: "backbone", color: rgb("#73000A")),
    group(from: "c", to: "d", label: "head", color: rgb("#466A9F")),
  ),
)
```

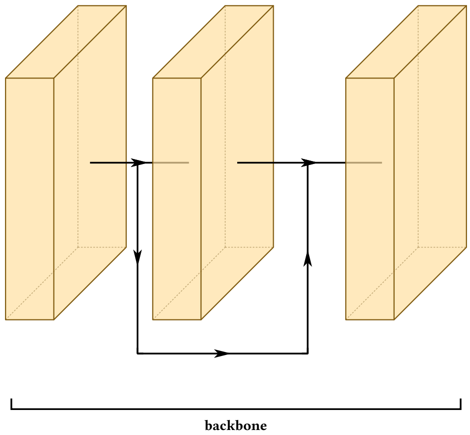


```
#draw-network(
  (conv(name: "a"), conv(name: "b"), conv(name: "c")),
  groups: (
    group(from: "a", to: "b", label: "stage 1"),
    group(from: "c", to: "c", label: "stage 2"),
    group(from: "a", to: "c", label: "whole network",
     offset: 2.1, color: rgb("#73000A")),
  ),
)
```



Brackets are placed below everything else in the figure, including a route running underneath the stack, so the two never collide:

```
#draw-network(
  (conv(name: "a"), conv(name: "b"), conv(name: "c")),
  connections: (connection(from: "a", to: "c", mode: "flat"),),
  groups: (group(from: "a", to: "c", label: "backbone"),),
)
```



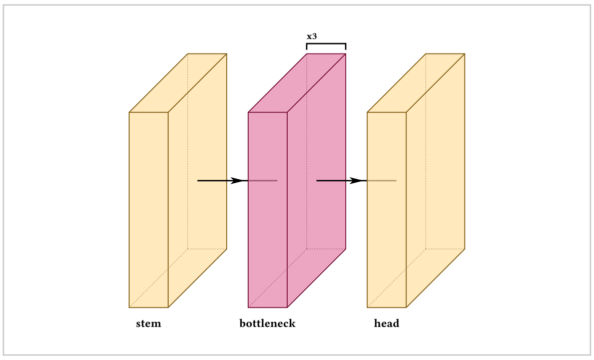
11.2 Repeated blocks

Depth-scaled models stack the same block several times. Writing that as extra entries in `widths` makes it **look** repeated, but the package has no idea it is: it cannot label the repeat or bracket it, and the figure claims one wide block rather than several.

`repeat` default 1

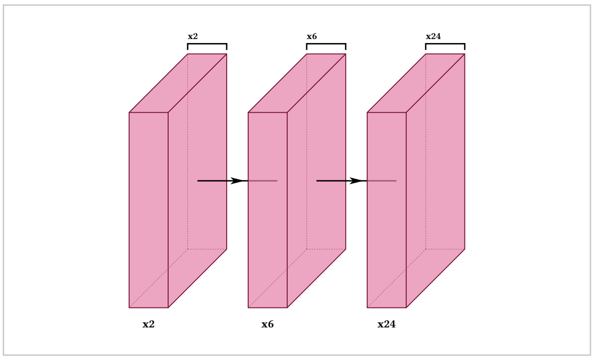
Marks the block as repeated in series. It is drawn once, with a bracket above it carrying the count. Ignored on `pool`, `unpool` and `sum`.

```
#draw-network((
  conv(label: "stem"),
  convres(label: "bottleneck", repeat: 3),
  conv(label: "head"),
))
```



The count costs no horizontal space, so it stays legible at any depth:

```
#draw-network((
  convres(label: "x2", repeat: 2),
  convres(label: "x6", repeat: 6),
  convres(label: "x24", repeat: 24),
))
```

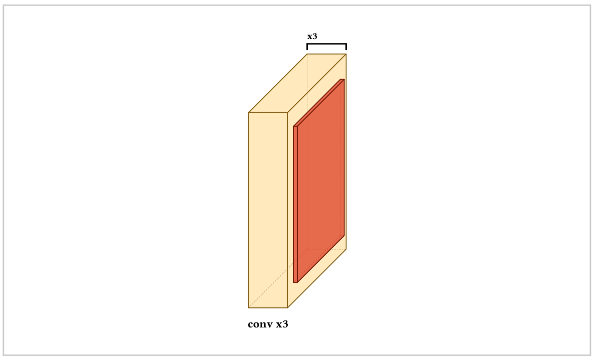


> Ghosted copies were tried first and rejected. An outline behind the block reads as an empty box rather than as another one of the same block, and drawing N of them is either misleading about the count or unreadable once N is large. A bracket states the count instead of depicting it.

The pool pitfall

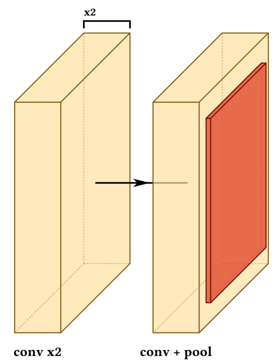
`repeat` describes one layer entry, not a run of them. Putting it on a convolution that has a pool attached reads as though the pool repeats too:

```
#draw-network((
  conv(label: "conv x3", repeat: 3),
  pool(),
))
```



Split it instead. Let the repeat cover the plain convolutions, and draw the last one on its own with the pool attached:

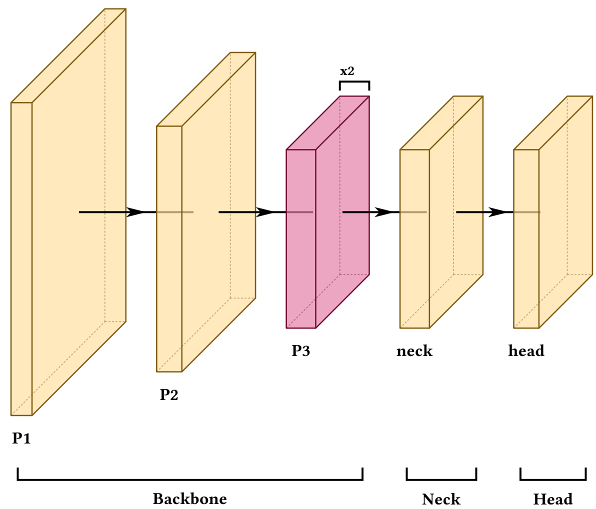
```
#draw-network((
  conv(label: "conv x2", repeat: 2),
  conv(label: "conv + pool", offset: 2),
  pool(),
))
```



For the same reason a repeated **pair**, such as attention-then-MLP, cannot be expressed with `repeat` at all. Section 15.5 shows what to do instead.

11.3 A composite

```
#draw-network(
  (
    conv(name: "s1", label: "P1", shape: (32, 160, 160)),
    conv(name: "s2", label: "P2", shape: (64, 80, 80)),
    convres(name: "s3", label: "P3", shape: (128, 40, 40), repeat: 2),
    conv(name: "n1", label: "neck", shape: (128, 40, 40)),
    conv(name: "h1", label: "head", shape: (64, 40, 40)),
  ),
  groups: (
    group(from: "s1", to: "s3", label: "Backbone"),
    group(from: "n1", to: "n1", label: "Neck"),
    group(from: "h1", to: "h1", label: "Head"),
  ),
)
```



CHAPTER 12

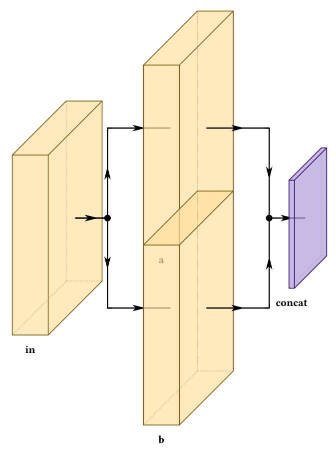
Parallel branches

`draw-network` advances one cursor along one axis, so anything genuinely parallel — two detection heads, the two paths inside a CSP block, a two-stream fusion — would have to be collapsed into a single block with arrows pointed at it. A `branch` entry draws it as it is.

`branches` default `required`

An array of layer arrays, one per parallel path. Each is walked exactly as the trunk is, so anything that works on the trunk works inside a branch.

```
#draw-network((
  conv(label: "in"),
  branch(spread: 5, branches: (
    (conv(label: "a"),),
    (conv(label: "b"),),
  )),
  concat(label: "concat"),
))
```



12.1 Separation and count

`spread` default `6`

Separation between paths, centred on the trunk. An odd count puts one path on the trunk line; an even count leaves it empty.

`lead` default `2.0`

How far the fan-out and rejoin arrows run.

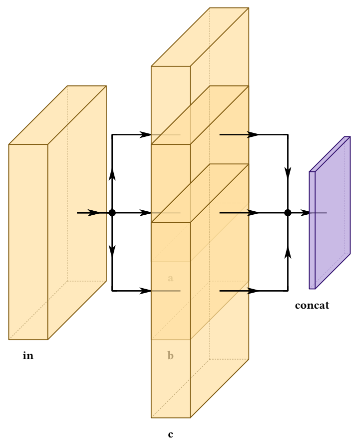
`entry-gap` default `0`

Extra flat run before the fan splits, without changing `lead` — more approach space, same fan shape. Negative removes run.

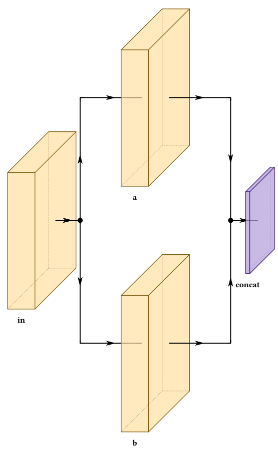
`exit-gap` default `0`

The same after the rejoin, before the next block.

```
#draw-network((
  conv(label: "in"),
  branch(spread: 4, branches: (
    (conv(label: "a"),),
    (conv(label: "b"),),
    (conv(label: "c"),),
  )),
  concat(label: "concat"),
))
```

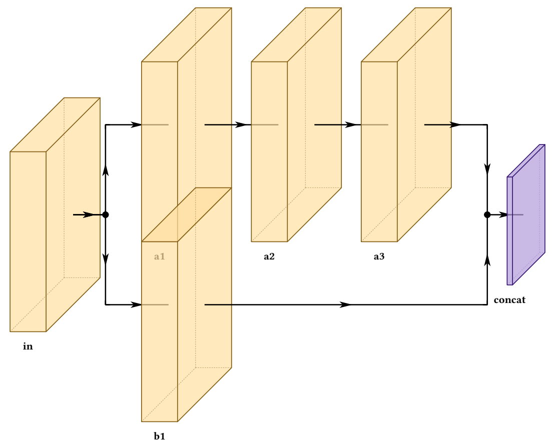



```
#draw-network((
  conv(label: "in"),
  branch(spread: 9, lead: 3, branches: (
    (conv(label: "a"),),
    (conv(label: "b"),),
  )),
  concat(label: "concat"),
))
```



Branches need not be the same length. The rejoin waits for the longest rather than cutting the others short:

```
#draw-network((
  conv(label: "in"),
  branch(spread: 5, branches: (
    (conv(label: "a1"), conv(label: "a2"), conv(label: "a3")),
    (conv(label: "b1"),),
  )),
  concat(label: "concat"),
))
```



A filled dot marks each point where flow divides or meets, and where a tooth leaves a spine that passes through. Merging routes terminate at the dot, and a single arrow leaves it for the next block.

12.2 Stacking along the projection

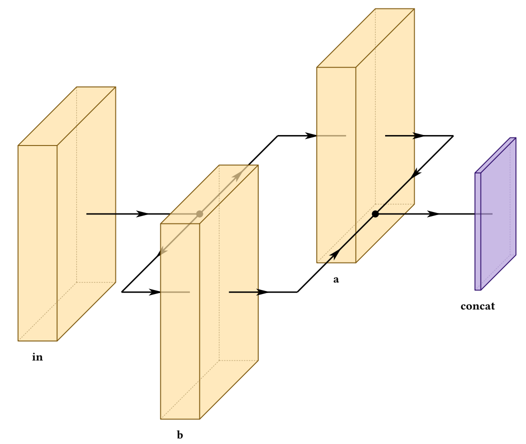
`spread-mode` default `"vertical"`

`"depth"` stacks the paths along the projection's 45-degree axis instead of straight up, so they read as parallel copies sitting behind one another.

`rejoin-lead` default `the value of lead`

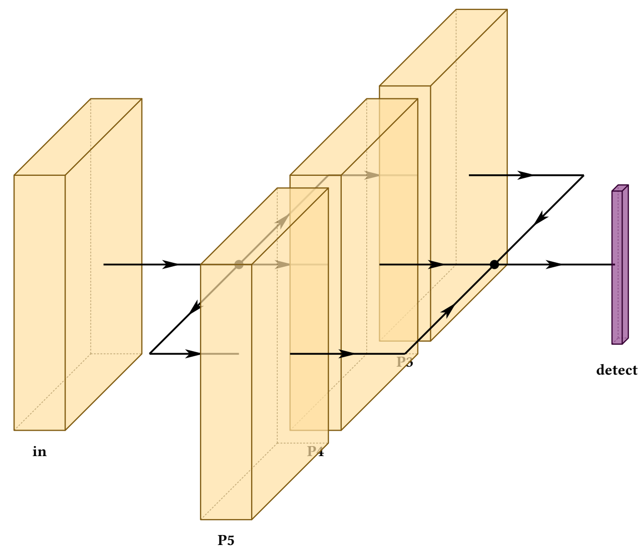
Widens the rejoin side alone. Depth mode often wants this, since its return spine descends past each block's lower-right corner, which is exactly where diagonal dimension labels sit.

```
#draw-network((
  conv(label: "in"),
  branch(spread: 4, spread-mode: "depth", branches: (
    (conv(label: "a"),),
    (conv(label: "b"),),
  )),
  concat(label: "concat"),
))
```



The fan-out and rejoin become two parallel spines with horizontal teeth — a parallelogram, rather than the mirrored hexagon a vertical split produces. Neither 45 is a free angle: it is the one the projection already uses. This is the natural way to draw a set of detection heads:

```
#draw-network((
  conv(label: "in"),
  branch(
    spread: 3.5, spread-mode: "depth", rejoin-lead: 3,
    branches: (
      (conv(label: "P3"),),
      (conv(label: "P4"),),
      (conv(label: "P5"),),
    ),
  ),
  output(label: "detect"),
))
```

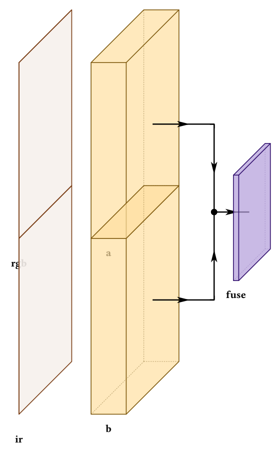


12.3 Open at one end

open default none

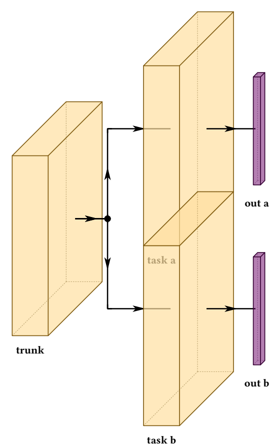
"start" draws no fan-out, "end" no rejoin. A branch with nothing before it is open at the start by definition. A branch open at the start is how a multi-input network begins:

```
#draw-network((
  branch(spread: 5, branches: (
    (input(label: "rgb"), conv(label: "a")),
    (input(label: "ir"), conv(label: "b")),
  )),
  concat(label: "fuse"),
))
```



Open at the end, one trunk fans out into independent outputs:

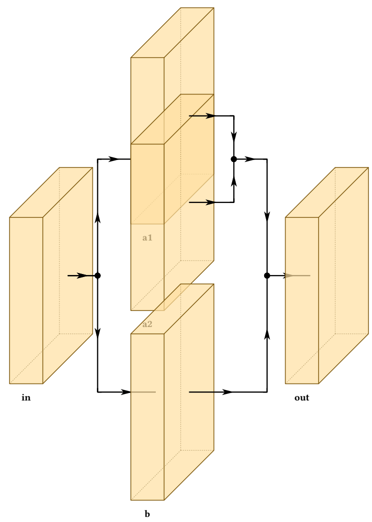
```
#draw-network((
  conv(label: "trunk"),
  branch(spread: 5, open: "end", branches: (
    (conv(label: "task a"), output(label: "out a")),
    (conv(label: "task b"), output(label: "out b")),
  )),
))
```



12.4 Nesting

A branch may contain branches, since walking a branch is the same operation as walking the trunk.

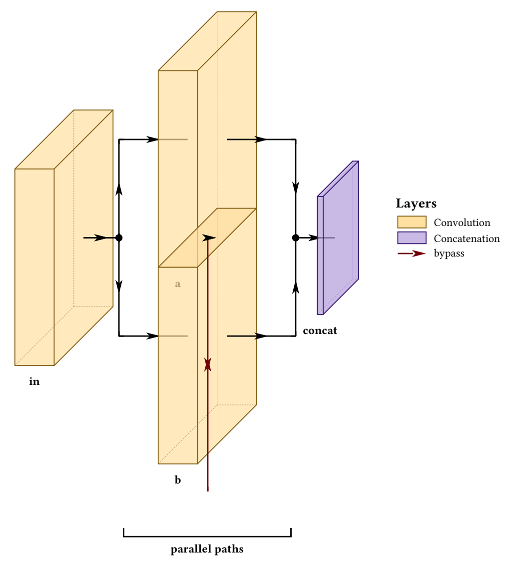
```
#draw-network((
  conv(label: "in"),
  branch(spread: 7, branches: (
    (
      branch(spread: 2.6, lead: 1.2, branches: (
        (conv(label: "a1"),),
        (conv(label: "a2"),),
      )),
    ),
    (conv(label: "b"),),
  )),
  conv(label: "out"),
))
```



12.5 Branches are not a separate world

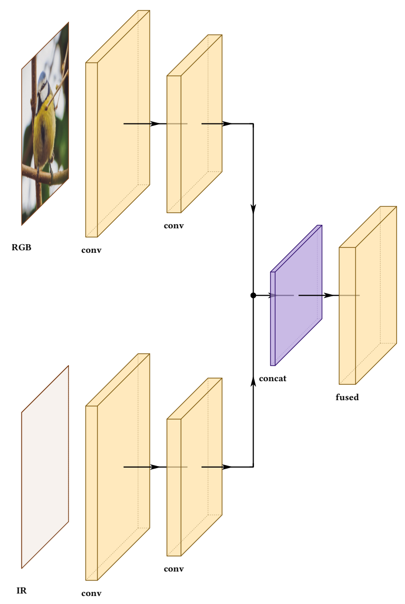
Named layers inside a branch join the same table the trunk uses, so connections, groups and automatic lane ranking all reach into them. A group naming any layer inside a branch widens to the branch's whole drawn extent, plumbing included.

```
#draw-network(
  (
    conv(name: "in", label: "in"),
    branch(spread: 5, branches: (
      (conv(name: "a", label: "a"),),
      (conv(name: "b", label: "b"),),
    )),
    concat(name: "cat", label: "concat"),
  ),
  connections: (
    connection(from: "in", to: "cat", mode: "flat",
     color: rgb("#73000A"), legend: "bypass"),
  ),
  groups: (group(from: "a", to: "b", label: "parallel paths"),),
  show-legend: true,
)
```

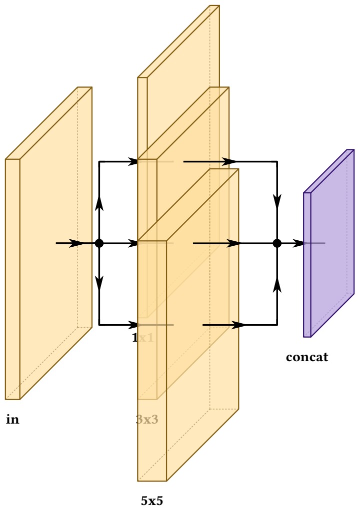


12.6 Two composites

```
#draw-network((
  branch(spread: 11, branches: (
    (
      input(label: "RGB", image: "default"),
      conv(name: "r1", label: "conv", shape: (32, 160, 160)),
      conv(name: "r2", label: "conv", shape: (64, 80, 80)),
    ),
    (
      input(label: "IR"),
      conv(name: "i1", label: "conv", shape: (32, 160, 160)),
      conv(name: "i2", label: "conv", shape: (64, 80, 80)),
    ),
  )),
  concat(label: "concat", depth: 5),
  conv(label: "fused", shape: (128, 80, 80)),
))
```



```
#draw-network((
  conv(label: "in", widths: (0.3,)),
  branch(spread: 3.4, lead: 1.6, branches: (
    (conv(label: "1x1", widths: (0.2,)),),
    (conv(label: "3x3", widths: (0.4,)),),
    (conv(label: "5x5", widths: (0.6,)),),
  )),
  concat(label: "concat"),
))
```



CHAPTER 13

The figure as a whole

13.1 Legends

show-legend default `false`

draw-network

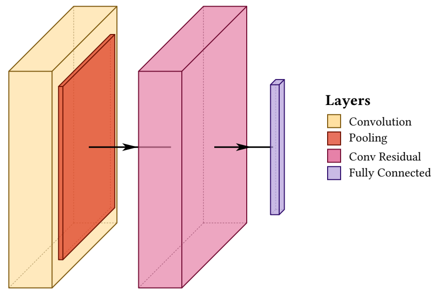
Collects every layer type used into a legend, in order of first appearance.

legend-title default `"Layers"`

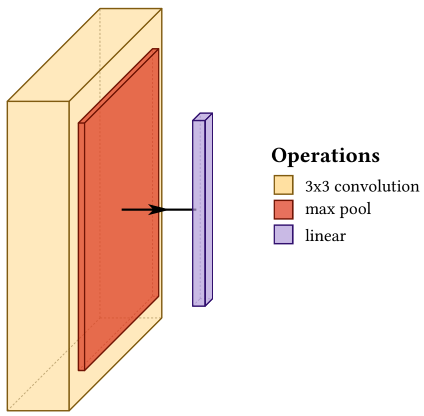
draw-network

Heading of the box.

```
#draw-network(
  (conv(), pool(), convres(), fc()),
  show-legend: true,
)
```

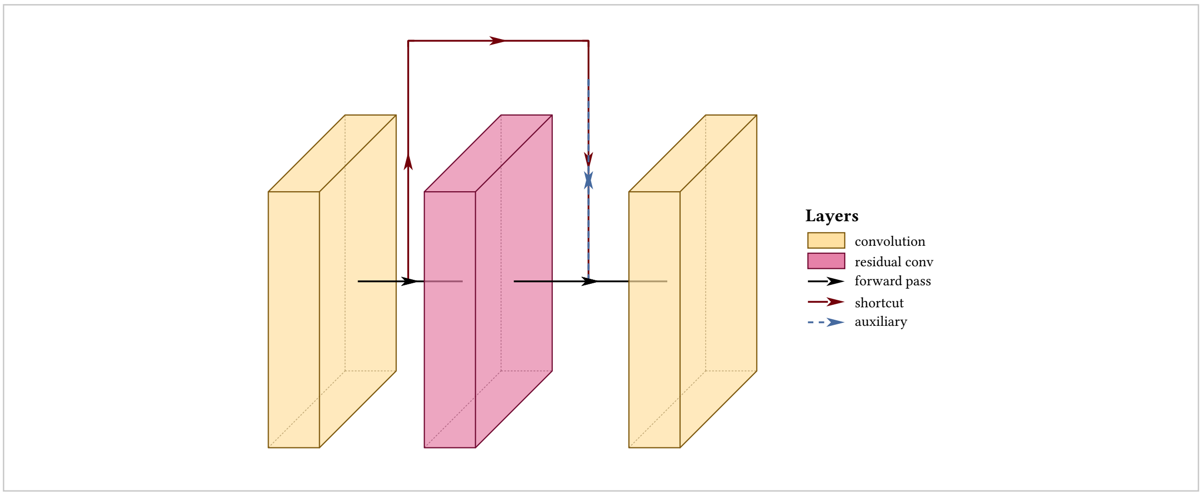



```
#draw-network(
  (
    conv(legend: "3x3 convolution"),
    pool(legend: "max pool"),
    fc(legend: "linear"),
  ),
  show-legend: true,
  legend-title: "Operations",
)
```



Layer swatches and route samples share one box. The sample column widens when any route entry is present, so a dash pattern has room to read, and the layer swatches widen with it so nothing floats away from its label:

```
#draw-network(
  (
    conv(name: "a", legend: "convolution"),
    convres(name: "b", legend: "residual conv"),
    conv(name: "c"),
  ),
  connections: (
    connection(from: "a", to: "c", color: rgb("#73000A"), legend: "shortcut"),
    connection(from: "b", to: "c", color: rgb("#466A9F"), dash: "dashed",
     legend: "auxiliary"),
  ),
  show-legend: true,
  main-legend: "forward pass",
)
```



13.2 Fitting the page

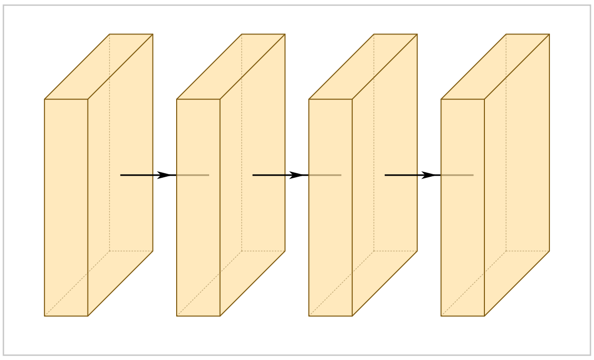
`scale`

default `100%`

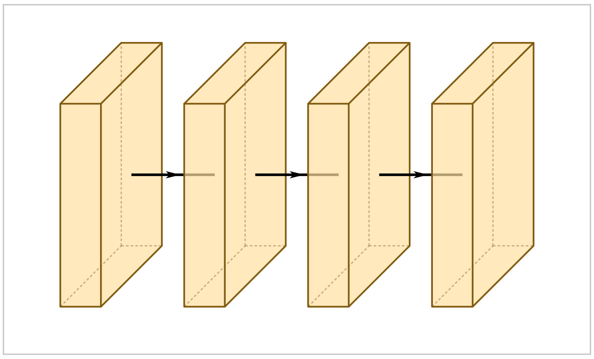
draw-network

Shrinks or grows the whole figure, text included.

```
#draw-network(
  (conv(), conv(), conv(), conv()),
  scale: 100%,
)
```



```
#draw-network(
  (conv(), conv(), conv(), conv()),
  scale: 55%,
)
```



A long network is wider than a page at its natural size, and `scale` is the first thing to reach for. If it makes the text too small to read, the next thing to try is `label-orient: "diagonal"` (section 6.5), which usually buys back enough horizontal space to raise the scale again.

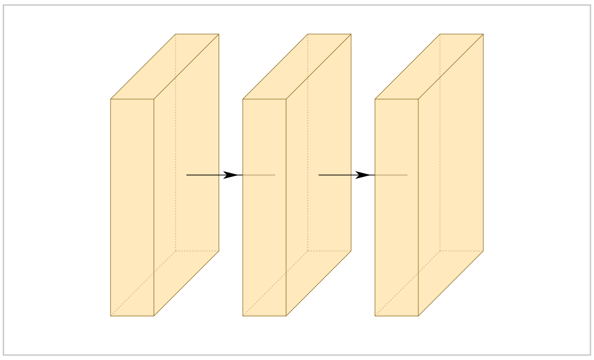
13.3 Line weight

`stroke-thickness` default 1

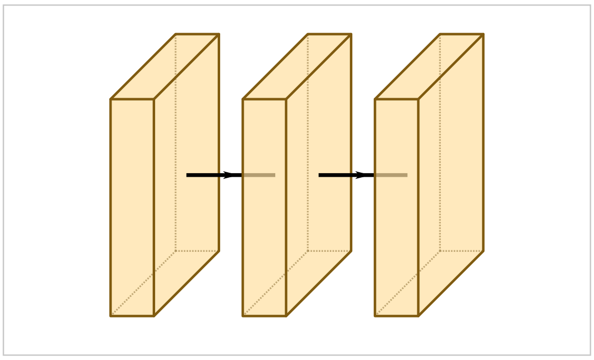
`draw-network`

Multiplies every stroke in the figure. Connection `thickness` multiplies on top of this.
Worth raising when a figure is reduced for a two-column layout, where hairlines start dropping out at print resolution:

```
#draw-network(
  (conv(), conv(), conv()),
  stroke-thickness: 0.5,
)
```



```
#draw-network(
  (conv(), conv(), conv()),
  stroke-thickness: 2.5,
)
```



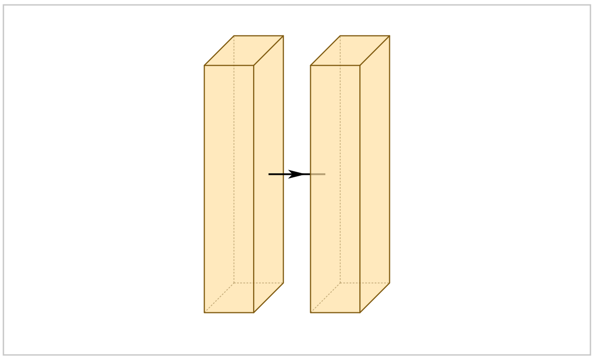
13.4 The projection

`depth-multiplier` default `0.3`

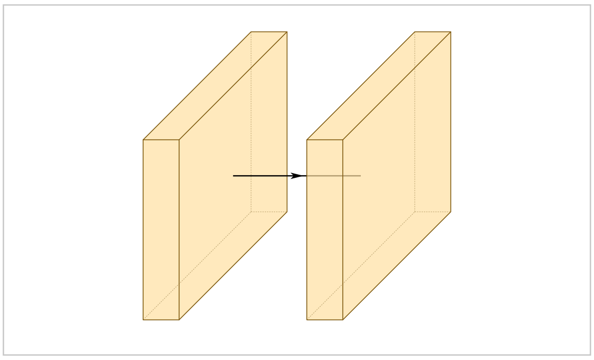
`draw-network`

How far blocks lean, which is the strength of the isometric projection. It also changes what `depth-shear` and `min-clear-offset` return, so pass the same value to those.

```
#draw-network(
  (conv(depth: 5), conv(depth: 5)),
  depth-multiplier: 0.12,
)
```



```
#draw-network(
  (conv(depth: 5), conv(depth: 5)),
  depth-multiplier: 0.6,
)
```

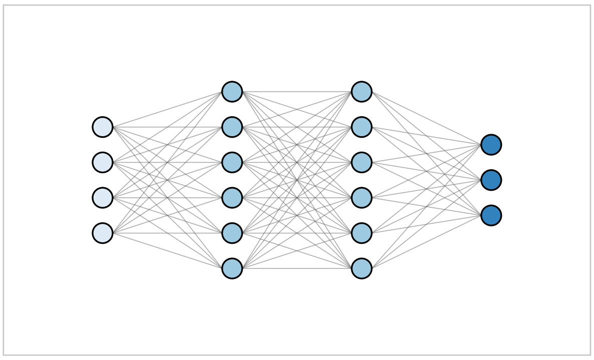


CHAPTER 14

Neuron diagrams

Everything so far has drawn a network as blocks. The other canonical figure draws it as neurons — a circle per unit, an edge per weight, the picture every introduction to the subject opens with. `draw-mlp` draws that figure. It is a second renderer, not a new mode: it shares the package's constructors, validation and palettes and nothing else, it opens its own canvas, and how it composes with `draw-network` in a single figure is deliberately not yet decided.

```
#draw-mlp((4, 6, 6, 3))
```



Four counts, no other options. This is the whole program.

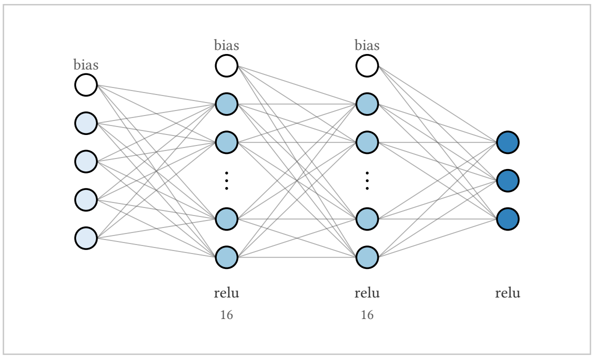
The zero-configuration output is the textbook look: circles, plain grey edges drawn behind the nodes, no arrowheads, no bias nodes, and the input, hidden and output roles told apart by fill. Every default follows a documented convention, and everything past the defaults is an option below.

> Indexing is 1-based everywhere. Layers count left to right and neurons top to bottom within a layer; under `direction: "up"` layers run bottom to top and neurons left to right. Every index an option takes — `dropped`, `exclude`, `edge-labels`, weight matrices, `mlp-edge` endpoints — means the **true** index of that neuron, whether or not its column is collapsed.

14.1 Four tiers

The layer list is an array whose entries are integers, `mlp-layer(..)` values or `mlp-gap(..)` values, mixed freely. An integer `n` is shorthand for `mlp-layer(n)`, and a plain dictionary is an error, exactly as in `draw-network`. The first tier past bare counts is figure-wide options:

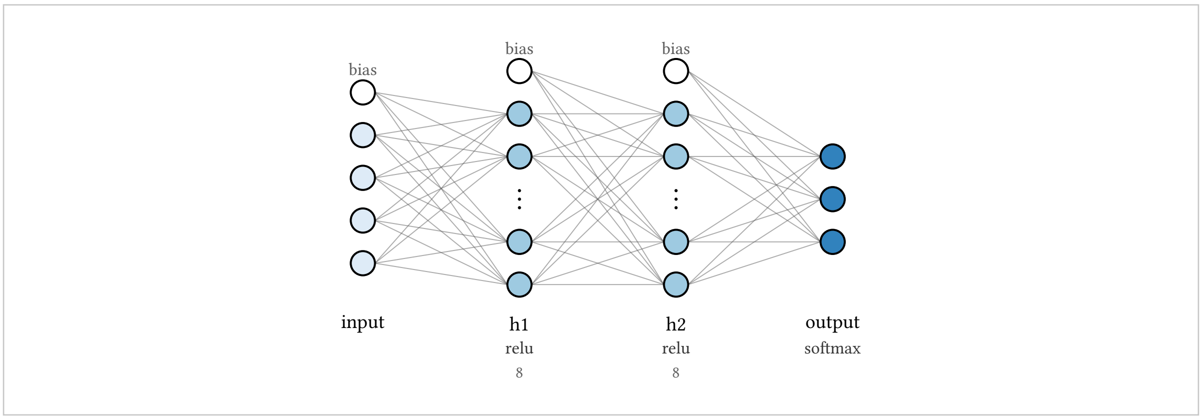
```
#draw-mlp((4, 16, 16, 3), activation: "relu", bias:
true, cutoff: 8)
```



Collapsed columns state their true width; bias nodes appear above each feeding layer.

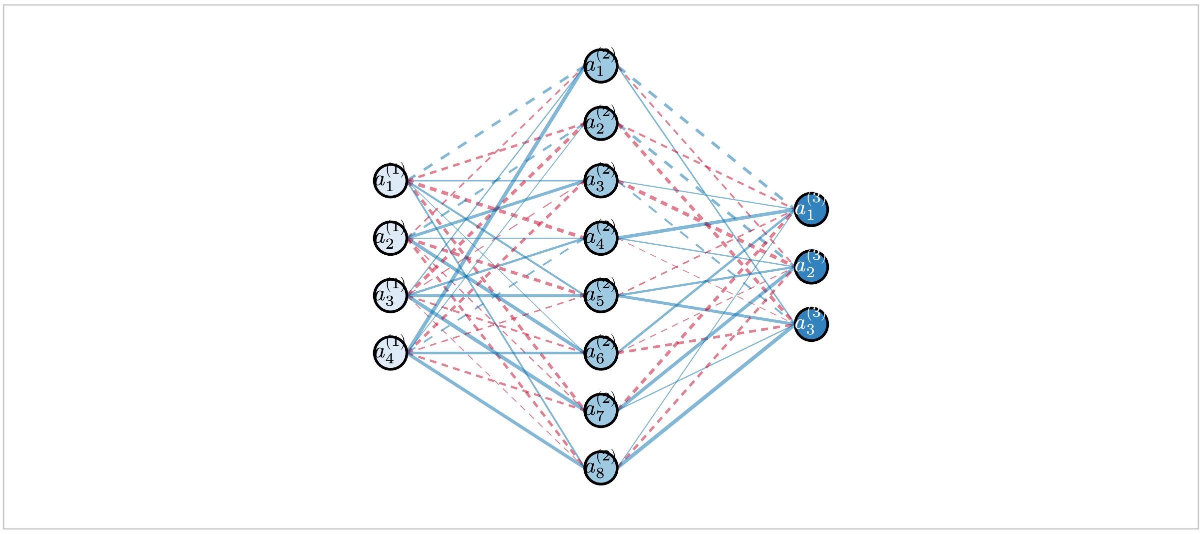
The second is per-layer constructors:

```
#draw-mlp((
  mlp-layer(4,  label: "input"),
  mlp-layer(8,  label: "h1", activation: "relu"),
  mlp-layer(8,  label: "h2", activation: "relu"),
  mlp-layer(3,  label: "output", activation: "softmax"),
), bias: true, cutoff: 6)
```



And the third is functions over indices. Weights come from a file, labels from a formula, styles from a predicate — the escape hatches are callbacks, not enumerations:

```
#draw-mlp((4, 8, 3),
  weights: json("w.json"),
  node-label: (l, i) => $a_#i^((#l))$,
  edge-style: (l, i, j) => if i == j { (dash: "dashed") },
)
```



`w.json` holds one matrix per inter-layer gap. Positive weights are blue, negative red and dashed, and magnitude is thickness.

Everything below is one of those tiers in detail.

14.2 Writing a column

`count` default `required`

`mlp-layer`

The layer's true width, as a positional integer of at least 1. Everything else about a column is named.

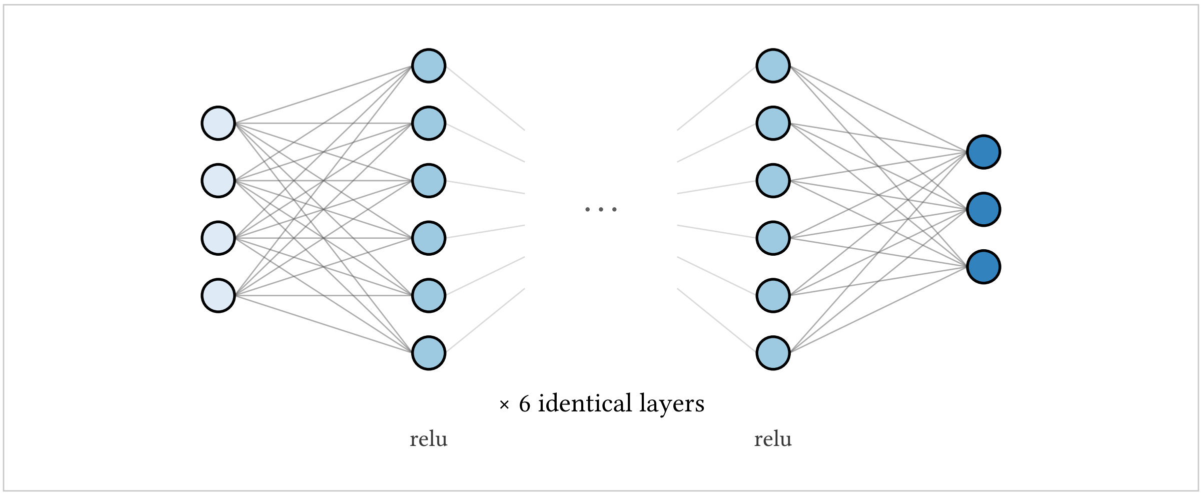
name default `auto`

mlp-layer

A reference for `mlp-edge`, never printed. Layers auto-name themselves `"l1"`, `"l2"`, … by position and gaps `"g1"`, `"g2"`, …; a user name replaces the auto name, shares its namespace, and collides loudly.

A gap column stands for elided depth — the “six more of these” of a network too deep to draw in full:

```
#draw-mlp((
  4,
  mlp-layer(6, activation: "relu"),
  mlp-gap(label: "× 6 identical layers", pitch: 1.4),
  mlp-layer(6, activation: "relu"),
  3,
))
```



label default `none`

mlp-gap

Caption in the label row, where the count usually goes.

pitch default `0.6 × layer-pitch`

mlp-gap

The width the gap column occupies.

A gap has no neurons. It cannot start or end an edge, it cannot sit first or last, and it breaks adjacency: the edge stubs entering and leaving it fade out rather than faking a connection across the elision. For the same reason a layer followed by a gap draws no bias node.

14.3 Layout

node-size default `0.17`

Node radius, in canvas units. Half the side length for squares. Also an `mlp-layer` option, so one layer can be drawn larger.

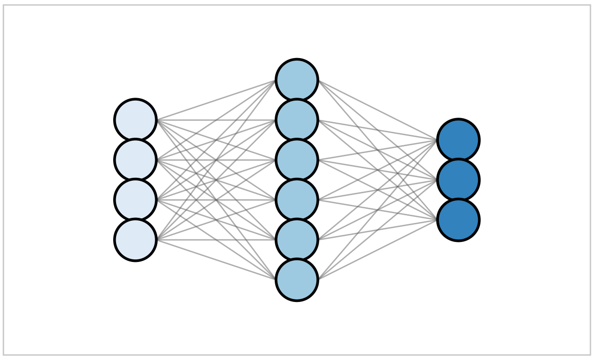
node-pitch default `0.6`

Spacing between neuron centers within a layer.

layer-pitch default `2.2`

Spacing between layer columns.

```
#draw-mlp((4, 6, 3),
  node-size: 0.22, node-pitch: 0.42, layer-pitch:
1.7)
```



Unequal layers are always centered on the flow axis. That is not a knob: every published version of this figure does it, and a version that does not reads as a mistake.

`unit` default `1cm`

`draw-mlp`

The canvas unit. It scales the geometry; text sizes are absolute, so a smaller unit buys a denser figure with the same size labels.

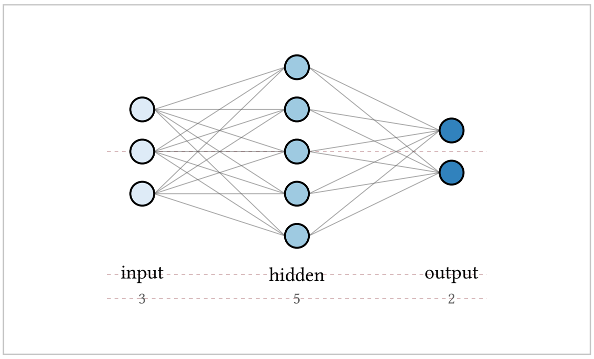
`title` default `none`

Centered above the figure.

`debug` default `false`

Draws the slot grid and the caption baseline rows in faint garnet. Useful for exactly one thing, which is seeing why something landed where it did.

```
#draw-mlp((3, 5, 2), layer-labels: auto, count-
badges: true,
  debug: true)
```

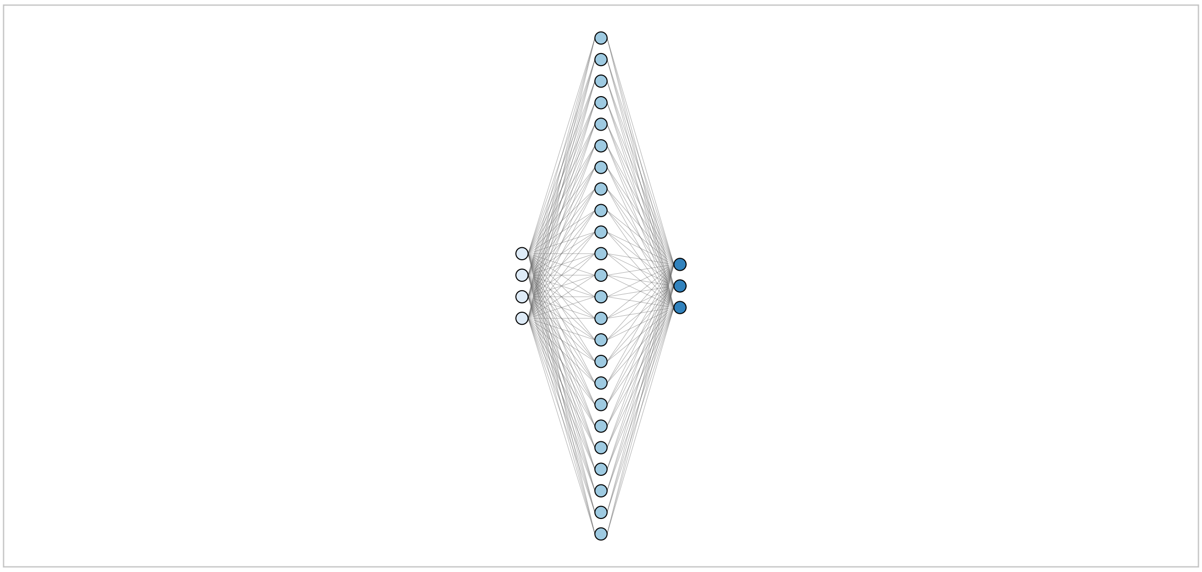


The dashed rows under the columns are shared, figure-wide baselines: one row for labels (with `sub` lines directly beneath), one for activation captions, one for count badges, in that order. Captions of a short column and a tall one sit level because the rows belong to the figure, not to the column, and a row that is empty for the whole figure takes no space.

14.4 Wide layers collapse

A 4–1024–10 classifier is a real network and an impossible figure. Drawing every neuron of a wide layer is the failure mode:

```
#draw-mlp((4, 24, 3), cutoff: none)
```



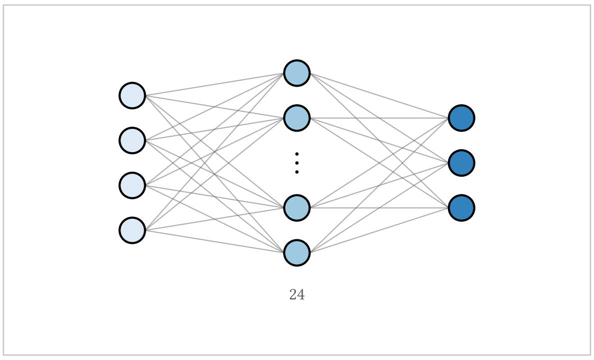
24 neurons drawn honestly, with `cutoff: none`. The figure is all layer and no network.

cutoff

default `10`

A layer with `count > cutoff` collapses: its first and last neurons drawn, a vertical ellipsis between them, and a badge stating the true count. `none` disables collapsing for the figure.

```
#draw-mlp((4, 24, 3))
```



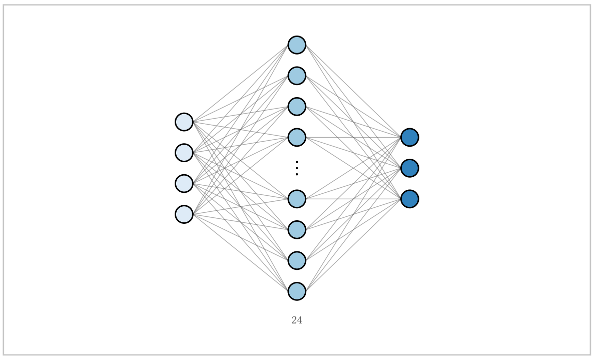
The same network with the default cutoff.

collapse-to

default `5`

How many slots a collapsed column occupies — odd and at least 3, since the middle slot holds the ellipsis. At `5`, the drawn neurons are 1, 2, count−1 and count. `cutoff` must exceed `collapse-to`, since collapsing a layer below that would not shorten it, and violating the pair is an error.

```
#draw-mlp((4, 24, 3), collapse-to: 9)
```



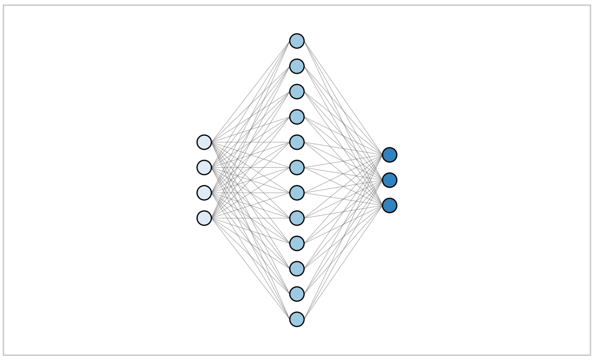
show-all

default `false`

mlp-layer

Exempts one layer from the figure `cutoff`.

```
#draw-mlp((4, mlp-layer(12, show-all: true), 3))
```

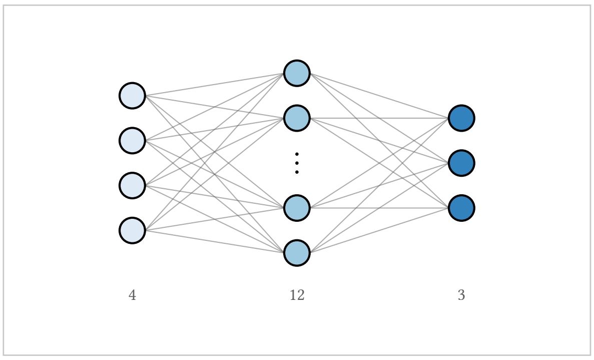


count-badges

default `auto`

`auto` badges collapsed columns only; `true` badges every column; `false` none. A collapsed column always states its true width unless you say `false`, because a figure that hides forty neurons and does not say so is lying about the model.

```
#draw-mlp((4, 12, 3), count-badges: true)
```



Edges attach only to drawn neurons. Naming an elided neuron — in an `mlp-edge` endpoint, say — is an error that lists the indices that **are** drawn.

14.5 Color

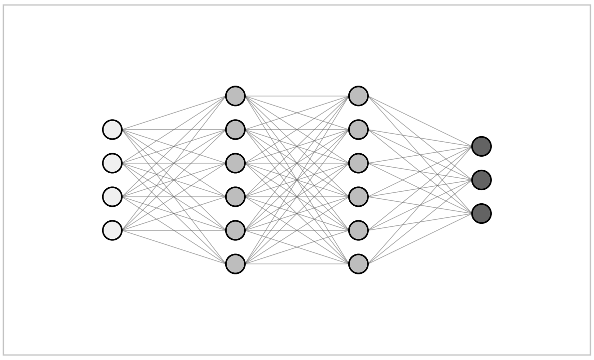
palette

default `"blues"`

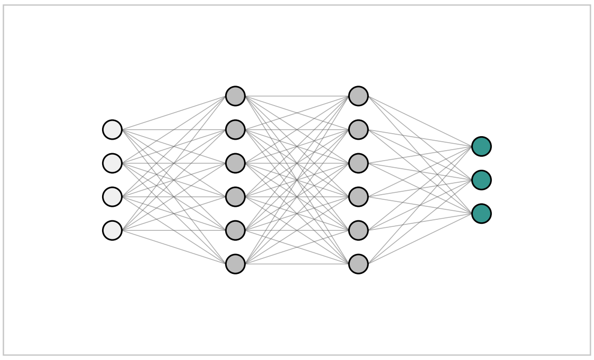
`"blues"`, `"classic"`, `"brand"`, `"greys"`, `"teal"`, `"lilaq"`, `"white"`, `"warm"`, `"cold"`, or a dictionary with keys among `input`, `hidden`, `output`, `bias` and `accent`, plus the optional ink keys `input-text`, `hidden-text`, `output-text` and `bias-text` described below. A partial dictionary overlays the blues palette; an unknown palette name is an error rather than a silent fallback.

The default is `blues` — light input, mid hidden, dark output, a single-hue ColorBrewer ladder, and the fills every figure in this chapter wears unless it says otherwise. The three roles are steps in luminance rather than changes of hue, so they stay apart when the figure is printed black-and-white or photocopied. `classic` is the green-input, blue-hidden, red-output TikZ convention most published MLP figures copy; its roles sit at near-equal luminance, which is exactly what a grayscale printer collapses, so it is a named option here rather than the default. `brand` is the package's house scheme in sand and grey with a garnet accent. `greys` is the ColorBrewer Greys ladder, pure grayscale for a venue that prints nothing else. `teal` keeps that grey ladder but moves the output to teal, so exactly one hue pops. `lilaq` takes the first colors of the lilaq plotting package's default cycle, so an MLP can sit beside lilaq plots without a second color language. `white` keeps every fill white with a black accent — the uncolored line-art figure textbooks print, its roles told apart by position and labels alone. `warm` and `cold` are derived from the isometric block palettes, so a document can keep its blocks and its neurons on one scheme.

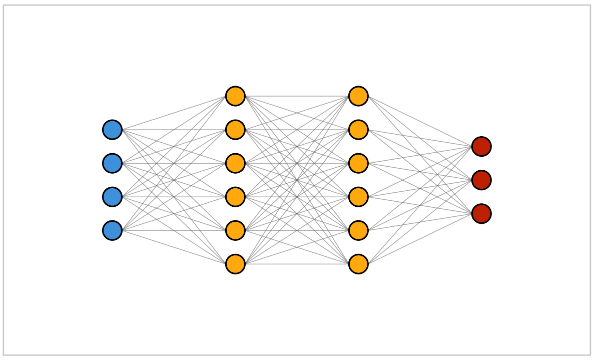
```
#draw-mlp((4, 6, 6, 3), palette: "greys")
```



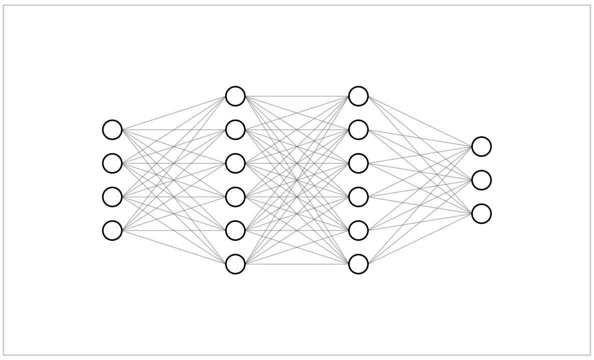
```
#draw-mlp((4, 6, 6, 3), palette: "teal")
```



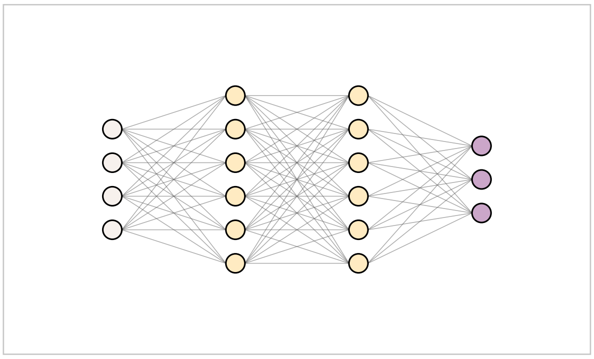
```
#draw-mlp((4, 6, 6, 3), palette: "lilaq")
```



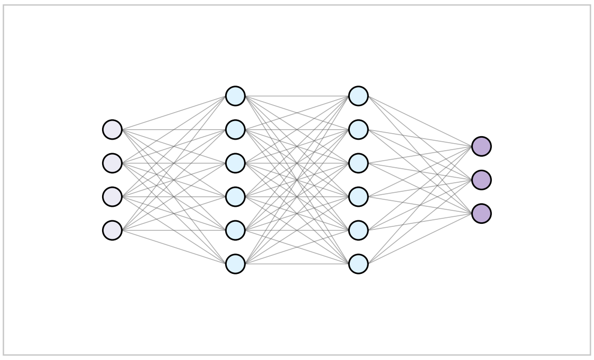
```
#draw-mlp((4, 6, 6, 3), palette: "white")
```



```
#draw-mlp((4, 6, 6, 3), palette: "warm")
```



```
#draw-mlp((4, 6, 6, 3), palette: "cold")
```



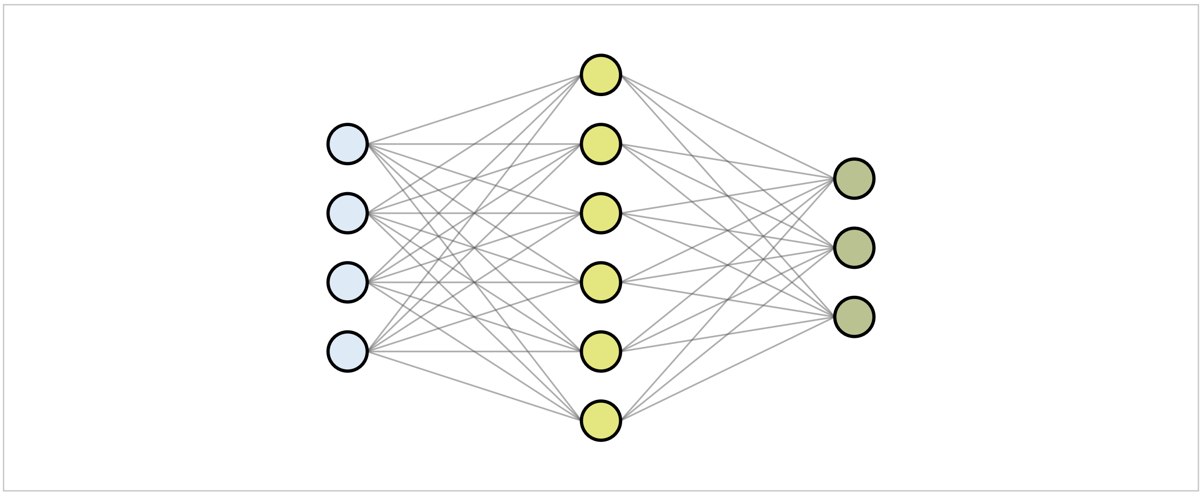
fill

default `role color`

`mlp-layer`

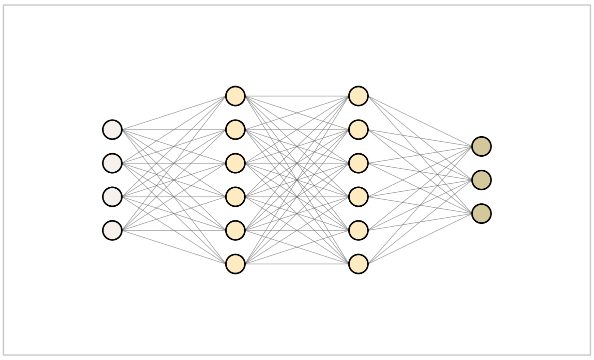
Node fill for one layer. The role assignment is positional: first layer input, last output, the rest hidden.

```
#draw-mlp((
  mlp-layer(4),
  mlp-layer(6, fill: rgb("#CED318").lighten(45%)),
  mlp-layer(3),
), palette: (
  output: rgb("#65780B").lighten(55%),
  accent: rgb("#65780B"),
))
```



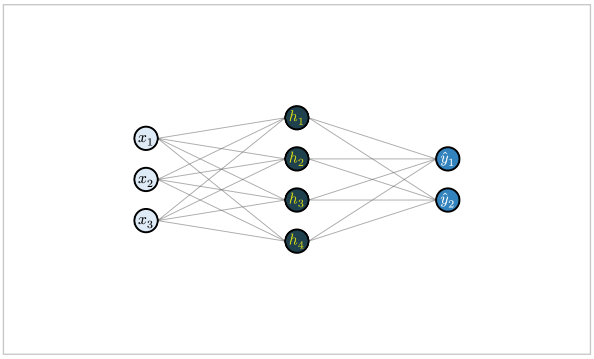
The named palettes are exported as `mlp-palettes`, so a scheme of your own can start from one instead of from nothing:

```
#draw-mlp((4, 6, 6, 3),
  palette: mlp-palettes.warm + (output:
rgb("#A49137").lighten(50%)))
```



Whatever the palette, everything drawn inside a node — an inside label, the Σ and f of a split node, an in-node glyph, value text, the inside bias label, the dropout X — picks black or white by the perceived luminance of the node's effective fill, so a dark scheme never buries its own ink. A palette dictionary can pin that ink per role instead, through the optional keys `input-text`, `hidden-text`, `output-text` and `bias-text`. A pinned color applies only while the node still wears the palette's own role fill; a per-layer `fill`, a `node-style` fill or a `values` ramp falls back to the automatic flip, since ink chosen against one fill cannot be trusted against another. No built-in palette sets these keys.

```
#draw-mlp((3, 4, 2), node-label: auto, palette: (
  hidden: rgb("#1F414D"),
  hidden-text: rgb("#CED318"),
))
```



The hidden fill is dark, so its ink would flip to white; the `hidden-text` key pins it instead. The other roles keep the automatic choice — black on the light input, white on the dark blues output.

Values as fill

`values` default `none`

`mlp-layer`

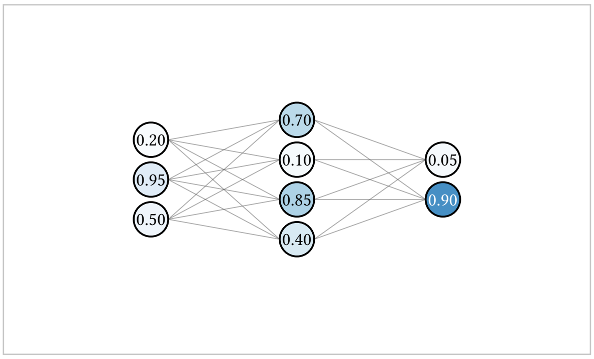
An array of floats in 0..1, one per neuron. The fill becomes a ramp from white to the layer's role color — the activations-as-brightness view 3Blue1Brown made standard.

`show-value-text` default `false`

`mlp-layer`

Prints the value, two decimals, inside the node. Requires `values`, and wins over an inside node label. Text flips to white on dark fills.

```
#draw-mlp((
  mlp-layer(3, values: (0.20, 0.95, 0.50), show-
value-text: true),
  mlp-layer(4, values: (0.70, 0.10, 0.85, 0.40),
show-value-text: true),
  mlp-layer(2, values: (0.05, 0.90), show-value-text:
true),
), node-size: 0.26)
```



14.6 Node shapes and the input

`node-shape` default `"circle"`

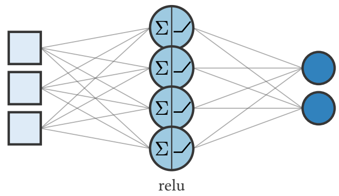
`"circle"`, `"square"` or `"split"`, for the whole figure. The per-layer `shape` option takes the same values and wins.

`shape` default `figure node-shape`

`mlp-layer`

`split` is the classic summation-then-squash view of a unit: a slightly larger circle with a vertical divider, Σ in the left half, the activation curve (or f when the layer has none) in the right.

```
#draw-mlp((
  mlp-layer(3, shape: "square"),
  mlp-layer(4, shape: "split", activation: "relu"),
  mlp-layer(2),
), node-size: 0.24,
   node-stroke: (paint: rgb("#363636"), thickness:
1pt))
```



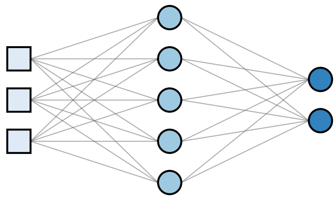
node-stroke default `(paint: black, thickness: 0.8pt)`

The node outline, for the whole figure; per-layer `stroke` wins.

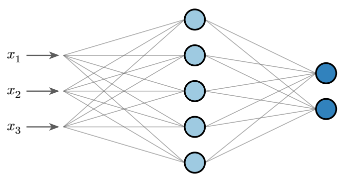
input-style default `"nodes"`

`"square"` draws layer 1 as squares, the AIMA convention for values-not-units. `"arrows"` replaces the layer-1 nodes with labelled stub arrows into layer 2, perceptron style; the layer-1 `count` still says how many.

```
#draw-mlp((3, 5, 2), input-style: "square")
```



```
#draw-mlp((3, 5, 2), input-style: "arrows")
```

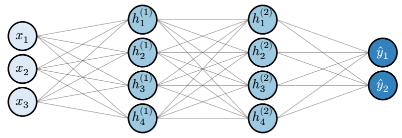


14.7 Node labels

node-label default `none`

`none`, `auto`, or a function. At the figure level the function takes `(l, i)`; on a layer it takes `(i)` and wins. `auto` is role-based math over true indices: x_i on the input, $h_i^{(k)}$ on hidden layer k (plain h_i when there is only one), $\hat{y}_i$ on the output — so a collapsed layer labels its last two drawn neurons $n - 1$ and n, not 4 and 5.

```
#draw-mlp((3, 4, 4, 2), node-label: auto, node-size:
0.26)
```

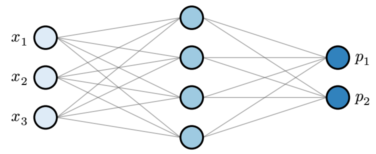


node-label-pos default `"inside"`

mlp-layer

`"inside"` shrinks the label to fit the node; `"left"` and `"right"` set it beside, which is where a label goes once nodes carry glyphs or values.

```
#draw-mlp((
  mlp-layer(3, node-label: i => $x_#i$, node-label-
pos: "left"),
  mlp-layer(4),
  mlp-layer(2, node-label: i => $p_#i$, node-label-
pos: "right"),
))
```



Inside labels yield gracefully: value text wins over an inside label, and the glyph activation style and the split shape suppress inside labels entirely rather than overprinting a curve.

14.8 Marking neurons

Four per-layer index arrays mark neurons, and a neuron may appear in at most one of them — overlap is an error, not a merge.

dropped default `()`

mlp-layer

An X through the node and its edges omitted — the dropout figure as Srivastava et al.
drew it.

dimmed default `()`

mlp-layer

The node and its edges at reduced opacity. De-emphasis without deletion.

highlighted default `()`

mlp-layer

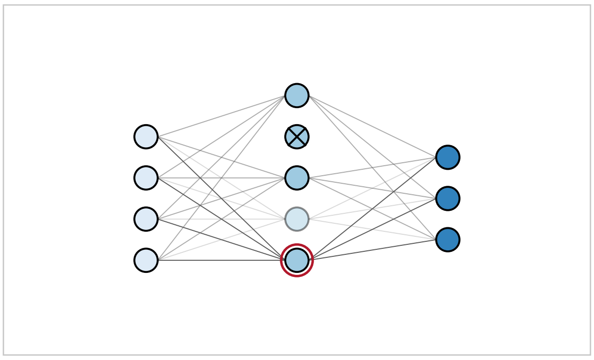
An accent ring around the node, its edges at full opacity. How you trace one path through a network.

exclude default `()`

mlp-layer

Not drawn at all, edges omitted. Referring to an excluded neuron from an edge endpoint is an error.

```
#draw-mlp((
  mlp-layer(4),
  mlp-layer(6, dropped: (2,), dimmed: (4,),
               highlighted: (5,), exclude: (6,)),
  mlp-layer(3),
))
```



In the hidden layer: neuron 2 dropped, 4 dimmed, 5 highlighted, and 6 excluded — the column is one node shorter than its count.

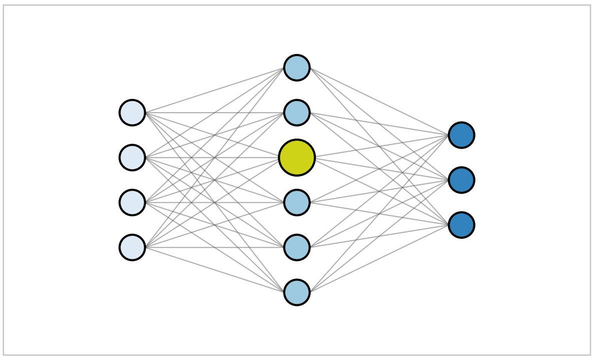
`node-style`

default `none`

`mlp-layer`

A function from neuron index to a dictionary of `fill`, `stroke`, `size`, `shape` — or `none` to leave the neuron alone. Applied last, after every other node option.

```
#draw-mlp((
  4,
  mlp-layer(6, node-style: i => if i == 3 {
    (fill: rgb("#CED318"), size: 0.24)
  }),
  3,
))
```



14.9 Edges

`edge-stroke`

default `(paint: #5C5C5C, thickness: 0.4pt)`

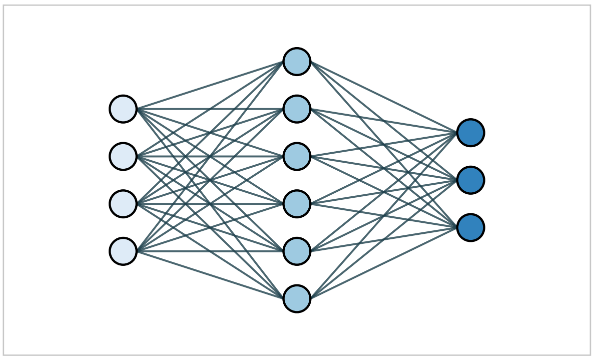
The base stroke of every adjacent-pair edge, before opacity.

`edge-opacity`

default `50%`

Applied to the stroke paint. Edges draw before nodes and the fan of a dense layer overlaps itself; at partial opacity it reads as texture rather than ink.

```
#draw-mlp((4, 6, 3),
  edge-stroke: (paint: rgb("#1F414D"), thickness:
0.7pt),
  edge-opacity: 80%)
```

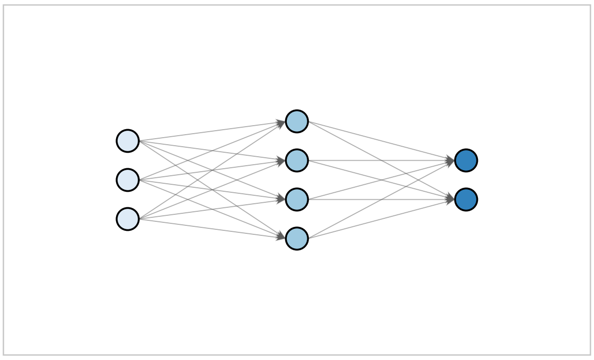


`arrows`

default `"none"`

`"all"` puts a small stealth head on every inter-layer edge. The textbook default is none: flow is left to right and saying so ninety-six times adds clutter, not information.

```
#draw-mlp((3, 4, 2), arrows: "all", layer-pitch: 2.6)
```



io-stubs

default `false`

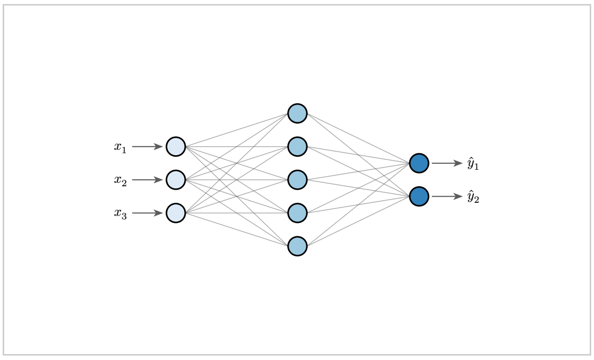
`true`, `"in"` or `"out"`: short stub arrows entering layer 1 and leaving layer L, one per drawn neuron. The input side is suppressed under `input-style: "arrows"`, which already draws them.

stub-labels

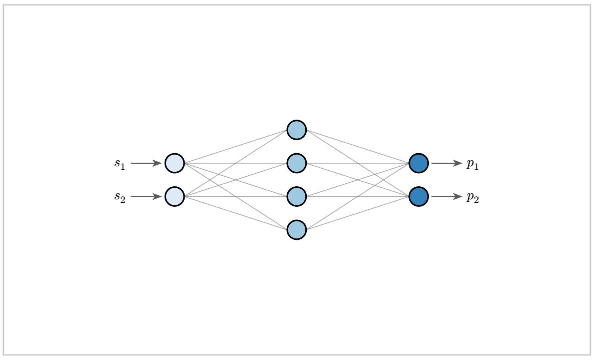
default `auto`

`auto` labels in-stubs x_i and out-stubs $\hat{y}_i$; `none` labels nothing; a function `(side, i)` labels however you like, with `side` one of `"in"` and `"out"`.

```
#draw-mlp((3, 5, 2), io-stubs: true)
```




```
#draw-mlp((2, 4, 2), io-stubs: true,
  stub-labels: (side, i) => if side == "in"
{ $s_#i$ } else { $p_#i$ })
```

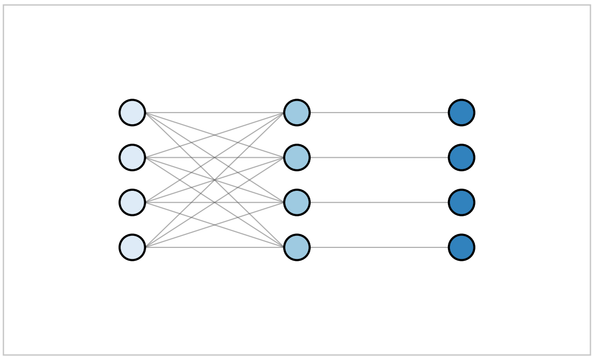


edge-filter

default `none`

A predicate `(l, i, j)` over the adjacent-pair edge set — gap `l` sits between layers `l` and `l+1`, `i` is the source neuron, `j` the target. Returning `false` drops the edge. Sparse connectivity as a function rather than an enumeration.

```
#draw-mlp((4, 4, 4),
  edge-filter: (l, i, j) => l == 1 or i == j)
```



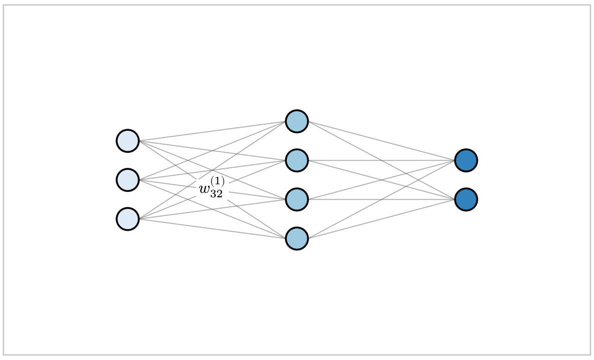
Full fan into the first gap, one-to-one in the second.

`edge-labels`

default `()`

An array of `(l: .., i: .., j: .., label: ..)` dictionaries naming edges to label; omitting `label` (or passing `auto`) typesets $w_{ji}^{(l)}$. This is deliberately per-edge: the convention is one representative label, not a hundred unreadable ones.

```
#draw-mlp((3, 4, 2), layer-pitch: 2.6,
  edge-labels: ((l: 1, i: 2, j: 3),))
```

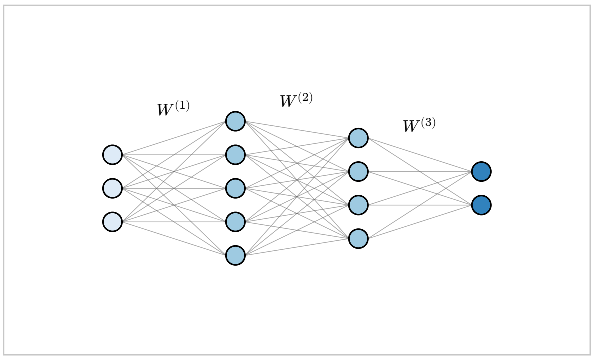


`matrix-labels`

default `false`

`true` centers $W^{(l)}$ in each inter-layer gap, above the fan — the Goodfellow compact annotation. A function `(l)` supplies your own; returning `none` skips a gap.

```
#draw-mlp((3, 5, 4, 2), matrix-labels: true)
```



14.10 Weights

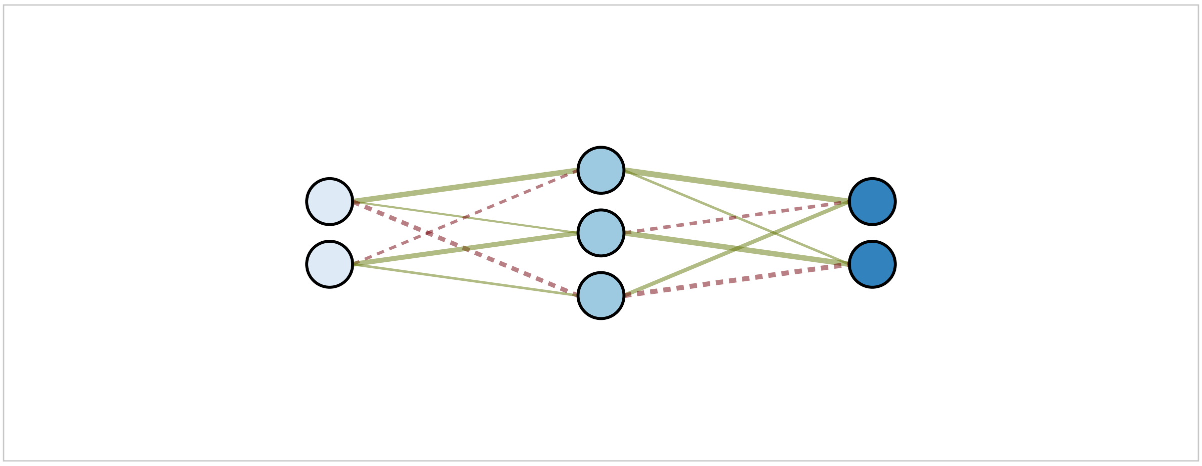
`weights`

default `none`

An array of matrices, a function `(l, i, j)` returning a float, or `"random"`. When set, edge appearance encodes the values.

Matrix orientation is the Wx convention and worth stating precisely. `weights.at(l - 1)` maps gap l, has `counts[l+1]` rows by `counts[l]` columns, and entry $[j][i]$ is the weight of the edge from neuron i of layer l into neuron j of layer $l + 1$. A mismatched matrix is an error stating both the expected and the received shape.

```
#draw-mlp((2, 3, 2),
  weights: (
    ((0.9, -0.4), (0.2, 0.8), (-0.7, 0.3)),
    ((1.0, -0.5, 0.6), (0.3, 0.9, -0.8)),
  ),
  weight-colors: (positive: rgb("#65780B"), negative: rgb("#73000A")),
  weight-thickness: (0.3pt, 1.6pt),
  node-size: 0.22, layer-pitch: 2.6)
```



`weight-encode` default `("color", "thickness", "dash")`

Which channels carry the value: any subset of `"color"`, `"thickness"`, `"opacity"`, `"dash"`. With $t = \min(|w| \ / \ \text{range}, 1)$, color paints by sign, thickness interpolates `weight-thickness` by t, opacity runs 15% to 100% by t in place of `edge-opacity`, and dash draws negative edges dashed, positive solid.

`weight-colors` default `(positive: #0571B0, negative: #CA0020)`

The sign colors — blue positive, red negative, the ColorBrewer endpoints the Playground and 3Blue1Brown made the convention. A partial dictionary overlays the default.

`weight-range` default `auto`

The $|w|$ that saturates the encoding. `auto` takes the maximum over the data: full matrices for the array form, collapsed entries included, so the visible encoding stays honest about the true scale; drawn edges only for the function and `"random"` forms.

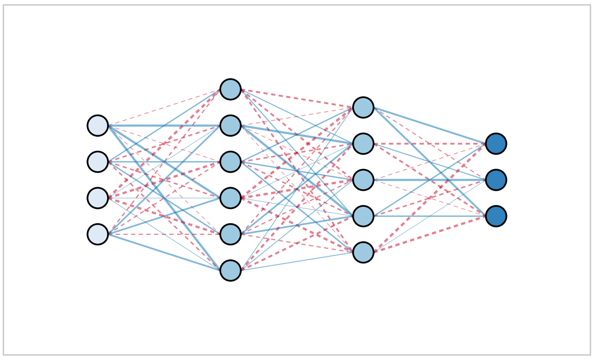
`weight-thickness` default `(0.2pt, 1.1pt)`

Thickness at zero and at saturation.

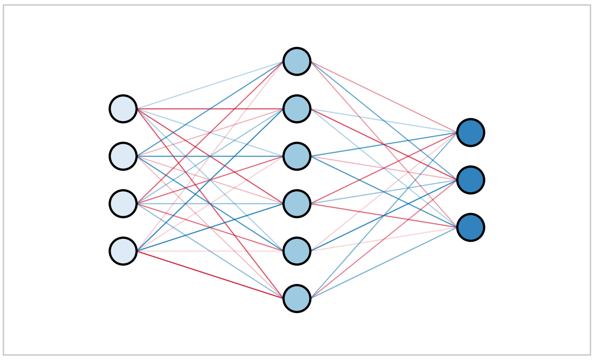
`seed` default `42`

For `weights: "random"`. Typst has no random source, so the values are a deterministic hash of the seed and the edge indices — the same seed always draws the same figure.

```
#draw-mlp((4, 6, 5, 3), weights: "random", seed: 7)
```

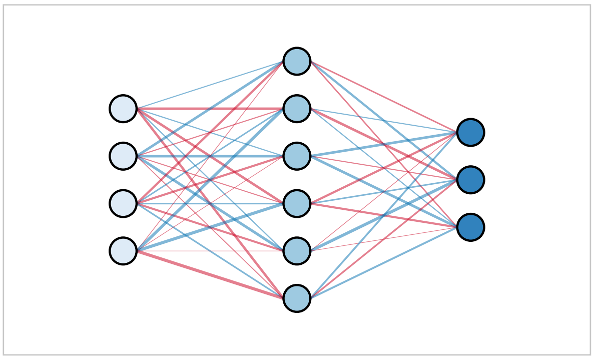


```
#draw-mlp((4, 6, 3), weights: "random",
  weight-encode: ("color", "opacity"), weight-range:
1.2)
```



Sign carried by color alone does not survive a grayscale printer or every reader's eyes. The `"dash"` member is the accessibility channel — negative edges render dashed, positive stay solid — and it sits in the default set, so the sign survives the photocopier without being asked for. Passing `weight-encode` without the member draws every edge solid, for the occasional dense fan where the dashes read as noise:

```
#draw-mlp((4, 6, 3), weights: "random",
  weight-encode: ("color", "thickness"))
```



> Bias edges are not covered by `weights` — those matrices describe W, not b — and `edge-filter` and `edge-style` do not see them either. Letting a weight matrix silently mis-size against `counts + 1` is the kind of trap this package exists to reject.

14.11 Bias nodes

`bias` default `false`

A bias node above each of layers 1 to $L - 1$, one `node-pitch` above the column's top slot, filled from the palette's `bias` role, with an edge to every drawn neuron of the next layer. Also an `mlp-layer` option, so one layer can opt out (or in).

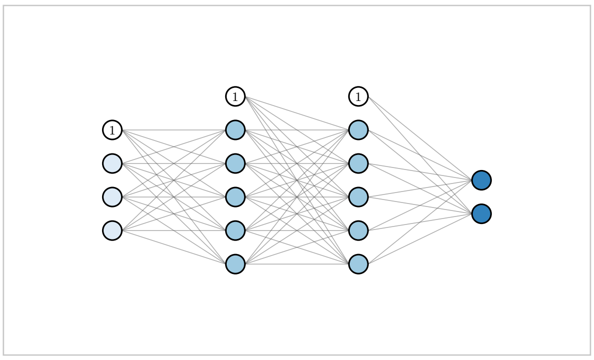
`bias-label` default `[bias]`

The label on every bias node. `none` suppresses it.

`bias-label-pos` default `"above"`

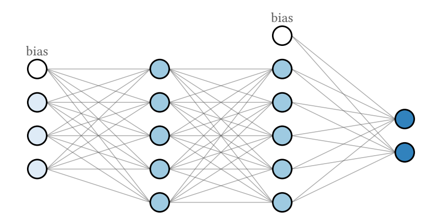
Where the label sits. `"above"`, the default, renders it over the node in small grey text, where a word fits; `"inside"` is the classic in-node placement for short content like `$1$`.

```
#draw-mlp((3, 5, 5, 2), bias: true, bias-label: $1$,
bias-label-pos: "inside")
```



`bias-label-pos: "inside"` restores the classic in-node 1.

```
#draw-mlp((
  3,
  mlp-layer(5, bias: false),
  5,
  2,
), bias: true)
```



`bias: true` on the figure, `bias: false` on layer 2.

`bias: true` on the last layer is an error — there is no next layer to feed — and a layer immediately followed by an `mlp-gap` draws none, since bias edges across an elision would fake adjacency.

14.12 Captions

`label`

default `none`

`mlp-layer`, `mlp-gap`

The column's caption in the label row. Any content.

`sub`

default `none`

`mlp-layer`

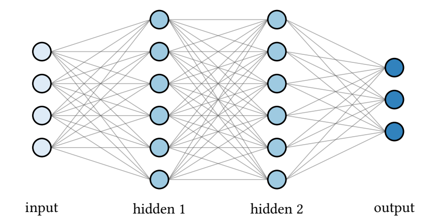
A second caption line, smaller and lighter, directly beneath the label.

`layer-labels`

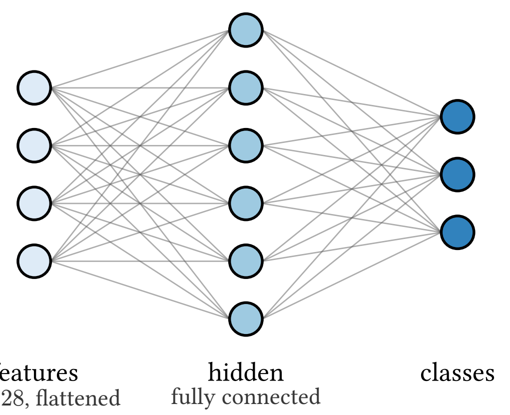
default `none`

`auto` captions the roles: "input", "hidden 1" … "hidden k", "output" (just "hidden" when there is one). An array gives one entry per layer, `none` entries included. A per-layer `label` wins over both.

```
#draw-mlp((4, 6, 6, 3), layer-labels: auto)
```



```
#draw-mlp((
  mlp-layer(4, label: "features", sub: [$28 times 28$, flattened]),
  mlp-layer(6, label: "hidden", sub: "fully connected"),
  mlp-layer(3, label: "classes"),
))
```

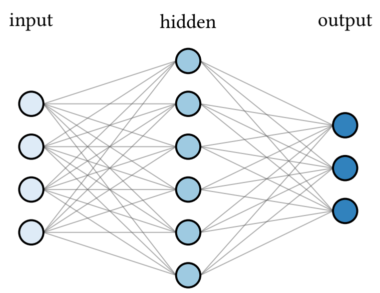


label-pos

default "below"

Which side of the columns the caption rows sit on. A `sub` still reads beneath its label either way.

```
#draw-mlp((4, 6, 3), layer-labels: auto, label-pos:
"above")
```



14.13 Activations

activation

default `none`

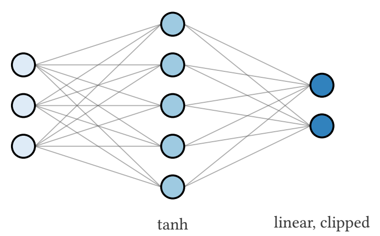
The figure-wide activation, applied to layers 2 to L; layer 1 never defaults to one. The per-layer `activation` wins, including `none` to exempt a layer and including layer 1 if you insist.

activation-style

default "caption"

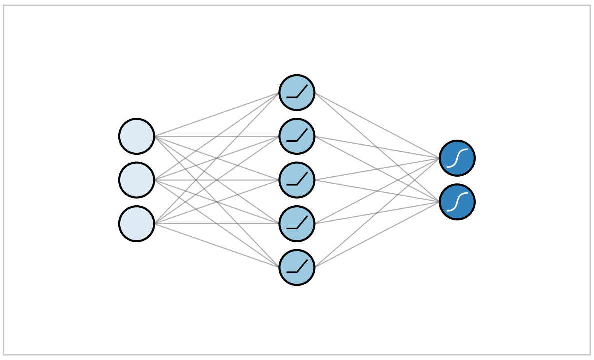
How activations are shown: "caption", "glyph", "block" or "split".
`caption` typesets the name, small, in the activation baseline row. Any string or content is accepted — nothing errors here, because anything renders as text:

```
#draw-mlp((
  3,
  mlp-layer(5, activation: "tanh"),
  mlp-layer(2, activation: [linear, clipped]),
))
```



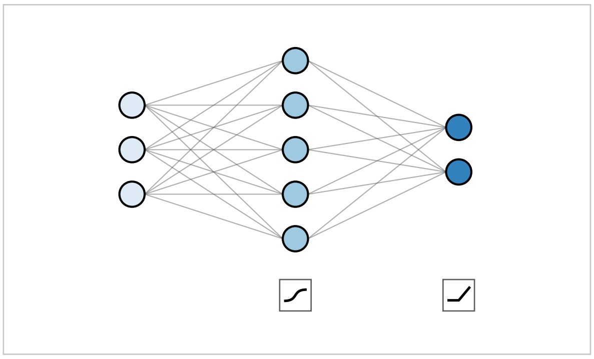
`glyph` draws the bare curve inside each neuron of the layer — curve only, no axes:

```
#draw-mlp((
  3,
  mlp-layer(5, activation: "relu"),
  mlp-layer(2, activation: "tanh"),
), activation-style: "glyph", node-size: 0.24)
```



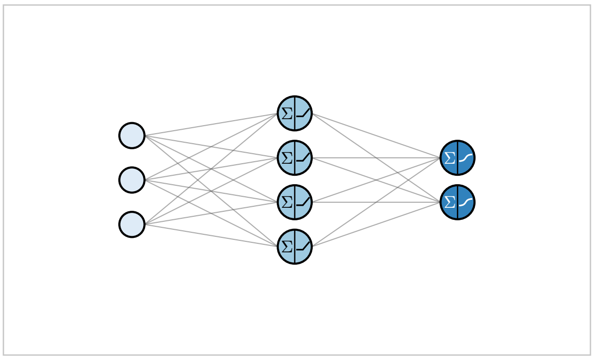
`block` draws one small transfer-function icon in the activation row under the column — the MATLAB/Hagan convention, placed where it cannot collide with edges:

```
#draw-mlp((
  3,
  mlp-layer(5, activation: "sigmoid"),
  mlp-layer(2, activation: "relu"),
), activation-style: "block")
```



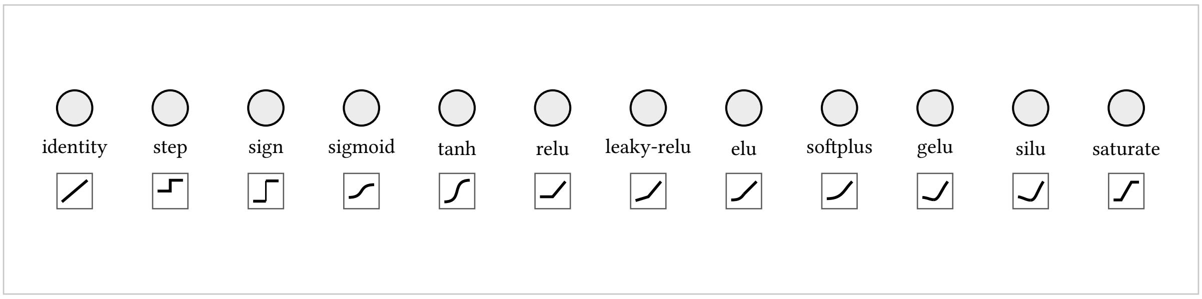
`split` forces the split node shape for layers with an activation — Σ then the curve:

```
#draw-mlp((
  3,
  mlp-layer(4, activation: "relu"),
  mlp-layer(2, activation: "sigmoid"),
), activation-style: "split")
```



The full catalog, one silhouette per name:

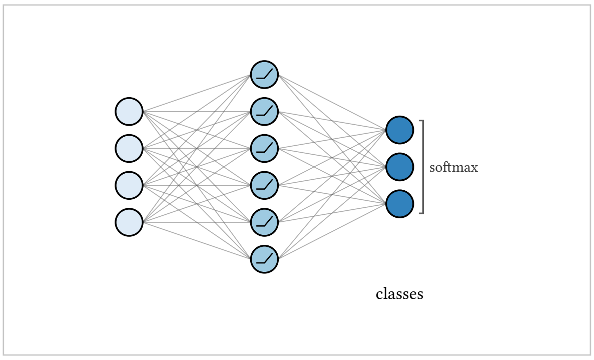
```
#let curves = ("identity", "step", "sign", "sigmoid", "tanh", "relu",
  "leaky-relu", "elu", "softplus", "gelu", "silu", "saturate")
#draw-mlp(
  curves.map(g => mlp-layer(1, activation: g, label: g,
    fill: rgb("#ECECEC"))),
  activation-style: "block", node-size: 0.24,
  edge-filter: (l, i, j) => false, layer-pitch: 1.3,
)
```



`linear`, `heaviside`, `logistic` and `swish` are aliases of `identity`, `step`, `sigmoid` and `silu`. The three smooth kinks differ deliberately: `elu` saturates left, `softplus` is everywhere-smooth, `gelu` and `silu` dip below zero before the rise.

`softmax` and `argmax` have no per-neuron curve — the function couples the whole layer — so in any glyph-bearing style they render as a right-side bracket spanning the layer, named beside it:

```
#draw-mlp((
  4,
  mlp-layer(6, activation: "relu"),
  mlp-layer(3, activation: "softmax", label:
"classes"),
), activation-style: "glyph", node-size: 0.22)
```

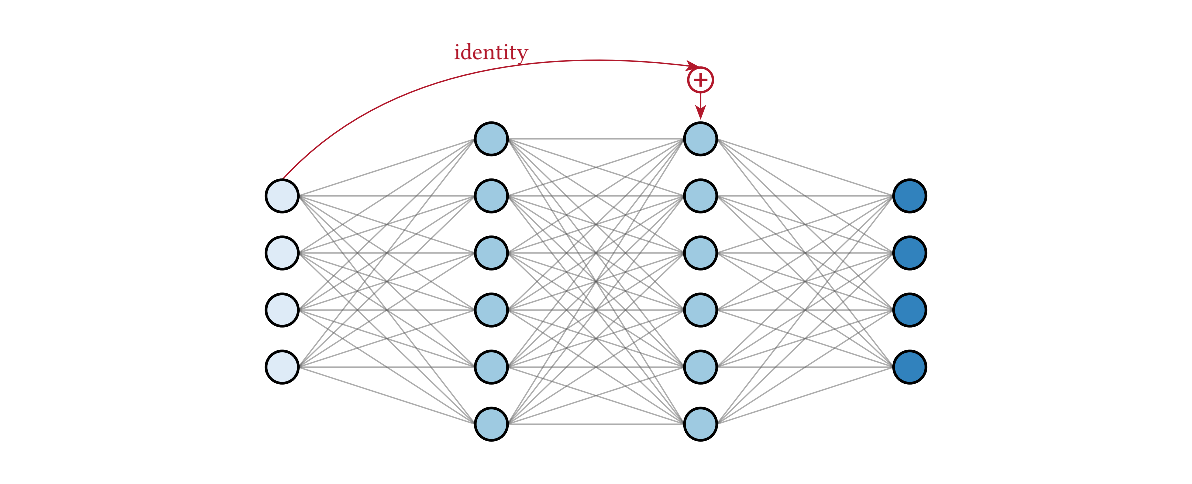


An activation name outside the catalog is an error under `"glyph"`, `"block"` and `"split"`, and the error lists the catalog. Under `"caption"` it is just text.

14.14 Extra edges

Everything beyond the adjacent-pair fan — a skip, a recurrent loop, one restyled edge — is an `mlp-edge(..)` in the figure's `edges` array. An endpoint is `"name.index"` for one neuron, `"name"` for a whole layer, or the positional forms `(layer, index)` and `(layer,)`. What the edge **is** follows from its endpoints. An adjacent node pair restyles the existing edge rather than duplicating it (and re-adds the single link if `edge-filter` removed it). A non-adjacent node pair is a new arc routed above the network, or below with `style: "arc-below"`. A non-adjacent layer pair is one layer-level skip arc, column top to column top. An adjacent layer pair restyles every edge in that gap. And `from` equal to `to`, layer-level, is a recurrent self-loop.

```
#draw-mlp((
  mlp-layer(4, name: "in"),
  mlp-layer(6, name: "h1"),
  mlp-layer(6, name: "h2"),
  mlp-layer(4, name: "out"),
), edges: (
  mlp-edge(from: "in", to: "h2", sum: true, label: "identity"),
))
```

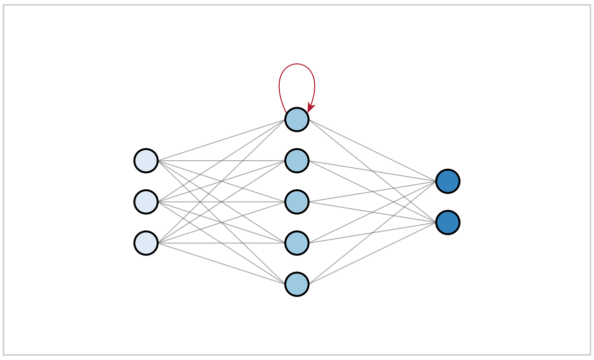


sum default `false`

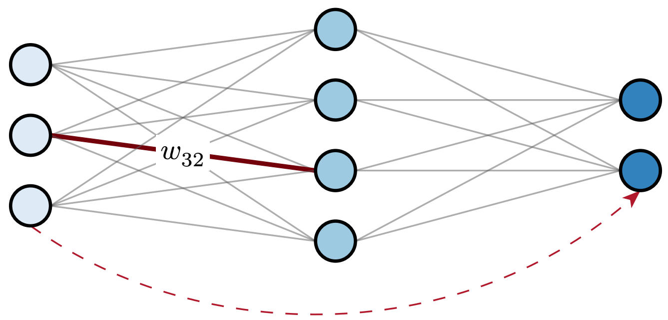
mlp-edge

Terminates a layer-level skip in a small circled plus before the target layer — the ResNet merge. On any other kind of edge it is an error.

```
#draw-mlp((
  mlp-layer(3, name: "x"),
  mlp-layer(5, name: "h"),
  mlp-layer(2, name: "y"),
), edges: (mlp-edge(from: "h", to: "h", style:
"loop"),))
```



```
#draw-mlp((3, 4, 2), layer-pitch: 2.6, edges: (
  mlp-edge(from: "l1.2", to: "l2.3", paint: rgb("#73000A"),
    thickness: 1.1pt, opacity: 100%, label: $w_(32)$),
  mlp-edge(from: "l1.3", to: "l3.2", style: "arc-below",
    dash: "dashed"),
))
```



An adjacent pair restyled and labelled, and a node-level skip routed underneath.

style default `auto`

`mlp-edge`

`"straight"`, `"arc-above"`, `"arc-below"` or `"loop"`. `auto` picks straight for adjacent endpoints, an arc above for skips, a loop when the endpoints match.

paint default `edge default; accent for skips and loops`

`mlp-edge`

thickness default `edge default`

`mlp-edge`

dash default `none`

`mlp-edge`

opacity default `edge default; 100% for skips and loops`

`mlp-edge`

arrow default `true for skips and loops, false for adjacent`

`mlp-edge`

A stealth head at `to`.

label default `none`

`mlp-edge`

At the arc apex, or the midpoint of a straight edge.

edge-style default `none`

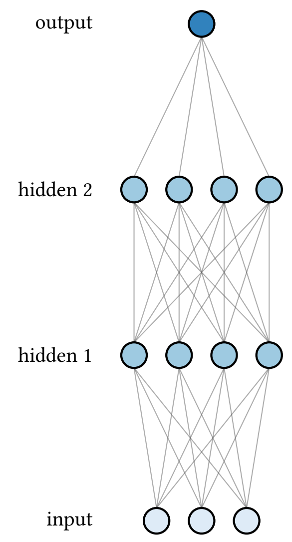
The last word on adjacent-pair edges: a function `(l, i, j)` returning a dictionary of `paint`, `thickness`, `dash`, `opacity` — or `none` to leave the edge alone. Style precedence over an edge is base stroke, then weight encoding, then dimmed and highlighted states, then `mlp-edge` overrides, then this hook.

14.15 Growing upward

direction default "right"

"up" is the Goodfellow/AIMA orientation: input at the bottom, output at the top. It is a coordinate swap after layout, so every option means what it meant. Captions move to the left of their rows, badges to the right, and label-pos is ignored.

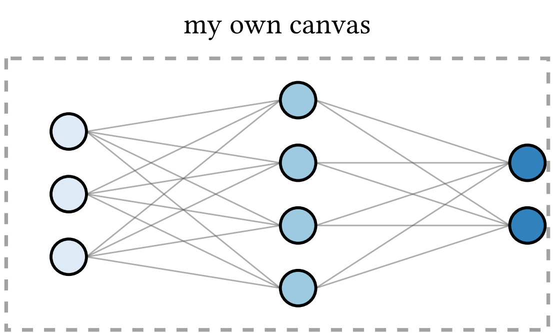
```
#draw-mlp((3, 4, 4, 1), direction: "up", layer-labels: auto)
```



14.16 Embedding in a canvas

draw-mlp is exactly canvas(length: unit, mlp-content(..)), and mlp-content is exported. Inside your own CeTZ canvas the figure is draw content like any other, with the first column centered at the origin and the flow axis on $y = 0$:

```
#canvas(length: 1cm, {
  mlp-content((3, 4, 2))
  draw.rect((-0.6, -1.3), (4.6, 1.3),
    stroke: (paint: rgb("#A2A2A2"), dash: "dashed"))
  draw.content((2, 1.6), text(size: 8pt)[my own canvas])
})
```

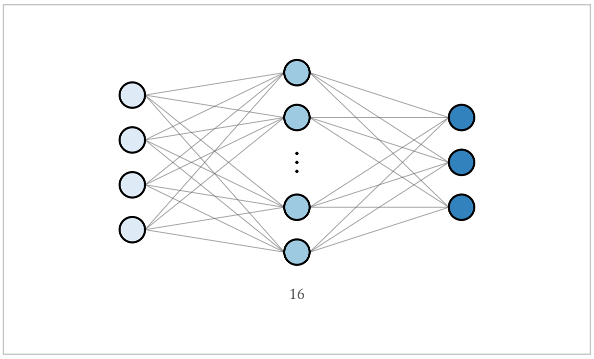


14.17 Typos are errors here too

The MLP renderer keeps the package's validation contract: unknown options on every constructor and on the figure, plain dictionaries, bad counts, gap placement, name collisions, index ranges and the mutual exclusivity of the marking arrays, values length, cutoff against collapse-to, palette names and keys, glyph names when a glyph is required, weight shapes per gap, `weight-encode` members, endpoint resolution, and `edge-labels` that name no drawn edge. The voice is the same, `did-you-mean` included:

```
error: unknown draw-mlp option "cutof" on the figure.
Did you mean "cutoff"? Options accepted here: activation,
activation-style, arrows, bias, bias-label, bias-label-pos, ...
```

```
#draw-mlp((4, 16, 3), cutoff: 8)
```



The fix. One letter, and the figure it was always meant to be.

`weights` matrix N (layers N -> N+1) must be R x C

The matrix facing a gap has the wrong shape. The message states the expected and the received shape, and restates the target-major orientation while it is at it.

activation "X" on layer N has no glyph

A glyph-bearing style needs a curve from the catalog; the message lists it, suggests the nearest name, and notes that under `"caption"` anything is text.

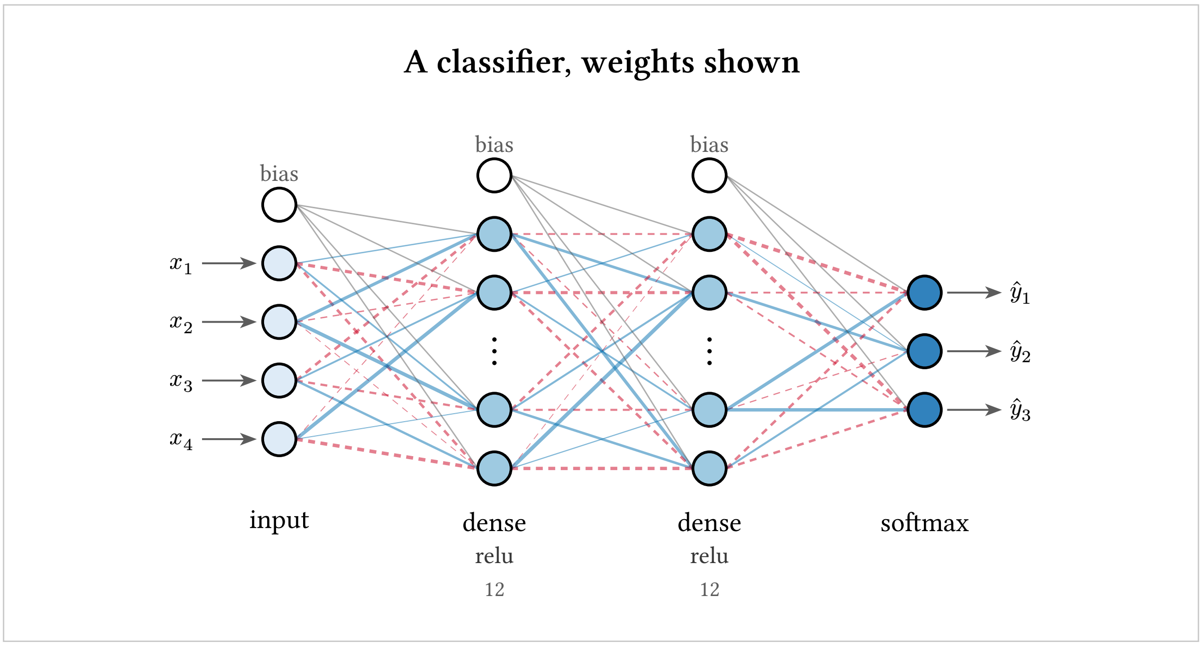
mlp-edge endpoint refers to "X", which is not the name of any layer

A dangling endpoint, with a `did-you-mean` over the names that exist. Gaps, excluded neurons and elided neurons get their own messages, each saying why the endpoint cannot take an edge.

14.18 A composite

Weights, bias, stubs, captions, a collapse and a title — everything this chapter covered, in one figure that is still eleven lines:

```
#draw-mlp((
  mlp-layer(4,  label: "input"),
  mlp-layer(12, label: "dense", activation: "relu"),
  mlp-layer(12, label: "dense", activation: "relu"),
  mlp-layer(3,  label: "softmax"),
), cutoff: 8, bias: true, weights: "random",
   io-stubs: true, title: "A classifier, weights shown")
```



CHAPTER 15

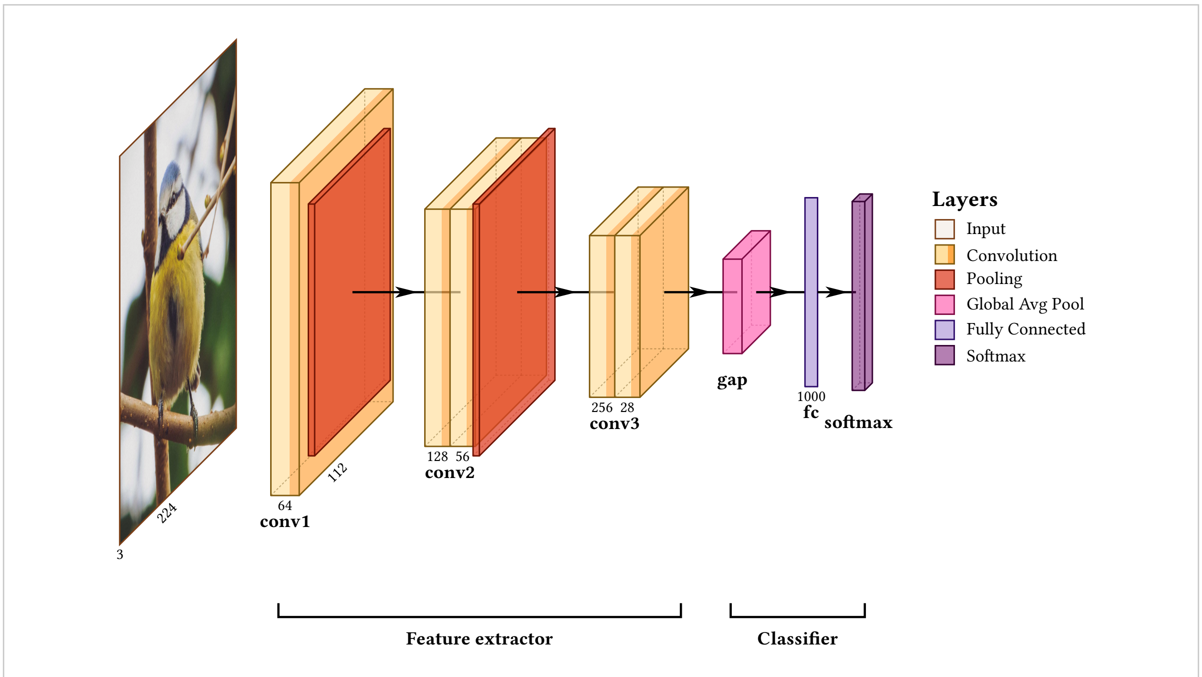
Patterns

Everything here is built only from what the preceding chapters covered. These are not architectures to copy so much as shapes to recognise: most figures are one of them with different labels.

15.1 A classifier

Types, shapes and two groups.

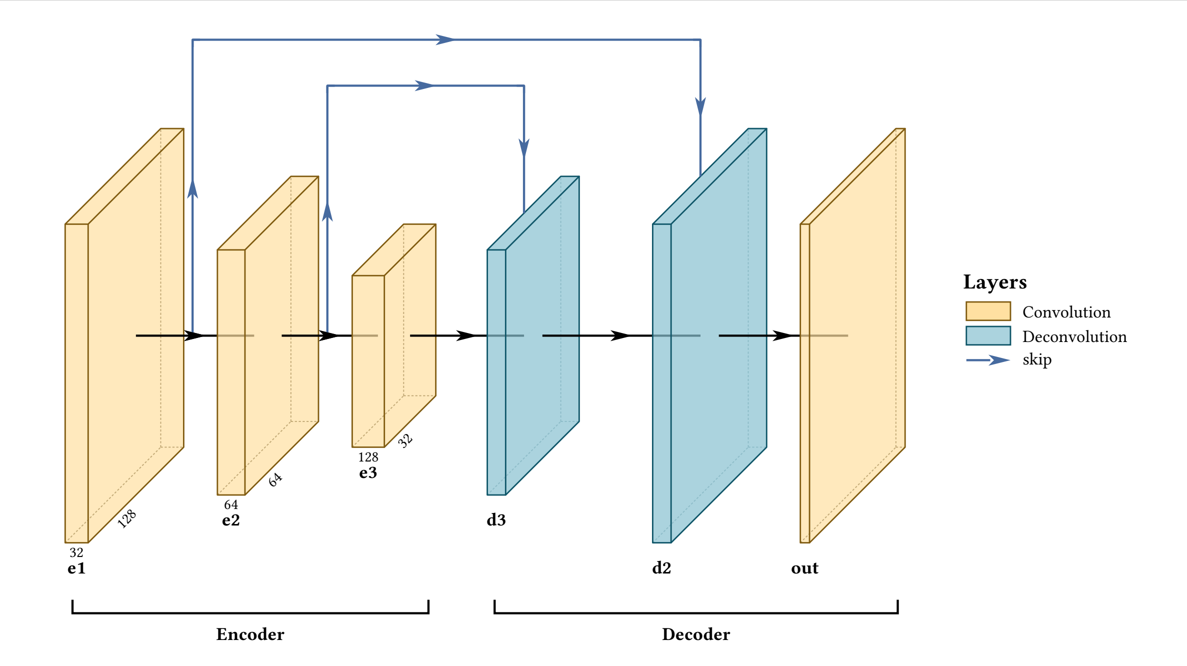
```
#draw-network(
  (
    input(name: "in", image: "default", shape: (3, 224, 224),
          channels: (3, 224)),
    conv(name: "c1", label: "conv1", shape: (64, 112, 112),
         channels: (64, 112)),
    pool(),
    conv(name: "c2", label: "conv2", shape: (128, 56, 56),
         channels: (128, 56), widths: (0.4, 0.4)),
    pool(),
    conv(name: "c3", label: "conv3", shape: (256, 28, 28),
         channels: (256, 28), widths: (0.4, 0.4)),
    gap(name: "g", label: "gap"),
    fc(name: "f", label: "fc", channels: (1000,), depth: 0),
    softmax(name: "s", label: "softmax"),
  ),
  groups: (
    group(from: "c1", to: "c3", label: "Feature extractor"),
    group(from: "g", to: "s", label: "Classifier"),
  ),
  show-legend: true,
  show-relu: true,
)
```



15.2 An encoder-decoder

A size pyramid down and back up, with skips arriving on their targets. The two `deconv` blocks take their size from `shape` exactly as the encoder does, so the figure is symmetric because the model is, not because it was measured.

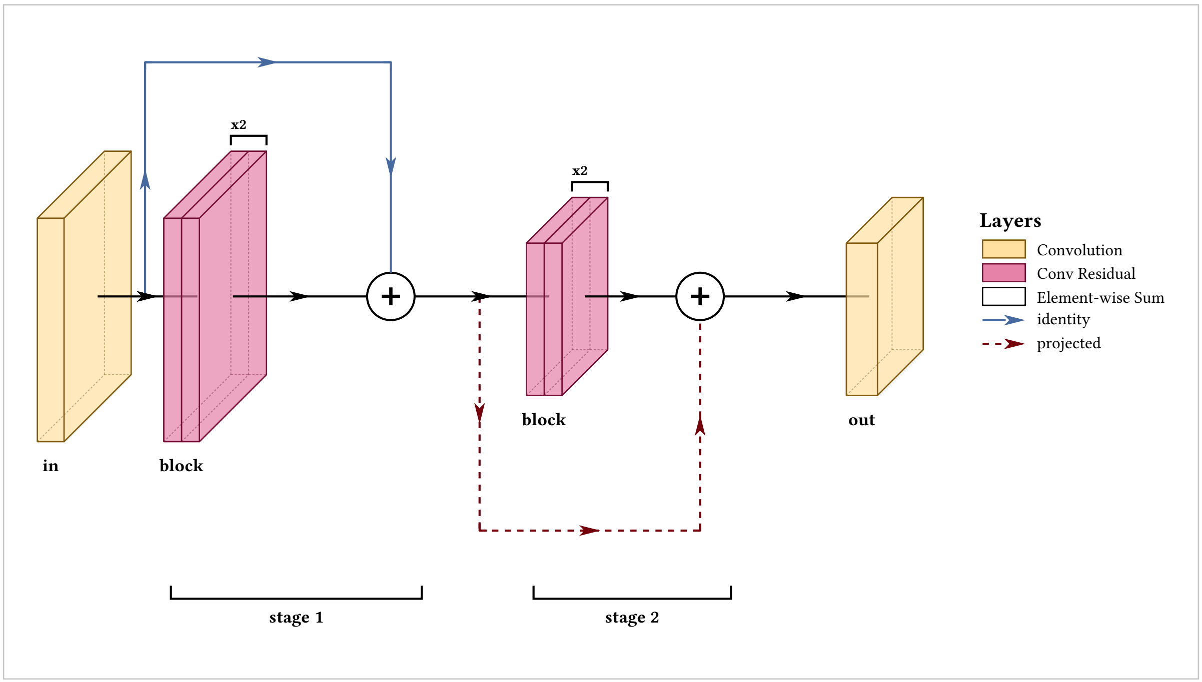
```
#let e(n, s) = conv(name: n, label: n, shape: s, channels: (s.at(0), s.at(1)))
#let d(n, s) = deconv(name: n, label: n, shape: s)

#draw-network(
  (
    e("e1", (32, 128, 128)),
    e("e2", (64, 64, 64)),
    e("e3", (128, 32, 32)),
    d("d3", (64, 64, 64)),
    d("d2", (32, 128, 128)),
    conv(name: "out", label: "out", shape: (1, 128, 128)),
  ),
  connections: (
    connection(from: "e2", to: "d3", touch-layer: true,
     color: rgb("#466A9F"), legend: "skip"),
    connection(from: "e1", to: "d2", touch-layer: true,
     color: rgb("#466A9F")),
  ),
  groups: (
    group(from: "e1", to: "e3", label: "Encoder"),
    group(from: "d3", to: "out", label: "Decoder"),
  ),
  show-legend: true,
)
```



15.3 Residual stages

`repeat` for the blocks inside a stage, a `sum` node for each add, and two route colours to separate an identity shortcut from a projected one.

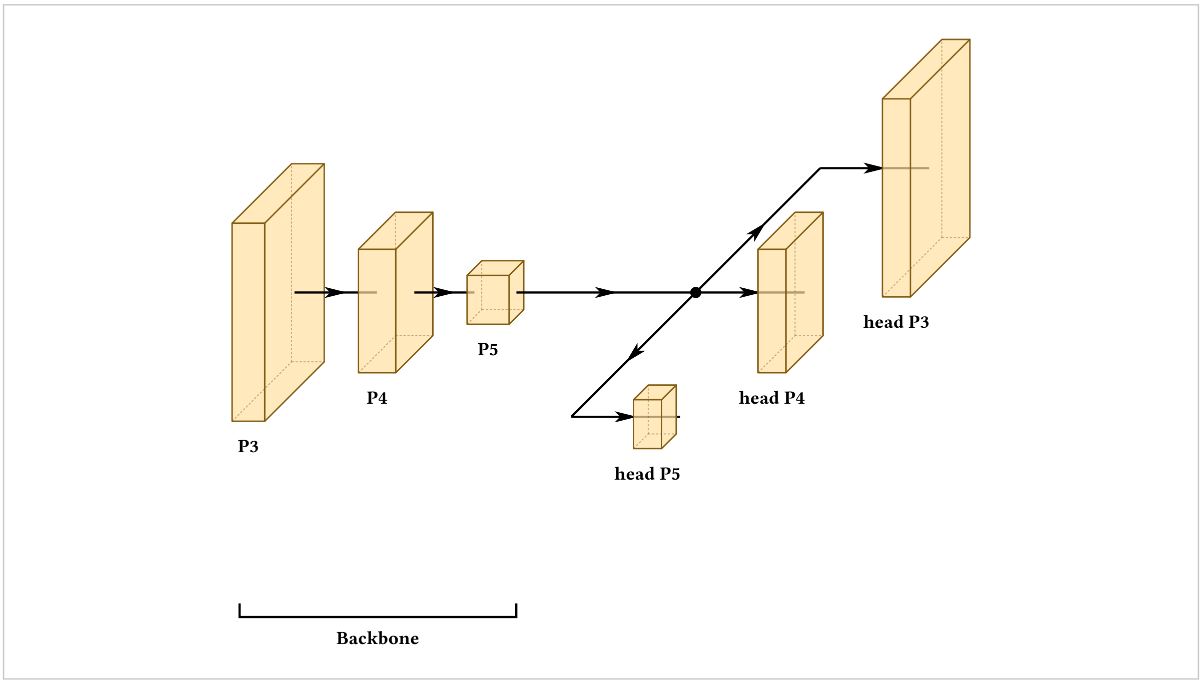
```
#draw-network(
  (
    conv(name: "in", label: "in", shape: (64, 56, 56)),
    convres(name: "b1", label: "block", shape: (64, 56, 56),
            widths: (0.3, 0.3), repeat: 2),
    sum(name: "a1"),
    convres(name: "b2", label: "block", shape: (128, 28, 28),
            widths: (0.3, 0.3), repeat: 2),
    sum(name: "a2"),
    conv(name: "out", label: "out", shape: (128, 28, 28)),
  ),
  connections: (
    connection(from: "in", to: "a1", color: rgb("#466A9F"),
     legend: "identity"),
    connection(from: "a1", to: "a2", color: rgb("#73000A"), dash: "dashed",
     legend: "projected"),
  ),
  groups: (
    group(from: "b1", to: "a1", label: "stage 1"),
    group(from: "b2", to: "a2", label: "stage 2"),
  ),
  show-legend: true,
)
```



15.4 A multi-scale head

A depth-mode branch, open at the end, is what a detector's heads look like.

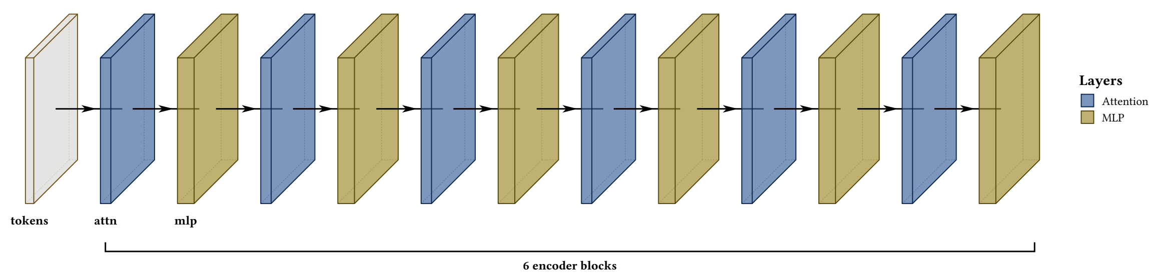
```
#draw-network(
  (
    conv(name: "p3", label: "P3", shape: (128, 40, 40)),
    conv(name: "p4", label: "P4", shape: (256, 20, 20)),
    conv(name: "p5", label: "P5", shape: (512, 10, 10)),
    branch(spread: 4, spread-mode: "depth", rejoin-lead: 3.2,
      open: "end", branches: (
        (conv(name: "h3", label: "head P3", shape: (64, 40, 40)),),
        (conv(name: "h4", label: "head P4", shape: (64, 20, 20)),),
        (conv(name: "h5", label: "head P5", shape: (64, 10, 10)),),
      )),
  ),
  groups: (group(from: "p3", to: "p5", label: "Backbone"),),
)
```



15.5 A repeated pair

`repeat` cannot express a repeated **pair** (section 11.3), so build the list instead. Twelve blocks are written once, and the count lives in the `range` rather than in the figure:

```
#let block(i) = (
  custom(name: "a" + str(i), label: if i == 1 { "attn" },
         fill: rgb("#466A9F"), width: 0.25, height: 3.5, depth: 3.5,
         legend: "Attention"),
  custom(name: "m" + str(i), label: if i == 1 { "mlp" },
         fill: rgb("#A49137"), width: 0.4, height: 3.5, depth: 3.5,
         legend: "MLP"),
)

#draw-network(
  (custom(name: "in", label: "tokens", width: 0.2,
          height: 3.5, depth: 3.5),)
    + range(1, 7).map(block).flatten(),
  groups: (group(from: "a1", to: "m6", label: "6 encoder blocks"),),
  show-legend: true,
)
```



This is the general answer to any figure that is getting long. The layer list is an array and a constructor is a function; anything Typst can do to those, you can do to a figure.

CHAPTER 16

Every option

The tables below are generated from the package itself rather than written out here, so an option cannot exist without appearing in this chapter.

16.1 Options by layer type

Type	Options
branch	branches, entry-gap, exit-gap, lead, name, open, rejoin-lead, spread, spread-mode
concat	channels, connection-label, depth, fill, height, image, label, label-anchor, label-angle, label-dx, label-dy, label-orient, legend, name, offset, opacity, repeat, shape, show-connection, width
conv	bandfill, channels, connection-label, depth, fill, height, image, label, label-anchor, label-angle, label-dx, label-dy, label-orient, legend, name, offset, opacity, repeat, shape, show-connection, show-relu, widths, xlabel, ylabel, zlabel
convres	bandfill, channels, connection-label, depth, fill, height, image, label, label-anchor, label-angle, label-dx, label-dy, label-orient, legend, name, offset, opacity, repeat, shape, show-connection, show-relu, widths, xlabel, ylabel, zlabel
convsoftmax	channels, connection-label, depth, fill, height, image, label, label-anchor, label-angle, label-dx, label-dy, label-orient, legend, name, offset, opacity, repeat, shape, show-connection, width
custom	bandfill, channels, connection-label, depth, fill, height, image, input-style, label, label-anchor, label-angle, label-dx, label-dy, label-orient, legend, name, offset, opacity, repeat, shape, show-connection, show-relu, width, widths, xlabel, ylabel, zlabel
deconv	channels, connection-label, depth, fill, height, image, label, label-anchor, label-angle, label-dx, label-dy, label-orient, legend, name, offset, opacity, repeat, shape, show-connection, width
fc	channels, connection-label, depth, fill, height, image, label, label-anchor, label-angle, label-dx, label-dy, label-orient, legend, name, offset, opacity, repeat, shape, show-connection
gap	channels, connection-label, depth, fill, height, image, label, label-anchor, label-angle, label-dx, label-dy, label-orient, legend, name, offset, opacity, repeat, shape, show-connection
input	channels, connection-label, depth, fill, height, image, input-style, label, label-anchor, label-angle, label-dx, label-dy, label-orient, legend, name, offset, opacity, repeat, shape, show-connection, width
output	channels, classes, connection-label, depth, fill, height, image, label, label-anchor, label-angle, label-dx, label-dy, label-orient, legend, name, offset, opacity, repeat, shape, show-connection, width
pool	channels, connection-label, depth, fill, height, image, label, label-anchor, label-angle, label-dx, label-dy, label-orient, legend, name, offset, opacity, repeat, shape, show-connection
softmax	channels, classes, connection-label, depth, fill, height, image, label, label-anchor, label-angle, label-dx, label-dy, label-orient, legend, name, offset, opacity, repeat, shape, show-connection, width
sum	channels, connection-label, depth, fill, height, label, label-anchor, label-angle, label-dx, label-dy, label-orient, legend, name, offset, opacity, radius, show-connection, stroke, symbol
unpool	channels, connection-label, depth, fill, height, image, label, label-anchor, label-angle, label-dx, label-dy, label-orient, legend, name, offset, opacity, repeat, shape, show-connection

16.2 Connection options

arrive-offset · clearance · color · dash · from · label · layers · legend · mode · opacity · pos · thickness · to · touch-layer · type

16.3 Group options

`color · from · label · offset · to`

16.4 Options of the MLP renderer

Constructor	Options
mlp-layer	activation, bias, count, dimmed, dropped, exclude, fill, highlighted, label, name, node-label, node-label-pos, node-size, node-style, shape, show-all, show-value-text, stroke, sub, values
mlp-gap	label, pitch
mlp-edge	arrow, dash, from, label, opacity, paint, style, sum, thickness, to
draw-mlp	activation, activation-style, arrows, bias, bias-label, bias-label-pos, collapse-to, count-badges, cutoff, debug, direction, edge-filter, edge-labels, edge-opacity, edge-stroke, edge-style, edges, input-style, io-stubs, label-pos, layer-labels, layer-pitch, matrix-labels, node-label, node-pitch, node-shape, node-size, node-stroke, palette, seed, stub-labels, title, unit, weight-colors, weight-encode, weight-range, weight-thickness, weights

Plus the exports themselves: `draw-mlp`, `mlp-content`, the `mlp-layer`, `mlp-gap` and `mlp-edge` constructors, and the `mlp-palettes` dictionary.

16.5 Arguments to draw-network

```
#draw-network(
  layers,
  connections: (),
  groups: (),
  palette: "warm",
  show-legend: false,
  legend-title: "Layers",
  main-legend: none,
  scale: 100%,
  stroke-thickness: 1,
  depth-multiplier: 0.3,
  lane-unit: 0.75,
  shape-scale: (spatial: (1.2, -3.2), channels: (0.075, 0.0)),
  auto-gap: 0.55,
  show-relu: false,
)
```

16.6 Also exported

```
depth-shear(depth, depth-multiplier: 0.3)
min-clear-offset(depth, depth-multiplier: 0.3)
```

Plus a constructor per layer type, and `connection(..)` and `group(..)`.

16.7 Coverage

> Every option exported by the package is described somewhere above.

> Every option of the MLP renderer is described somewhere above.

CHAPTER 17

Errors

Every message the package raises, and what causes it.

`unknown layer option "X" on layer N (type "T")`

`X` is not an option of that type. The message lists what is, and suggests the nearest match. A correctly spelled option belonging to a **different** type gives this too — `widths` on a `pool`, `width` on a `conv`.

`unknown layer type "X" on layer N`

Not one of the fourteen. See section 16.1.

`layer N is a plain dictionary`

Every layer comes from a constructor. Write `(type: "conv", ..)` as `conv(..)`; a dictionary of options spreads into it as `conv(..opts)`.

``connection N refers to "X", which is not the `name` of any layer``

Either the target has no `name`, or the reference is misspelled. Groups give the same message.

``connection N has no `from` ``

A route names both layers it runs between.

`` `branches` on layer N must be an array of layer arrays``

A common slip is passing one layer list rather than a list of them. Two branches of one layer each is `((conv(),), (conv(),))`.

`branch open must be "start" or "end"`

`label-orient must be "horizontal", "diagonal" or "vertical"`

`Use either label-orient or label-angle, not both`

The right anchor for an arbitrary rotation depends on where you want the text, so the package will not guess.

`shape must be (channels, height, width)`

Three numbers, in that order.

`array index out of bounds (index: 1, len: 1)`

Raised from inside `src/lib.typ` rather than as a message of the package's own. Almost always `channels` with more entries than the layer has bands, plus one. Either add matching `widths` entries or remove channel entries.

`unexpected argument: X`

Typst's own message, from a constructor given an option that does not exist. It is raised before the package sees the call, which is one reason to prefer constructors.

CHAPTER 18

Where to go next

Complete architectures are in `examples/` in the repository, and their rendered output in `gallery/`. They are ordinary Typst files, meant to be read and copied:

`examples/networks/` AlexNet, LeNet-5, VGG16, VGG19, ResNet18, U-Net, FCN-8 and YOLO26-n, plus five RGB-IR fusion variants of the last one.

`examples/features/` every predefined layer type side by side in both palettes, and a figure exercising every feature at once.

`examples/mlp/` the neuron diagrams of chapter 14 — the classic three-layer figure, a Playground-style random-weight network, a Rosenblatt perceptron, a Goodfellow-orientation textbook figure, a dropout figure, and a deep network with a gap, a skip and a recurrent loop.

`examples/imported/` figures generated from a traced PyTorch model rather than written by hand, using `tools/import_model.py`.

The YOLO26-n example is the one to read if you want to see the automatic machinery carrying a whole figure. Every block states its tensor shape, every gap is automatic, every lane height is automatic, and the three detection heads are a depth-mode branch. Nothing in it is sized or positioned by hand.

This manual is itself plain text. `doc/manual.typ` and the 179 files in `doc/examples/` are the whole of it, each example a standalone figure that compiles on its own. That source, rather than this PDF, is what to point a language model at.